STAT201 · STATISTICS · LECTURE 1

Data, Descriptive Statistics and Visualization

Which numbers and which graphs you need to describe a data set honestly

Syllabus topic 1 · week 1 (sessions 1 and 2) · 4 academic hours

LECTURE GOALS

- determine the measurement scale of a variable and know which statistics are admissible for it;
- compute the mean, median and mode by hand and choose a suitable measure of centre;
- compute the range, variance, standard deviation and coefficient of variation and interpret them in the units of the data;
- find quartiles, the interquartile range and the five-number summary;
- build a frequency table, histogram, box plot and scatter plot and read them correctly;
- apply the $1.5 \times$ IQR rule to find outliers and report honestly what was done with them;
- tell an honest graph from a misleading one.

1 Why we need statistics and how to use these notes

Statistics is the science of drawing well-founded conclusions from data when the outcome is uncertain. A bank decides whether to grant a loan; a coffee-shop chain decides where to open a new branch; a ministry announces "the average household income". Behind each decision stand numbers, and the question is always the same: *what do these numbers really say, and what do they not say?*

The course has four parts, and this week's lecture serves as a vocabulary for the other three.

Weeks 1–2	**Describe**	What do the data contain, and how likely is a given outcome?
Weeks 3–7	**Model**	Probability distributions, expected value, the central limit theorem.
Weeks 9–11	**Infer**	Confidence intervals, hypothesis tests, p-values.
Weeks 12–15	**Predict**	Regression and the difference between correlation and causation.

How to use these notes. Read a section through, then close the text and recompute the worked example by hand without peeking. Only then move on to the problems at the end of the lecture. The glossary tables also give the Russian term, because the course also runs in Russian. The rule of the semester: **every number we compute, we must be able to explain in one sentence.**

CONVENTIONS OF THIS COURSE

Decimals are written with a point (35.75, not 35,75): this is how calculators, Excel and R print them. An answer without units is not an answer: “35.75” means nothing, “35.75 million UZS per month per branch” does.

2 Data and measurement scales

2.1 Unit of observation, variable, population and sample

BASIC CONCEPTS

Unit of observation — the object about which data are collected (a coffee shop, a customer, a region, a month). **Variable** — a characteristic of a unit that can take different values (revenue, age, region). **Population** — all the units about which we want to draw a conclusion; its size is denoted N. **Sample** — the part of the population that we actually measured; its size is denoted n.

Statistics distinguishes between population *parameters* and sample *statistics*. Parameters are written with Greek letters (μ — population mean, σ — its standard deviation), sample statistics with Latin letters ($\bar{x}$, s). The formulas are often the same and only the letter changes, but the letter tells the reader whether you measured everyone or only a part. Why this matters we shall see in weeks 7 and 9.

2.2 Four measurement scales

Not every number in a table is a number in the mathematical sense. The scale on which a variable is measured determines which calculations with it are meaningful.

Scale	What it means	Example	Admissible measure of centre
Nominal	Names only, no order	Region, industry, bank name	Mode
Ordinal	There is an order, distances between values are unequal	Credit rating, satisfaction score 1–5	Mode, median
Interval	Distances are equal, no true zero	Temperature in °C, calendar year	Mode, median, mean
Ratio	Distances are equal and there is a true zero	Revenue, price, hours worked	All three; ratios are also meaningful (“twice as much”)

COMMON MISTAKE

A number written in a table is not yet a quantitative variable. The postal code 100084 is a nominal variable: its mean makes no sense. The same holds for an employee number or a region code. A test question: “Does it make sense to add two such values?”

Likert-scale scores (“1 — strongly disagree, …, 5 — strongly agree”) are formally ordinal. In practice the mean is often computed for them, but this is an assumption and it must be stated aloud.

Quantitative variables are further divided into *discrete* (the result of counting: the number of customers per day, 0, 1, 2, …) and *continuous* (the result of measuring: revenue, weight, waiting time).

2.3 Three forms of data

- **Cross-section data**: many units at one point in time. Example: the revenue of 40 coffee shops in September 2026. We assume this form in weeks 1–11.
- **Time series**: one unit at many points in time. Example: monthly inflation in Uzbekistan over 2015–2026. The order of observations matters, so observations are not independent.
- **Panel data**: many units and many periods. Example: all 14 regions, every year for ten years. This is the main object of the econometrics course ECMT301.

3 Running example: twelve coffee shops

We illustrate all the material of the lecture with the same data. These are the monthly revenues of twelve branches of one coffee-shop chain in Tashkent for August 2026, in millions of UZS. The data are already sorted in ascending order; branch 12 is located in a shopping mall.

Number	1	2	3	4	5	6	7	8	9	10	11	12
Revenue x_i	18	22	25	27	28	30	30	33	36	40	44	**96**

Here $n = 12$, the sum is $\sum x_i = 429$, the smallest value is 18, the largest 96, and the scale is ratio. The unit of observation is a branch, the variable is monthly revenue.

4 Measures of centre

A measure of centre answers the question: “which single number best describes the typical value?” There are three such numbers, and they answer it differently.

4.1 Arithmetic mean

SAMPLE MEAN

$$\bar{x} = \frac{x_1 + x_2 + \dots + x_n}{n} = \frac{1}{n}\sum_{i=1}^{n} x_i. \tag{4.1}$$

For a population the same formula gives μ, with the sum running over all N units.

EXAMPLE 1. AVERAGE BRANCH REVENUE

$\bar{x} = \dfrac{429}{12} = 35.75$ million UZS per branch per month.

The mean is the *balance point* of the data: if you picture the values as weights on a ruler, the mean is the point at which the ruler balances. This is written as an exact property:

$$\sum_{i=1}^{n}(x_i - \bar{x}) = \sum x_i - n\bar{x} = n\bar{x} - n\bar{x} = 0. \tag{4.2}$$

Deviations on one side of the mean exactly offset the deviations on the other. In our data the deviation of branch 12 is $96 - 35.75 = +60.25$, and the deviations of the other eleven branches sum to -60.25: one large value balances all the others, and that is exactly why it pulls the mean so hard.

4.2 Median

MEDIAN

The value that splits an ordered series in half: half of the observations are no larger, half are no smaller. Algorithm:

1. sort the values in ascending order;
2. find the position $(n+1)/2$;
3. if n is odd, the median is the value in that position; if n is even, the median is the average of the two values on either side of the position.

EXAMPLE 2. MEDIAN REVENUE

For $n = 12$ the position is $(12+1)/2 = 6.5$ — between the sixth and seventh values. Both equal 30, so $\text{Me} = (30+30)/2 = 30.0$ million UZS. Half of the branches earn less than 30 million, half earn more. The median ignores *how much* more, which is exactly why branch 12 does not move it.

4.3 Mode

MODE

The most frequently occurring value. For continuous data one speaks of the modal interval — the tallest bar of the histogram. There may be several modes, or none at all.

In our data the value 30 occurs twice and the rest once each, so $\text{Mo} = 30$. The mode is the only measure of centre admissible for nominal data: “the most frequent region of customers”.

4.4 Weighted mean

If the values have different importance (weight), the ordinary mean is misleading.

$$\bar{x}_w = \frac{\sum_{i=1}^{n} w_i x_i}{\sum_{i=1}^{n} w_i}. \tag{4.3}$$

EXAMPLE 3. AVERAGE PRICE OF A LOAF

Three shops sold 200, 500 and 300 loaves in a week at prices of 4 000, 4 500 and 5 000 UZS. The simple average of the prices is $4\,500$, but it ignores sales volumes. The weighted mean with quantities as weights:

$$\bar{x}_w = \frac{200 \cdot 4000 + 500 \cdot 4500 + 300 \cdot 5000}{200 + 500 + 300} = \frac{4\,550\,000}{1\,000} = 4\,550 \text{ UZS}.$$

This is the actual average price paid by a customer.

4.5 Outliers and the choice of a measure of centre

What happens if we exclude branch 12 from the shopping mall?

	Mean	Median	Mode
All 12 branches	35.75	30.0	30
Without branch 12 ($n = 11$)	30.27	30.0	30
Change	-5.48	**0.0**	**0**

One observation out of twelve moved the mean by 5.48 million UZS (15% of its value), while the median did not change. A statistic that barely reacts to extreme values is called **robust**. The median is robust, the mean is not. Neither is "wrong", but reporting the mean of 35.75 would describe a chain to which none of the other eleven branches belongs.

Use	When
Mean	The distribution is roughly symmetric and you need to add up or forecast totals (total revenue $= n\bar{x}$).
Median	The data are skewed or contain outliers: incomes, house prices, waiting times.
Mode	The variable is categorical or you need the most frequent option.

COMMON MISTAKE

The mean need not resemble any real observation: 35.75 is the revenue of none of the twelve branches. The "average person" often does not exist.

5 Measures of spread

One number for the centre is not enough. Consider two stalls with the same average weekly revenue of 30 million UZS:

	Weekly sales	s
Stall A	29, 30, 30, 31, 30	0.71
Stall B	5, 55, 12, 48, 30	21.78

At stall A staffing and inventory plan themselves; at stall B every week is a gamble. The mean and median are the same, the only difference is the spread.

5.1 Range

$$R = x_{\max} - x_{\min} = 96 - 18 = 78.$$

The range uses only two observations, so a single error in the data ruins it. It is worth reporting, but should not be relied on.

5.2 Variance and standard deviation

A natural idea: average the deviations from the mean. But by formula (4.2) they sum to zero. To keep the deviations from cancelling each other, they are squared.

SAMPLE VARIANCE AND STANDARD DEVIATION

$$s^2 = \frac{\sum_{i=1}^{n} (x_i - \bar{x})^2}{n-1}, \qquad s = \sqrt{s^2}. \tag{5.1}$$

The variance is measured in squared units of the data ("million UZS squared") and is an intermediate step, not a result. The standard deviation s returns to the original units. It is read as follows: a typical branch lies about s away from the mean.

EXAMPLE 4. FIVE BRANCHES BY HAND

The number of customers per day at one branch: 20, 25, 30, 35, 40. The mean is $\bar{x} = 150/5 = 30$.

x_i	$x_i - \bar{x}$	$(x_i - \bar{x})^2$
20	-10	100
25	-5	25
30	0	0
35	$+5$	25
40	$+10$	100
$\sum = 150$	$\sum = 0$ (always)	$\sum = 250$

$$s^2 = \frac{250}{5-1} = 62.5, \quad s = \sqrt{62.5} = 7.91 \text{ customers.}$$

For the twelve coffee shops, computing from the definition is inconvenient, so we use the working formula, equivalent to (5.1):

$$s^2 = \frac{\sum x_i^2 - n\bar{x}^2}{n-1}. \tag{5.2}$$

Here $\sum x_i^2 = 19\,883$ and $n\bar{x}^2 = 12 \cdot 35.75^2 = 15\,336.75$, so

$$s^2 = \frac{19\,883 - 15\,336.75}{11} = \frac{4\,546.25}{11} = 413.30, \qquad s = \sqrt{413.30} = 20.33 \text{ million UZS.}$$

Why $n-1$ in the denominator and not n

The mean $\bar{x}$ is computed from the same sample, so the data are on average closer to $\bar{x}$ than to the unknown μ, and dividing by n systematically understates the spread. In addition, by formula (4.2) the deviations from $\bar{x}$ are tied by one condition (they sum to zero), so only $n-1$ of them vary freely. The number $n-1$ is called the number of *degrees of freedom*.

EXAMPLE 5. A CHECK ON A TINY POPULATION

Let the population consist of two equally likely values, 0 and 2. Its variance is $\sigma^2 = 1$. Take samples of two values with replacement: there are four and they are equally likely.

Sample	$\bar{x}$	$\sum(x_i-\bar{x})^2$
(0, 0)	0	0
(0, 2)	1	2
(2, 0)	1	2
(2, 2)	2	0

The average of $\sum(x_i-\bar{x})^2$ over the four samples is $(0+2+2+0)/4 = 1$. Dividing by $n-1=1$ gives 1 on average — correct, equal to σ^2. Dividing by $n=2$ gives 0.5 on average — half of the truth. Conclusion: with $n-1$ the sample variance is an *unbiased* estimator. More on this in week 9.

5.3 Coefficient of variation

Standard deviations cannot be compared if the data are measured in different units or on different scales. A relative measure serves this purpose:

$$CV = \frac{s}{\bar{x}} \times 100\%. \tag{5.3}$$

EXAMPLE 6. COFFEE SHOPS AND A BANK

Coffee-shop chain: $\bar{x} = 35.75$, $s = 20.33$, so $CV = 56.9\%$. Bank branches: $\bar{x} = 1\,240$ million, $s = 186$ million, so $CV = 15.0\%$. The bank's standard deviation in absolute money is nine times larger, but relative to their own scale its branches are far more homogeneous.

COMMON MISTAKE

Never compare two standard deviations measured in different units: compare coefficients of variation. But CV makes sense only for ratio-scale variables with a positive mean (for temperature in °C, for example, it is meaningless: the zero is arbitrary).

5.4 What happens to the measures under a data transformation

Let each value be recomputed by the rule $y_i = a + b\,x_i$ (change of units, a bonus, indexation).

LINEAR TRANSFORMATION

$$\bar{y} = a + b\bar{x}, \qquad \mathrm{Me}_y = a + b\,\mathrm{Me}_x, \qquad s_y = |b|\,s_x.$$

Adding a constant a shifts the centre but **does not change the spread**. Multiplying by b stretches both the centre and the spread. The coefficient of variation does not change under multiplication ($a = 0$) but does change under a shift.

6 Measures of position: quartiles and the five-number summary

Quartiles split ordered data into four parts with equal numbers of observations. Q_2 is the median; Q_1 is the median of the lower half; Q_3 is the median of the upper half. If n is odd, the median itself belongs to neither half.

EXAMPLE 7. QUARTILES OF THE COFFEE SHOPS

Lower half: 18, 22, 25, 27, 28, 30. Its median is $Q_1 = (25 + 27)/2 = 26$.
Upper half: 30, 33, 36, 40, 44, 96. Its median is $Q_3 = (36 + 40)/2 = 38$.
Interquartile range $\mathrm{IQR} = Q_3 - Q_1 = 38 - 26 = 12$.

FIVE-NUMBER SUMMARY

$$x_{\min}, \quad Q_1, \quad \mathrm{Me}, \quad Q_3, \quad x_{\max} \qquad \Rightarrow \qquad 18,\ 26,\ 30,\ 38,\ 96.$$

The middle 50% of branches earn between 26 and 38 million UZS, i.e. lie in a band 12 million wide; compare this with the range of 78. These numbers alone show that one observation behaves unusually.

QUARTILES IN SOFTWARE

There are several ways to compute quartiles and they give slightly different numbers. For our data: the halves method (used in this course) gives $Q_1 = 26$ and $Q_3 = 38$; the Excel function `QUARTILE.INC` and R by default give 26.5 and 37; the function `QUARTILE.EXC` gives 25.5 and 39. For large n the difference is small. On paper in this course we use the halves method, and when working with software we always state which method was used.

A *percentile* generalizes the quartile: the p-th percentile is the value below which $p\%$ of the observations lie. Quartiles are the 25th, 50th and 75th percentiles.

7 Frequency distributions and histograms

When there are many observations, a list of values is unreadable and the data are grouped into intervals.

Construction algorithm. (1) Choose the interval width. (2) Define intervals closed on the left: $[15, 25)$ includes 15 but not 25. (3) Count the observations in each interval — the *frequency* f. (4) Find the *relative frequency* f/n and the *cumulative frequency*.

Interval, million UZS	Frequency f	Rel. frequency	Cumulative
[15, 25)	2	16.7%	2
[25, 35)	6	50.0%	8
[35, 45)	3	25.0%	11
[45, 55)	0	0.0%	11
[55, 65)	0	0.0%	11
[65, 75)	0	0.0%	11
[75, 85)	0	0.0%	11
[85, 95)	0	0.0%	11
[95, 105)	1	8.3%	12
Total	**12**	**100%**	

Check: the frequencies sum to $n = 12$, the relative frequencies to 100%.

A **histogram** is the graph of this table: intervals along the horizontal axis, frequencies along the vertical. The bars touch because revenue is continuous. The main decision in constructing it is the interval width.

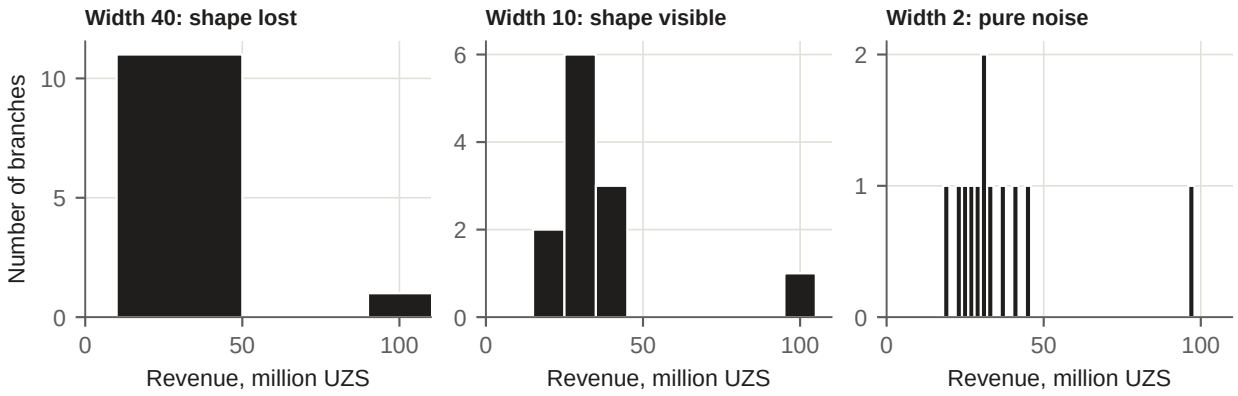


Fig. 1. The same 12 values with different interval widths. Intervals that are too wide erase the shape, intervals that are too narrow leave noise.

Rule of thumb: start with a number of intervals close to $\sqrt{n}$ or to $1 + \log_2 n$ (Sturges' rule) and adjust until the shape stops changing; always state the interval width in the caption. For $n = 12$ any histogram is crude; the rule works well from about $n \approx 50$. In the figure, with width 10 you can clearly see the core (25–45 million), the long right tail and the empty gap before the lone bar at 96.

8 Shape of a distribution

Shape describes symmetry and tails. Three main situations:

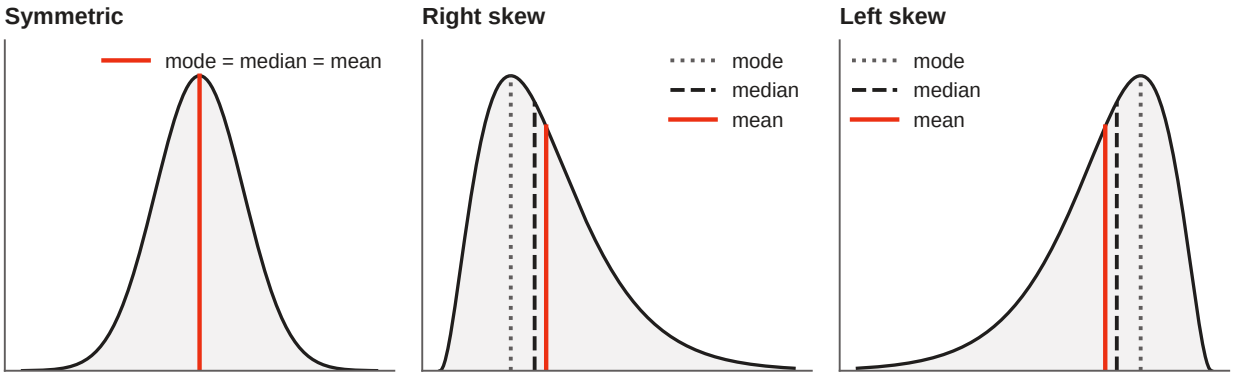


Fig. 2. Positions of the mode, median and mean for three distribution shapes.

- **Symmetric**: mean = median = mode. Human height, measurement errors, a well-designed exam.
- **Right-skewed** (long tail to the right): mode < median < mean. Incomes, house prices, waiting times, our coffee-shop chain.
- **Left-skewed** (long tail to the left): mean < median < mode. Age at death, scores on an easy test.

The mean is "dragged" along with the tail while the median hardly moves — so the order of mode, median and mean hints at the shape.

Skewness can be measured by a number. The simplest coefficient (Pearson's second coefficient):

$$Sk = \frac{3(\bar{x} - \mathrm{Me})}{s}. \tag{8.1}$$

For the coffee shops: $Sk = \dfrac{3(35.75 - 30)}{20.33} = \dfrac{17.25}{20.33} = +0.85$. Guidelines: $Sk \approx 0$ — symmetry; from 0.5 to 1 — moderate right skew; above 1 — strong; a negative value — tail to the left. Software usually reports a different coefficient (based on the third moment); its sign is the same.

9 The box plot and outliers

A **box plot** draws the five-number summary: a box from Q_1 to Q_3, a line inside the box at the median (not the mean), "whiskers" extending to the most distant observations that do not go beyond the fences, and observations beyond the fences marked as separate points.

THE $1.5 \times$ IQR RULE

$$\text{lower fence} = Q_1 - 1.5 \cdot \mathrm{IQR}, \tag{9.1}$$

$$\text{upper fence} = Q_3 + 1.5 \cdot \mathrm{IQR}. \tag{9.2}$$

An observation beyond the fences is called an **outlier**.

EXAMPLE 8. IS BRANCH 12 AN OUTLIER?

$\mathrm{IQR} = 12$, so $1.5 \cdot 12 = 18$. The fences are $26 - 18 = \mathbf{8}$ and $38 + 18 = \mathbf{56}$. Since $96 > 56$, branch 12 is an outlier. The whiskers go to 18 and 44 — the most extreme values inside the fences.

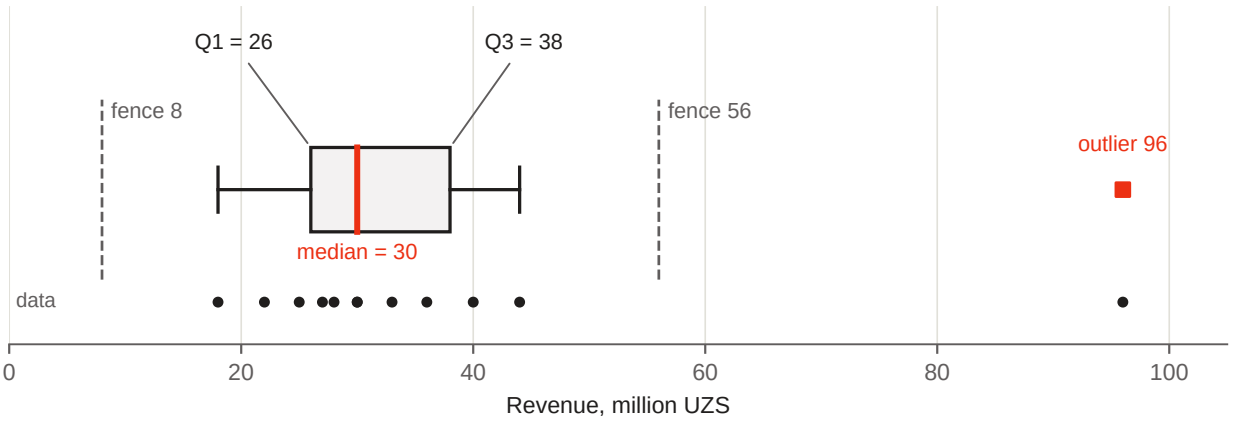


Fig. 3. Box plot of the revenue of the twelve branches. The data themselves are shown below the box.

An outlier is a definition, not an opinion. The rule gives every analyst the same answer.

9.1 What to do with an outlier

The rule only *flags* an observation for inspection; it does not say the observation is wrong. There are four admissible actions:

1. **Check**: is it a typo (96 instead of 9.6)? Is it a different unit of measurement?
2. **Explain**: a branch in a shopping mall has a different business model.
3. **Report both results**: "mean 35.75 million for all branches; 30.27 million without the mall branch".
4. **Change the statistic**: use the median and IQR.

> **AN INADMISSIBLE ACTION**
>
> Quietly deleting an outlier so the data "look better". Any exclusion of observations must be recorded in the text of the report.

9.2 Missing values

An outlier is not the only data problem. If revenue is not recorded for 3 of 40 coffee shops, you must report how many observations were actually used and think about *why* the values are missing. If the gaps are random, the loss is small. But if mostly loss-making branches are missing because they did not file their reports, the mean of the remaining ones is systematically too high. The rule is the same as for outliers: hide nothing and describe how missing values were handled.

10 The scatter plot

When each unit has two numerical variables, their relationship is shown by a **scatter plot**: one point per unit. The explanatory variable is plotted horizontally, the response vertically. Let us add to our data the branches' advertising spend x (million UZS) and revenue y (million UZS):

Branch	1	2	3	4	5	6	7	8	9	10	11	12
Advertising x	1.0	1.4	1.6	1.9	2.0	2.3	2.4	2.8	3.1	3.5	3.9	5.6
Revenue y	18	22	25	27	28	30	30	33	36	40	44	96

(Illustrative data.)

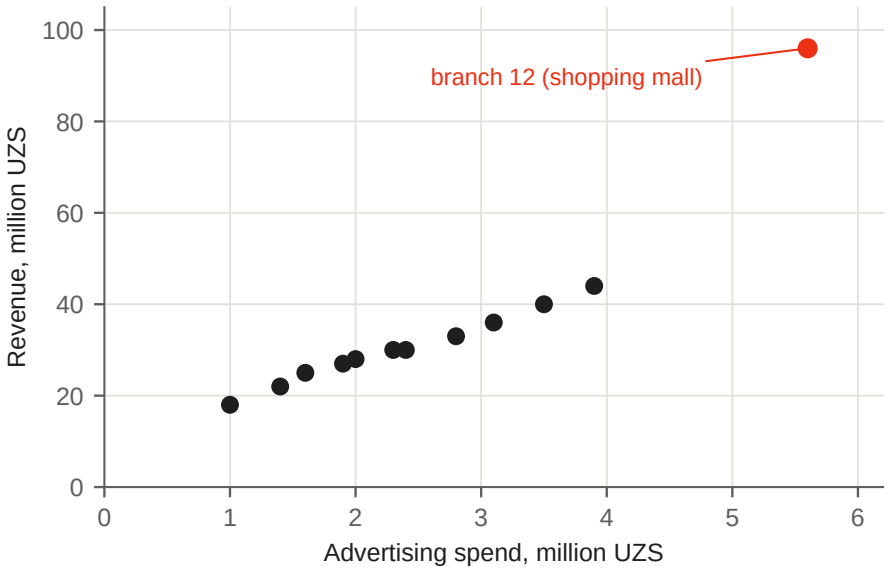


Fig. 4. Advertising spend and revenue of the twelve branches.

Describing a scatter plot, one speaks of four things:

- **Direction**: positive (as advertising grows, revenue grows).
- **Form**: roughly linear. A linear model is justified only for a straight band of points.
- **Strength**: strong. In week 6 we shall learn to measure it by a number — the correlation coefficient r (for these data $r = 0.93$, and without branch 12 it is $r = 0.996$).
- **Unusual points**: branch 12.

> **COMMON MISTAKE**
>
> That advertising and revenue grow together does not prove that advertising *causes* revenue to grow: perhaps large branches simply spend more on advertising. We shall examine the difference between correlation and causation in week 13.

11 How to choose a graph and not deceive the reader

What you have	Which graph	Why
One categorical variable	Bar chart	Length is the easiest thing to compare.
One numerical variable	Histogram or box plot	Shape and outliers are visible at once.
One numerical variable, several groups	Side-by-side box plots	Centre, spread and shape of the groups in one frame.
Two numerical variables	Scatter plot	You see the joint picture, not two separate summaries.
A numerical variable over time	Line chart	Order matters and the line conveys it.
Parts of a whole	Still a bar chart	Angles are hard to read; a pie chart is unsuitable with more than three shares.

11.1 Honest numbers, dishonest picture

- **A truncated axis.** If the y-axis does not start at zero, a small increase looks like a cliff. Bar charts must start at zero.
- **Area instead of length.** Doubling both height and width gives four times the "ink" for a doubled value.
- **A pie chart with seven shares.** Nobody can rank the shares by size; a sorted bar chart answers that question at once.
- **Two axes on one graph.** By choosing the scales one can make lines cross and create a relationship that does not exist.

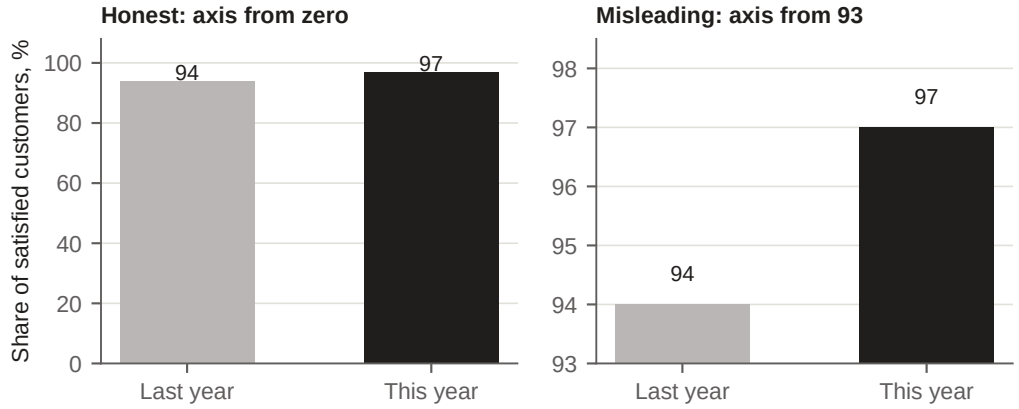


Fig. 5. The same data: the share of satisfied customers rose from 94% to 97%, i.e. by 3.2%. In the right-hand graph this year's bar looks four times taller than last year's.

A **graph caption** states the conclusion, not the type of graph. Bad: "Box plot of revenue". Good: "One mall branch earns three times the median".

12 Key terms

Term	*In Russian*	Meaning
Population	*Генеральная совокупность*	all units about which the conclusion is drawn (N, μ, σ)
Sample	*Выборка*	the measured part of the population (n, $\bar{x}$, s)
Scale of measurement	*Шкала измерения*	nominal, ordinal, interval, ratio
Mean	*Среднее*	the sum divided by the number of values; the balance point
Median	*Медиана*	the middle of the ordered series
Mode	*Мода*	the most frequent value
Robust	*Устойчивый*	hardly dependent on extreme values
Range	*Размах*	$x_{\max} - x_{\min}$
Variance	*Дисперсия*	the sum of squared deviations divided by $n-1$
Standard deviation	*Стандартное отклонение, СКО*	the square root of the variance, in the units of the data
Coefficient of variation	*Коэффициент вариации*	$s/\bar{x}$, a dimensionless relative measure of spread
Quartile	*Квартиль*	points dividing the data into quarters
IQR (interquartile range)	*Межквартильный размах*	$Q_3 - Q_1$
Skewness	*Асимметрия*	lack of symmetry of a distribution
Box plot	*Ящик с усами*	graph of the five-number summary
Outlier, fence	*Выброс, граница выброса*	a value beyond $Q_1 - 1.5\text{IQR}$, $Q_3 + 1.5\text{IQR}$
Scatter plot	*Диаграмма рассеяния*	points (x_i, y_i) for two variables
Histogram	*Гистограмма*	graph of frequencies by interval

Formulas of Lecture 1

$$\bar{x} = \frac{1}{n}\sum x_i, \qquad \bar{x}_w = \frac{\sum w_i x_i}{\sum w_i}, \qquad \sum (x_i - \bar{x}) = 0,$$

$$s^2 = \frac{\sum (x_i - \bar{x})^2}{n-1} = \frac{\sum x_i^2 - n\bar{x}^2}{n-1}, \qquad s = \sqrt{s^2}, \qquad CV = \frac{s}{\bar{x}} \cdot 100\%,$$

$$R = x_{\max} - x_{\min}, \qquad \text{IQR} = Q_3 - Q_1, \qquad Sk = \frac{3(\bar{x} - \text{Me})}{s},$$

outlier fences: $Q_1 - 1.5\,\text{IQR}$, $Q_3 + 1.5\,\text{IQR}$,

$y = a + bx \;\Rightarrow\; \bar{y} = a + b\bar{x}, \;\; s_y = |b|\, s_x.$

13 Practice problems

> Problem level is shown in square brackets: **[A]** — basic, tests understanding of definitions; **[B]** — standard computation; **[C]** — advanced, requires a derivation or an interpretation. Decimals are written with a point, as in a calculator, Excel and R. If a problem needs tables, use the appendix "Statistical Tables".

Problem 1 [A] For each variable state the measurement scale and which measures of centre (mode, median, mean) are admissible:

(a) employee number;
(b) years of education;
(c) satisfaction with a service on a scale of 1–5;
(d) monthly salary in UZS;
(e) region of residence;
(f) year of birth;
(g) air temperature in °C;
(h) place achieved in a competition (1st, 2nd, …).

Problem 2 [A] True or false? Justify briefly.

(a) The median does not depend on the size of the extreme values.
(b) If the mean is greater than the median, the distribution is, as a rule, right-skewed.
(c) The sum of deviations from the median is always zero.
(d) The variance can be negative.
(e) If all values are multiplied by 2, the standard deviation doubles and the coefficient of variation does not change.
(f) The line inside the box of a box plot shows the mean.

Problem 3 [B] A courier's number of deliveries over seven days: 12, 15, 15, 18, 22, 25, 61. Find the mean, median, mode, range, variance, standard deviation and coefficient of variation. Use the $1.5 \times$ IQR rule to check for outliers. Which measure of centre would you publish and why? Answer in one sentence.

Problem 4 [B]

(a) A bank issued three loans: 50 million UZS at 18%, 30 million at 22% and 20 million at 24%. Find the average interest rate of the portfolio (simple and weighted) and explain the difference.
(b) A student got 85 points in a 5-credit course, 70 points in a 4-credit course and 90 points in a 3-credit course. Find the weighted average score.

Problem 5 [B] Monthly rent of 15 apartments (million UZS): 2.4, 2.6, 2.8, 3.0, 3.1, 3.2, 3.5, 3.6, 3.8, 4.0, 4.2, 4.5, 4.8, 5.0, 9.5.

(a) Find the five-number summary and the IQR.
(b) Find the outlier fences and identify the outliers.
(c) Describe the box plot: where do the whiskers end?

(d) Compare the mean and the median. What do they say about the shape of the distribution? Which measure would you report?

Problem 6 [B] Number of orders at a coffee shop over 30 days: 14, 18, 19, 21, 22, 23, 24, 24, 25, 26, 27, 27, 28, 29, 29, 30, 31, 31, 32, 33, 34, 35, 36, 37, 38, 40, 42, 45, 49, 58.

(a) Build a frequency table with intervals $[10, 20), [20, 30), \dots, [50, 60)$: frequency, relative frequency, cumulative frequency.
(b) On what share of days were there fewer than 30 orders?
(c) State the modal interval.
(d) Find the mean and median from the raw data and compare them. What does this say about the shape?

Problem 7 [B]

(a) The average salary in firm A is 2.4 million UZS with a standard deviation of 0.6 million; in firm B it is 5.0 million with a standard deviation of 1.0 million. Where are salaries more homogeneous?
(b) The return on stock X: mean 8%, standard deviation 12%; on stock Y: mean 5%, standard deviation 6%. Which stock has less risk per unit of return?

Problem 8 [C]

(a) Prove that $\sum(x_i - \bar{x}) = 0$.
(b) The salaries of five employees (million UZS): 2.0, 2.5, 3.0, 3.5, 4.0. Find $\bar{x}$, s and CV. Then each salary is raised by 10% and a bonus of 0.2 million is added: $y = 1.1x + 0.2$. Find $\bar{y}$, s_y, CV_y and explain why CV changed.

Problem 9 [C] Monthly incomes of 10 households (million UZS): 2.1, 2.4, 2.6, 2.8, 3.0, 3.1, 3.4, 3.9, 4.5, 18.2. The ministry reports “average income 4.6 million”, the trade union “typical income 3.05 million”. Both use the same data honestly.

(a) Verify both numbers.
(b) How many of the ten households have an income below the “average”?
(c) Check for outliers.
(d) Which number should a family planning its budget use, and why?

Problem 10 [C] A company reports that the share of satisfied customers rose from 94% to 97%. In the bar chart the y-axis starts at 93%.

(a) By how many percentage points and by what relative percentage did the share grow?
(b) How many times taller does this year’s bar look than last year’s?
(c) How would you fix the graph? Write a caption for the corrected graph.

Problem 11 [C] Hours of preparation x and score y for eight students:
$(1, 45), (2, 52), (3, 58), (4, 61), (5, 70), (6, 74), (7, 79), (8, 60)$.

(a) Draw the scatter plot.
(b) Describe it in four words: direction, form, strength, unusual points.
(c) What will you do with the point $(8, 60)$ before drawing conclusions?
(d) Can we claim that more hours of preparation *cause* a higher score?

STAT201 · STATISTICS · LECTURE 2

Foundations of Probability Theory

Events, Kolmogorov's axioms, the addition and multiplication rules, independence and combinatorics

Syllabus topic 2 · week 2 · 4 academic hours

LECTURE GOALS

- write down a sample space and express events as its subsets;
- state the three Kolmogorov axioms and use them to reject impossible statements;
- apply the complement, addition and multiplication rules to business data;
- read joint, marginal and conditional probabilities from a contingency table and check independence;
- distinguish mutually exclusive events from independent events;
- count the number of ways using factorials, permutations, arrangements and combinations and apply them to compute probabilities.

1 From description to probability

Last week we described data that already exist: what happened. Now we turn to the question of what *may happen*. Probability is the language in which statistics speaks about uncertainty, and all the conclusions of the last weeks of the course depend on it: confidence intervals and p-values (weeks 9–13) are statements about probability.

Probability theory is built like geometry: first a few simple definitions and axioms, then the rules that follow from them. Once you master this lecture, you will be able to check any probability calculation for consistency.

2 Random experiment, sample space, events

BASIC CONCEPTS

Random experiment — a procedure whose result is not known in advance. **Sample space** S — the set of all possible results, each written once. **Event** A — any subset of S (a set of outcomes that we ask about).

EXAMPLE 1. ONE AND TWO DICE

One die: $S = \{1, 2, 3, 4, 5, 6\}$; the event "an even number came up" is the subset $A = \{2, 4, 6\}$. Two dice: S consists of 36 ordered pairs (i, j). The event "the sum equals 7" consists of six pairs: $(1, 6), (2, 5), (3, 4), (4, 3), (5, 2), (6, 1)$.

1st \ 2nd	1	2	3	4	5	6
1	2	3	4	5	6	**7**
2	3	4	5	6	**7**	8
3	4	5	6	**7**	8	9
4	5	6	**7**	8	9	10
5	6	**7**	8	9	10	11
6	**7**	8	9	10	11	12

If all outcomes are equally likely, the probability of an event equals the share of favourable outcomes:

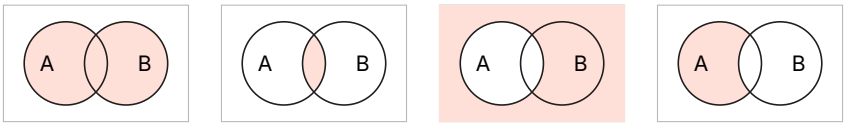
$$P(A) = \frac{\text{number of favourable outcomes}}{\text{total number of outcomes}}. \tag{2.1}$$

From the table: $P(\text{sum} = 7) = 6/36 = 0.167$; $P(\text{sum} = 12) = 1/36 = 0.028$; $P(\text{sum} > 9) = 6/36 = 0.167$; $P(\text{both even}) = 9/36 = 0.250$.

COMMON MISTAKE

The 36 ordered pairs are equally likely, but the 11 possible sums are not. A sum of 7 occurs six times as often as a sum of 12 (Fig. 1). Formula (2.1) may be applied only after checking that the outcomes really are equally likely.

2.1 Operations on events



$A \cup B$ — A or B (or both) | $A \cap B$ — A and B together | A' — not A | $A \cap B'$ — A but not B

- **Union** $A \cup B$ — A occurred, or B, or both.
- **Intersection** $A \cap B$ (also written AB) — both A and B occurred.
- **Complement** A' — everything that is not in A occurred.
- Events are **mutually exclusive** if they cannot occur at the same time: $A \cap B = \emptyset$. The decision on one loan application cannot be both "approved" and "rejected".

"Neither A nor B" is written as $(A \cup B)' = A' \cap B'$.

3 Kolmogorov's axioms and sources of probability

KOLMOGOROV'S AXIOMS, 1933

1. $0 \le P(A) \le 1$ for any event A.
2. $P(S) = 1$: a certain event has probability 1.
3. If A and B are mutually exclusive, then $P(A \cup B) = P(A) + P(B)$.

Everything else follows from these three axioms. For example, $P(\emptyset) = 0$, and if A is part of B, then $P(A) \le P(B)$. The main practical consequence: any statement about probabilities that violates the axioms is impossible. For example, one cannot have $P(A) = 0.20$ and $P(A \cap B) = 0.35$, because the intersection is part of A and cannot be more probable than it.

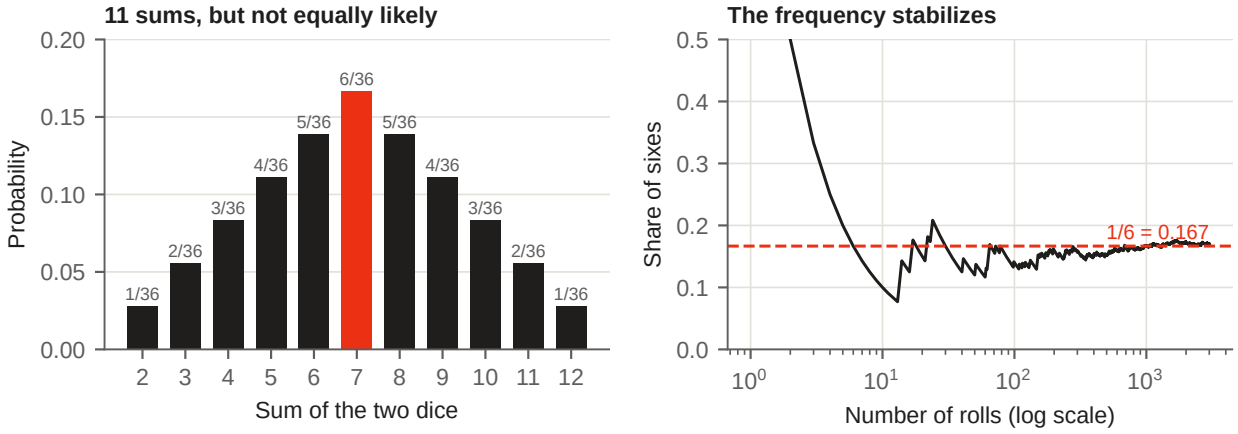


Fig. 1. Left: the sum of two dice, the 11 values are not equally likely. Right: the share of sixes in successive rolls of one die fluctuates at the start but over time approaches $1/6$.

3.1 Three sources of probability

- **Classical**: $P(A) =$ favourable/all; requires equally likely outcomes (dice, cards, a draw).
- **Frequentist**: $P(A) \approx$ number of occurrences/number of trials. Example: of 400 applications the bank approved 260, so $P(\text{approval}) \approx 0.65$. The right-hand panel of Fig. 1 shows why this works: the more trials, the closer the frequency to the "true" probability (the law of large numbers, week 7).
- **Subjective**: a weighted degree of belief ("a 70% probability of a rate cut"). It too must obey the axioms.

4 The complement and addition rules

COMPLEMENT RULE

$$P(A') = 1 - P(A). \tag{4.1}$$

Especially useful for "at least one" questions: it is easier to find the probability that *none occurs* and subtract it from 1.

EXAMPLE 2. CALL CENTRE

8% of calls are lost (the caller hangs up). Calls are independent; four came in. The probability that none is lost is $0.92^4 = 0.716$, so

$$P(\text{at least one lost}) = 1 - 0.92^4 = 1 - 0.716 = \mathbf{0.284}.$$

ADDITION RULE

$$P(A \cup B) = P(A) + P(B) - P(A \cap B). \tag{4.2}$$

The subtraction is needed so as not to count the intersection twice. For mutually exclusive events $P(A \cap B) = 0$ and what remains is $P(A \cup B) = P(A) + P(B)$.

EXAMPLE 3. BANK CUSTOMERS

Of 200 bank customers, 120 hold a debit card (D), 60 a credit card (C), 30 both. Then $P(D) = 0.60$, $P(C) = 0.30$, $P(D \cap C) = 0.15$ and

$$P(D \cup C) = 0.60 + 0.30 - 0.15 = \mathbf{0.75}, \qquad P(\text{no cards}) = 1 - 0.75 = \mathbf{0.25}\ (50 \text{ customers}).$$

If we had simply added $0.60 + 0.30 = 0.90$, the customers with two cards would have been counted twice.

COMMON MISTAKE

Probabilities may be added without subtracting the intersection only for mutually exclusive events. A typical mistake: "the default probability of each of three borrowers is 5%, so the probability that at least one goes bankrupt is 15%". This is wrong: the events are not mutually exclusive (several defaults may occur at once), and the sum overstates the answer.

5 Contingency table: joint, marginal and conditional probabilities

Data on two categorical attributes are conveniently written in a contingency table. Here are 400 loan applications: T — the applicant is from Tashkent, A — the application was approved.

	Approved (A)	Rejected (A')	Total
Tashkent (T)	168	72	240
Other regions (T')	92	68	160
Total	**260**	**140**	**400**

Three kinds of probabilities can be read from a single table:

- **Marginal** (from the total rows and columns): $P(A) = 260/400 = 0.65$; $P(T) = 240/400 = 0.60$.
- **Joint** (from the cells): $P(A \cap T) = 168/400 = 0.42$; $P(A' \cap T') = 68/400 = 0.17$.
- **Conditional** (within a row or column): $P(A \mid T) = 168/240 = 0.70$; $P(A \mid T') = 92/160 = 0.575$.

CONDITIONAL PROBABILITY

$P(A \mid B)$ is read "the probability of A given B" and is computed inside B:

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}, \qquad P(B) > 0. \tag{5.1}$$

In the table: $P(A \mid T) = \dfrac{168/400}{240/400} = \dfrac{168}{240} = 0.70$. The denominator narrowed from 400 applications to the 240 from Tashkent. We shall study conditional probability in detail next week.

6 The multiplication rule and independence

Let us rewrite (5.1) in a form convenient for calculation:

$$P(A \cap B) = P(A) \cdot P(B \mid A). \tag{6.1}$$

"For both events to occur, the first must occur, and then the second, given that the first occurred."

INDEPENDENT EVENTS

Events A and B are **independent** if the occurrence of one does not change the probability of the other:

$$P(A \cap B) = P(A)\,P(B), \qquad \text{equivalently} \quad P(A \mid B) = P(A). \tag{6.2}$$

EXAMPLE 4. ARE REGION AND APPROVAL INDEPENDENT?

From the table $P(A) \cdot P(T) = 0.65 \times 0.60 = 0.39$, while in fact $P(A \cap T) = 0.42$. Since $0.42 \neq 0.39$, the events are *dependent*: approval is more likely for applicants from Tashkent (0.70 against the overall 0.65).

MUTUALLY EXCLUSIVE $\neq$ INDEPENDENT

Mutually exclusive events cannot occur together: $P(A \cap B) = 0$. This answers the question "is it possible for both at once?". **Independent** events carry no information about each other: $P(A \cap B) = P(A)P(B)$. This answers the question "does one change the probability of the other?". If $P(A) > 0$, $P(B) > 0$ and the events are mutually exclusive, they are *necessarily dependent*: having learned that A occurred, we know for certain that B did not.

For several independent events the probabilities are simply multiplied: $P(A_1 \cap \cdots \cap A_k) = P(A_1) \cdots P(A_k)$.

EXAMPLE 5. THREE BORROWERS

The default probability of each of three independent borrowers is 0.05. The probability that nobody defaults is $0.95^3 = 0.857$. Therefore

$$P(\text{at least one default}) = 1 - 0.95^3 = \mathbf{0.143} \quad (\text{and not } 0.15).$$

The naive sum $3 \times 0.05 = 0.15$ slightly overstates the answer: the probability of two and three simultaneous defaults has been counted an extra time. The more borrowers or the higher the probability, the larger the error.

THE GAMBLER'S FALLACY

A coin has come up heads five times in a row. The probability of heads on the sixth toss is still 0.5: the coin does not remember past tosses (tosses are independent). But the probability of six heads in a row, in advance, is $0.5^6 = 1/64 = 0.0156$. These are two different questions.

7 Combinatorics: how to count the number of ways

Formula (2.1) requires us to be able to count favourable and all outcomes. Combinatorics serves this purpose.

7.1 The multiplication principle

MULTIPLICATION PRINCIPLE

If the first choice can be made in n_1 ways, after it the second in n_2 ways, and so on, the total number of ways is $n_1 \cdot n_2 \cdots n_k$.

EXAMPLE 6

A menu of 3 starters, 4 main courses and 2 desserts gives $3 \cdot 4 \cdot 2 = 24$ different meals. A four-digit PIN (digits may repeat): $10^4 = 10\,000$ options; with distinct digits: $10 \cdot 9 \cdot 8 \cdot 7 = 5\,040$.

7.2 Factorial and permutations

Factorial: $n! = n \cdot (n-1) \cdots 2 \cdot 1$, by definition $0! = 1$. The number of ways to line up n distinct objects in a row is $n!$ (*permutations*). Five applications can be reviewed in $5! = 120$ different orders.

7.3 Arrangements and combinations

We choose r objects out of n distinct ones. Two questions: does the order of selection matter? If yes, these are *arrangements* (ordered selections); if not, *combinations*.

Arrangements and combinations

Arrangements (order matters): $$P(n,r) = \frac{n!}{(n-r)!},$$

Combinations (order does not matter): $$C(n,r) = \binom{n}{r} = \frac{n!}{r!\,(n-r)!}.$$

Relationships: $\binom{n}{r} = P(n,r)/r!$, and $\binom{n}{r} = \binom{n}{n-r}$. Russian-language textbooks use the notation A_n^r and C_n^r; English ones (Newbold, OpenStax) use $P(n,r)$ and $\binom{n}{r}$.

EXAMPLE 7. ARRANGEMENT OR COMBINATION?

From 10 analysts we choose a **president, vice-president and treasurer**: the posts are different, order matters, $P(10,3) = 10 \cdot 9 \cdot 8 = 720$. From 12 employees we choose a **committee of 4**: order does not matter, $\binom{12}{4} = \dfrac{12 \cdot 11 \cdot 10 \cdot 9}{4!} = 495$.

7.4 Permutations with repeated elements

If among n objects n_1 are of one kind, n_2 of another, etc., the number of distinguishable arrangements is $\dfrac{n!}{n_1!\,n_2!\cdots n_k!}$. The word “STATISTICS” (10 letters: S×3, T×3, A×1, I×2, C×1) can be formed in $\dfrac{10!}{3!\,3!\,1!\,2!\,1!} = \dfrac{3\,628\,800}{72} = 50\,400$ ways.

7.5 Probability through combinatorics

EXAMPLE 8. QUALITY CONTROL

A batch has 20 items, 3 of them defective. An inspector picks 4 items at random (without replacement). All $\binom{20}{4} = 4\,845$ sets are equally likely.

- No defectives: we choose all four from the 17 good ones, $\binom{17}{4} = 2\,380$, so $P = 2\,380/4\,845 = \mathbf{0.491}$.
- Exactly one defective: $\binom{3}{1}\binom{17}{3} = 3 \cdot 680 = 2\,040$, $P = 2\,040/4\,845 = \mathbf{0.421}$.
- At least one defective: $1 - 0.491 = \mathbf{0.509}$.

General formula: $P = \dfrac{\binom{K}{k}\binom{N-K}{n-k}}{\binom{N}{n}}$, where N is the batch size, K the number of defectives, n the sample size, k the number of defectives in the sample.

EXAMPLE 9. THE “6 OUT OF 45” LOTTERY

The winning combination is one of $\binom{45}{6} = 8\,145\,060$ equally likely sets. The probability of winning with one ticket is $1/8\,145\,060 \approx 1.2 \cdot 10^{-7}$.

EXAMPLE 10. THE BIRTHDAY PARADOX

What is the probability that in a group of n people at least two share a birthday? We count through the complement:

$$P = 1 - \frac{365 \cdot 364 \cdots (365 - n + 1)}{365^n}.$$

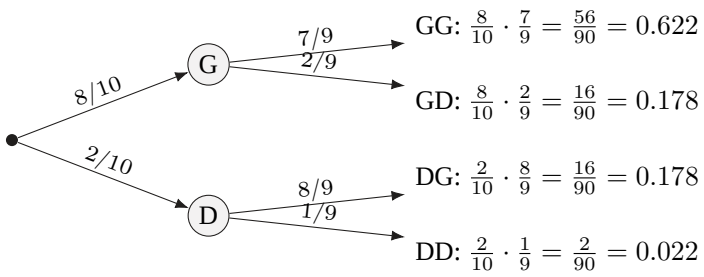
For $n = 23$ we get 0.507, for $n = 30$ we get 0.706. Already in a group of 23 a match is more likely than not, although intuition suggests much higher numbers.

8 The probability tree

When events occur in sequence, it is convenient to write the probabilities on a tree: along a branch the probabilities are *multiplied* (rule (6.1)), the probabilities of different leaves are *added*.

EXAMPLE 11. SELECTION WITHOUT REPLACEMENT

A batch has 10 items, 2 defective (D), 8 good (G). We take two items in a row without replacement.



Check: $56 + 16 + 16 + 2 = 90$, i.e. the leaf probabilities sum to 1. The probability of exactly one defective $= \frac{16}{90} + \frac{16}{90} = \frac{32}{90} = 0.356$. Combinatorics gives the same probability of DD: $\binom{2}{2}/\binom{10}{2} = 1/45 = 0.022$.

9 Key terms

Term	*In Russian*	Meaning
Random experiment	*Случайный эксперимент*	a procedure with an uncertain result
Sample space	*Пространство исходов*	the set of all possible outcomes, S
Event	*Событие*	a subset of S
Union	*Объединение*	$A \cup B$: A or B
Intersection	*Пересечение*	$A \cap B$: A and B together
Complement	*Дополнение*	A': not A
Mutually exclusive	*Несовместные*	cannot occur together
Independent	*Независимые*	$P(A \cap B) = P(A)P(B)$
Joint probability	*Совместная вероятность*	$P(A \cap B)$
Marginal probability	*Маргинальная вероятность*	$P(A)$, from the table totals
Conditional probability	*Условная вероятность*	$P(A \mid B)$, "A given B"
Factorial	*Факториал*	$n!$
Permutation (ordered selection)	*Размещение*	ordered choice of r from n
Combination	*Сочетание*	unordered choice of r from n

Formulas of Lecture 2

$$0 \le P(A) \le 1, \quad P(S) = 1, \quad P(A \cup B) = P(A) + P(B) \text{ (mutually exclusive)},$$

$$P(A') = 1 - P(A), \qquad P(A \cup B) = P(A) + P(B) - P(A \cap B),$$

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}, \qquad P(A \cap B) = P(A)P(B \mid A),$$

$$\text{independence: } P(A \cap B) = P(A)P(B) \iff P(A \mid B) = P(A),$$

$$n!, \qquad P(n, r) = \frac{n!}{(n-r)!}, \qquad C(n, r) = \frac{n!}{r!(n-r)!}, \qquad \frac{n!}{n_1! \cdots n_k!}.$$

10 Practice problems

Problem level is shown in square brackets: **[A]** — basic, tests understanding of definitions; **[B]** — standard computation; **[C]** — advanced, requires a derivation or an interpretation. Decimals are written with a point, as in a calculator, Excel and R. If a problem needs tables, use the appendix "Statistical Tables".

Problem 1 [A] A questionnaire has two questions, each of which can be answered "yes" (Y) or "no" (N).

(a) Write down the sample space S.
(b) Write down the event "at least one answer is yes". How many outcomes does it contain?
(c) Find its probability, assuming all outcomes are equally likely.

Problem 2 [A]

(a) $P(\text{late}) = 0.30$. Find $P(\text{not late})$.
(b) $P(A) = 0.40$, $P(B) = 0.30$, $P(A \cap B) = 0.15$. Find $P(A \cup B)$.
(c) It is known that $P(A) = 0.20$ and $P(A \cap B) = 0.35$. What can you say about these numbers?
(d) Two dice are rolled. Find the probability that at least one six came up (in two ways).

Problem 3 [B] Of 500 firms, 300 export, 200 import and 120 do both. Find the probabilities that a randomly chosen firm: (a) exports or imports; (b) does neither; (c) only exports; (d) only imports.

Problem 4 [B] Of 300 students, 180 study Finance and 120 study Marketing. In the retake exam 90 finance students and 40 marketing students passed.

(a) Build the contingency table.
(b) Find $P(\text{passed})$, $P(\text{passed} \mid \text{finance})$, $P(\text{passed} \mid \text{marketing})$.
(c) Check whether the programme and passing are independent.

Problem 5 [B] An analyst claims: "the default probability of each of three borrowers is 0.05, so the probability that at least one of them goes bankrupt is $3 \times 0.05 = 0.15$". Point out two errors in this reasoning and find the correct value assuming the borrowers are independent.

Problem 6 [B] In how many ways can one:

(a) form a four-digit PIN code from distinct digits;
(b) choose a committee of 4 from 12 employees;
(c) choose a president, vice-president and treasurer from 10 analysts;
(d) rearrange the letters of the word "ECONOMICS"?

Problem 7 [B] A batch has 30 phones, 5 of them faulty. For inspection 3 are chosen at random (without replacement). Find the probabilities that among those inspected: (a) none is faulty; (b) exactly one is faulty; (c) at least one is faulty.

Problem 8 [B] (a) Find the probability that among 25 people there are at least two with a common birthday (assume the year has 365 days and all dates are equally likely). (b) Derive the formula and explain why it is easier to compute through the complement. (c) Find the answer for a group of 40 people.

Problem 9 [C] A bank issued 600 loans: 360 to employees, of whom 18 stopped paying, and 240 to entrepreneurs, of whom 36 stopped paying.

(a) Build the contingency table "borrower category $\times$ default".
(b) Find $P(\text{default})$, $P(\text{default} \mid \text{entrepreneur})$, $P(\text{default} \mid \text{employee})$.
(c) Are borrower category and default independent?
(d) Find $P(\text{entrepreneur} \mid \text{default})$ and explain how this probability differs from $P(\text{default} \mid \text{entrepreneur})$.

Problem 10 [C] For three events $P(A) = 0.50$, $P(B) = 0.40$, $P(C) = 0.30$, $P(A \cap B) = 0.20$, $P(A \cap C) = 0.15$, $P(B \cap C) = 0.10$, $P(A \cap B \cap C) = 0.05$.

(a) Find $P(A \cup B \cup C)$ by the inclusion–exclusion formula

$$P(A \cup B \cup C) = P(A) + P(B) + P(C) - P(A \cap B) \\ - P(A \cap C) - P(B \cap C) + P(A \cap B \cap C).$$

(b) Find the probability that none of the three events occurs.

(c) Explain why the triple intersection is added at the end.

Problem 11 [C] It is known that $P(A) = 0.30$ and $P(B) = 0.50$.

(a) Let A and B be mutually exclusive. Find $P(A \cup B)$. Can they be independent?

(b) Let A and B be independent. Find $P(A \cap B)$ and $P(A \cup B)$.

(c) Can events be mutually exclusive and independent at the same time?

Problem 12 [C] In the 17th century the Chevalier de Méré considered two bets equivalent. (a) What is the probability of rolling at least one six in 4 rolls of one die? (b) What is the probability of at least one "double six" in 24 rolls of two dice? Which bet is more favourable? (c) A coin has landed heads 5 times in a row. What is the probability of heads on the next toss, and why is it not > 0.5?

STAT201 · STATISTICS · LECTURE 3

Conditional Probability and Bayes' Theorem

How to update probabilities with new data, and why an “accurate” test can be wrong most of the time

Syllabus topic 3 · week 3 · 4 academic hours

LECTURE GOALS

- compute conditional probability from the definition and from a contingency table;
- apply the law of total probability and build a probability tree;
- derive and apply Bayes’ theorem, expressing the result through natural frequencies;
- explain why the base rate (the share of a rare event) determines the result, and not confuse $P(A \mid B)$ with $P(B \mid A)$;
- update a probability sequentially from several signals and understand when the signals may be treated as independent;
- make a decision by a threshold, comparing expected losses.

1 Conditional probability

In Lecture 2 we defined conditional probability through the intersection. Let us recall the definition and understand what it means.

CONDITIONAL PROBABILITY

The probability of event A given that event B has occurred:

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}, \qquad P(B) > 0. \tag{1.1}$$

The meaning of the formula: having learned that B occurred, we *narrow* the sample space to B. The denominator becomes $P(B)$, and the numerator keeps only the outcomes lying in both A and B. The event after the bar determines the new denominator.

EXAMPLE 1. MOBILE APP AND DEPOSITS

Of 1 000 bank customers, 400 use the mobile app (M); 250 of them opened a deposit (D). Among the other 600, 120 opened a deposit.

	Deposit (D)	No deposit (D')	Total
App (M)	250	150	400
No app (M')	120	480	600
Total	**370**	**630**	**1 000**

$P(D) = 370/1000 = 0.37$. $P(D \mid M) = 250/400 = 0.625$. $P(D \mid M') = 120/600 = 0.20$.

App users open a deposit three times as often. And now the reverse probability: $P(M \mid D) = 250/370 = 0.676$ — this is the share of app users *among those who opened a deposit*. It is a different number with a different denominator.

$P(A \mid B) \neq P(B \mid A)$

In general $P(A \mid B) \neq P(B \mid A)$. The probability that a person with a fever has the flu and the probability that a person with the flu has a fever are different numbers. Mixing them up is the most common mistake in working with probabilities; this whole lecture is about how to move correctly from one to the other.

The definition immediately gives the multiplication rule $P(A \cap B) = P(A)\,P(B \mid A) = P(B)\,P(A \mid B)$ (Lecture 2). Two events are independent if and only if $P(A \mid B) = P(A)$.

2 The law of total probability

Often an event B can come about in several mutually exclusive ways. Events $A_1, \dots, A_k$ are said to form a *partition* of the sample space if they are pairwise mutually exclusive and together make up all of S ($A_1 \cup \dots \cup A_k = S$).

LAW OF TOTAL PROBABILITY

$$P(B) = \sum_{i=1}^{k} P(B \mid A_i)\,P(A_i) = P(B \mid A_1)P(A_1) + \dots + P(B \mid A_k)P(A_k). \qquad (2.1)$$

The probability of B is a weighted average of the conditional probabilities, and the weights are the probabilities of the paths.

EXAMPLE 2. THREE MACHINES

A part is produced on machine A_1 (50% of output, 2% defective), on A_2 (30%, 3% defective) or on A_3 (20%, 5% defective). The probability that a random part is defective (B):

$$P(B) = 0.50 \cdot 0.02 + 0.30 \cdot 0.03 + 0.20 \cdot 0.05 = 0.010 + 0.009 + 0.010 = \mathbf{0.029}.$$

The overall defect rate of 2.9% lies between 2% and 5%, as it should for a weighted average.

3 Bayes' theorem

The law of total probability goes "from causes to effects": knowing each machine's defect rate, it gives the overall defect rate. Bayes' theorem solves the inverse problem: *a defective part was found; what is the probability it was produced by machine A_3?*

BAYES' THEOREM

From the two expressions for $P(A_i \cap B)$ we obtain

$$P(A_i \mid B) = \frac{P(B \mid A_i)\,P(A_i)}{P(B)} = \frac{P(B \mid A_i)\,P(A_i)}{\sum_j P(B \mid A_j)\,P(A_j)}. \tag{3.1}$$

The **prior** probability $P(A_i)$ is before observing B; the **posterior** $P(A_i \mid B)$ is after. The factor $P(B \mid A_i)$ is called the *likelihood*.

Derivation in one line: $P(A_i \mid B)\,P(B) = P(A_i \cap B) = P(B \mid A_i)\,P(A_i)$; divide by $P(B)$ and substitute (2.1).

EXAMPLE 3. WHICH MACHINE IS TO BLAME?

The part is defective. The probability that it came from the third machine:

$$P(A_3 \mid B) = \frac{0.20 \cdot 0.05}{0.029} = \frac{0.010}{0.029} = \mathbf{0.345}.$$

Similarly $P(A_1 \mid B) = 0.010/0.029 = 0.345$ and $P(A_2 \mid B) = 0.009/0.029 = 0.310$. The sum is $0.345 + 0.310 + 0.345 = 1$. Although machine A_3 makes only 20% of the parts, because of its high defect rate it is responsible for 34.5% of all defects.

3.1 The main example: an alert from a fraud-monitoring system

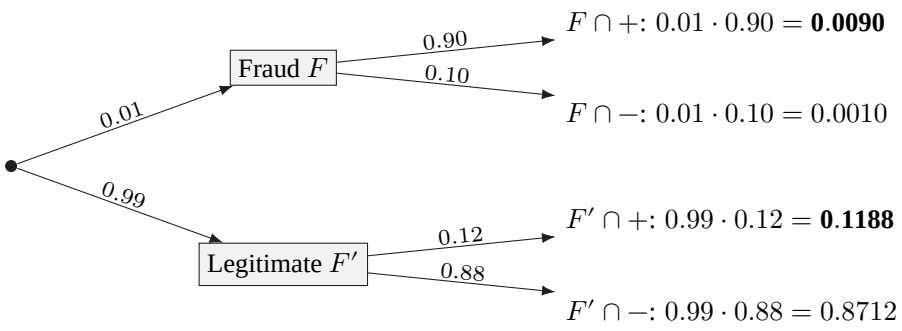
A bank uses a system that flags suspicious transactions. It is known that:

- **base rate**: 1% of transactions are fraudulent, $P(F) = 0.01$;
- **sensitivity**: the system catches 90% of fraudulent transactions, $P(+ \mid F) = 0.90$;
- **false-alarm rate**: the system wrongly flags 12% of legitimate transactions, $P(+ \mid F') = 0.12$.

A transaction is flagged. What is the probability that it is really fraudulent, $P(F \mid +)$?

Before computing, estimate by eye: is the answer closer to 7%, 50% or 90%? Most people say 90%.

The probability tree.



The alert occurs in two leaves: $P(+) = 0.0090 + 0.1188 = 0.1278$. Then

$$P(F \mid +) = \frac{0.0090}{0.0090 + 0.1188} = \frac{0.0090}{0.1278} = \mathbf{0.0704} \approx 7.0\%.$$

Natural frequencies (the same calculation without probabilities; it is usually more vivid). Take 10 000 transactions.

	Flagged	Not flagged	Total
Fraudulent (1%)	$100 \cdot 0.90 = 90$	10	100
Legitimate (99%)	$9\,900 \cdot 0.12 = 1\,188$	8 712	9 900
Total	**1 278**	**8 722**	**10 000**

Of the 1 278 flagged transactions only 90 are really fraudulent, i.e. $90/1\,278 = 7.04\%$. The system catches 90% of fraudsters, but almost 93% of its signals are false.

THE BASE RATE DECIDES EVERYTHING

Fraud is rare (1%), so there are a hundred times as many legitimate transactions. Even a small false-alarm rate (12%), applied to the huge mass of legitimate transactions, produces more signals than the 90% sensitivity applied to the small mass of fraudsters. Neglecting the base rate is called the *base rate fallacy*.

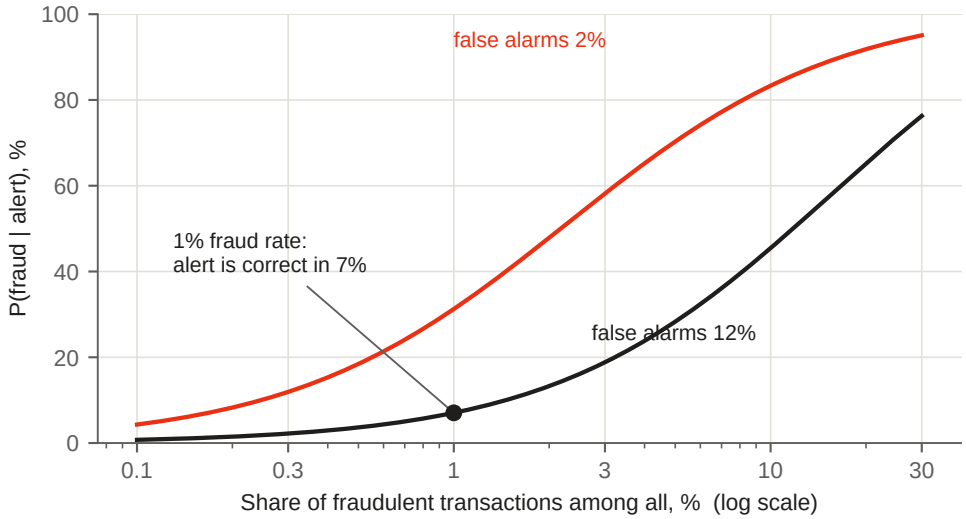


Fig. 1. Probability of fraud after an alert as a function of the base rate (sensitivity 90%). With 1% fraud and 12% false alarms the alert is right in only 7% of cases; reducing false alarms to 2% raises the answer to 31%.

Term	Notation	Meaning
Sensitivity	$P(+ \mid F)$	the share of real cases that the test finds
Specificity	$P(- \mid F') = 1 - P(+ \mid F')$	the share of legitimate transactions that the test lets through (here 0.88)
Positive predictive value	$P(F \mid +)$	the share of correct signals among all signals (here 0.0704)

3.2 An algorithm for Bayes problems

1. Name the events and write out *which* conditional probability is being asked for.
2. Write out the knowns: prior probabilities and likelihoods.
3. Build the tree (or a table of 10 000 objects) and find the leaf probabilities.
4. Add up the leaves in which the condition holds and divide the required leaf by this sum.
5. Check: the posterior probabilities over all hypotheses sum to 1; compare the answer with the prior probability.

4 Sequential updating and the odds form

Today's posterior probability becomes tomorrow's prior: new evidence arrives and we apply Bayes' theorem again. For such chains the notation through *odds* is convenient.

BAYES' RULE IN ODDS FORM

The odds of an event F: $\text{odds}(F) = \dfrac{P(F)}{1-P(F)}$; conversely, $P(F) = \dfrac{\text{odds}}{1+\text{odds}}$. Then

$$\underbrace{\frac{P(F \mid +)}{P(F' \mid +)}}_{\text{posterior odds}} = \underbrace{\frac{P(F)}{P(F')}}_{\text{prior odds}} \times \underbrace{\frac{P(+ \mid F)}{P(+ \mid F')}}_{LR_+}, \qquad LR_+ = \frac{\text{sensitivity}}{\text{false-alarm rate}}. \tag{4.1}$$

The likelihood ratio LR_+ shows how many times more likely a positive result is with fraud than without it.

EXAMPLE 4. ONE AND TWO ALERTS

Here $LR_+ = 0.90/0.12 = 7.5$. The prior odds are $0.01/0.99 = 0.0101$.
After one alert: $0.0101 \cdot 7.5 = 0.0758$, hence $P = 0.0758/1.0758 = 0.0704$. The answer coincides with the calculation above.
After a second independent alert: $0.0101 \cdot 7.5^2 = 0.568$, hence $P = 0.568/1.568 = \mathbf{0.362}$.
Conclusion: two independent signals raise the probability from 7% to 36%, but not to 90%.

DOUBLE COUNTING

Likelihood ratios may be multiplied only if the signals are *conditionally independent* given the state (under fraud and without it). Two models trained on the same customer data often output the same thing, and the second alert carries almost no new information. An automatic model and a subsequent document check are much closer to independent. In the extreme case, if the second alert simply repeats the first, the probability remains 7.04%, not 36.2%.

5 A decision by threshold

A probability by itself makes no decision. The costs of errors must be compared. Let the possible loss from fraud be $L = 10$ million UZS and the cost of an investigation $c = 0.5$ million UZS. Investigating

pays off if the *expected loss prevented* exceeds the cost of the investigation:

$$p \cdot L > c \quad \Longleftrightarrow \quad p > \frac{c}{L} = \frac{0.5}{10} = \mathbf{5}\%. \tag{5.1}$$

Both probabilities, 7.04% after one alert and 36.2% after two, exceed the 5% threshold, so an investigation is justified already after the first signal. Notice that the 5% threshold is far lower than the "intuitive" 50%: if the cost of a miss is high and checking is cheap, it is worth investigating even at a low probability.

6 A bridge to the next lecture: from events to random variables

An event says *what happened*. A random variable says *how much*. The same outcome can be evaluated by different numbers. For a single loan the outcomes "repaid", "overdue", "default" give the bank's profit X (million UZS):

Outcome	x, million UZS	$P(X = x)$
Repaid	$+1.2$	0.80
Overdue	$+0.4$	0.15
Default	-8.0	0.05

The average we expect in the long run is $\mathrm{E}[X] = 1.2 \cdot 0.80 + 0.4 \cdot 0.15 - 8 \cdot 0.05 = \mathbf{0.62}$ million UZS. This is *not* the profit on a particular loan, but the average over a set of similar loans. This apparatus is the subject of Lecture 4.

7 Key terms

Term	*In Russian*	Meaning
Conditional probability	*Условная вероятность*	$P(A \mid B) = P(A \cap B)/P(B)$
Partition	*Разбиение*	pairwise mutually exclusive events making up all of S
Law of total probability	*Формула полной вероятности*	$P(B) = \sum P(B \mid A_i)P(A_i)$
Bayes' theorem	*Теорема Байеса*	$P(A_i \mid B) = \dfrac{P(B \mid A_i)P(A_i)}{P(B)}$
Prior	*Априорная вероятность*	before the evidence is observed
Posterior	*Апостериорная вероятность*	after the evidence is observed
Likelihood	*Правдоподобие*	$P(\text{data} \mid \text{hypothesis})$
Base rate	*Базовая частота*	the share of the event in the original population
Sensitivity	*Чувствительность*	$P(+ \mid \text{present})$
Specificity	*Специфичность*	$P(- \mid \text{absent})$
Odds	*Шансы*	$p/(1-p)$
Likelihood ratio	*Отношение правдоподобия*	$LR_+ = \text{sens.}/\text{false-alarm rate}$

Formulas of Lecture 3

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}, \qquad P(B) = \sum_i P(B \mid A_i)P(A_i),$$

$$P(A_i \mid B) = \frac{P(B \mid A_i)P(A_i)}{\sum_j P(B \mid A_j)P(A_j)},$$

$$\text{odds:}\ \ \text{odds} = \frac{p}{1-p}, \quad p = \frac{\text{odds}}{1+\text{odds}}, \quad \text{odds}_{\text{post}} = \text{odds}_{\text{prior}} \times LR, \quad LR_+ = \frac{\text{sens.}}{\text{false-alarm rate}},$$

$$\text{investigate if } \ p > \frac{c}{L}.$$

8 Practice problems

Problem level is shown in square brackets: **[A]** — basic, tests understanding of definitions; **[B]** — standard computation; **[C]** — advanced, requires a derivation or an interpretation. Decimals are written with a point, as in a calculator, Excel and R. If a problem needs tables, use the appendix "Statistical Tables".

Problem 1 [A] It is known that $P(A \cap B) = 0.10$, $P(A) = 0.40$, $P(B) = 0.50$.

(a) Find $P(A \mid B)$ and $P(B \mid A)$.
(b) A student claims: "Since $P(A \mid B) = 0.8$, also $P(B \mid A) = 0.8$." Is he right?

Problem 2 [A] Of 1000 bank customers, 400 use the mobile app (M). Of these, 250 hold a deposit (D); among the other 600 customers, 120 hold a deposit.

(a) Find $P(\mathrm{D})$.
(b) Find $P(\mathrm{D} \mid \mathrm{M})$ and $P(\mathrm{D} \mid \overline{\mathrm{M}})$.
(c) Find $P(\mathrm{M} \mid \mathrm{D})$.

Problem 3 [B] A factory makes parts on three machines A, B, C with output shares of 50%, 30%, 20%. The defect rates are 2%, 3%, 5% respectively.

(a) Find the probability that a random part is defective.
(b) A part turns out to be defective. Find the probability that it was made on machine C; on A; on B.
(c) Check that the three posterior probabilities sum to 1.

Problem 4 [B] A bank catches fraudulent transactions. The share of fraudulent transactions is 2%. The system flags 85% of fraudulent transactions and wrongly flags 10% of normal ones.

(a) Find the probability that a flagged transaction is really fraudulent.
(b) Solve the same problem by the natural-frequency method on 10 000 transactions.
(c) Explain why the result is much smaller than the 85% sensitivity.

Problem 5 [B] A disease occurs in 0.5% of the population. A test detects it in 98% of the sick and gives a false positive in 5% of the healthy.

(a) What is the probability of disease given a positive result?
(b) The test was repeated independently (given health status) and is again positive. What is the probability of disease now?

Problem 6 [B] Continue the fraud problem (share 2%, sensitivity 85%, false alarms 10%). Suppose k independent (given fraud / no fraud) systems have raised a signal on the same transaction.

(a) Find the prior odds and the likelihood ratio LR^{+}.
(b) Find the probability of fraud for $k = 1,\ 2,\ 3$ signals.
(c) What happens to the result if the signals are not independent but fully duplicate each other?

Problem 7 [B] For a transaction in the fraud problem, the loss from a missed fraud is 8 million UZS, an investigation costs 0.4 million UZS (and completely eliminates the loss).

(a) Find the threshold probability above which an investigation pays off.
(b) Is it worthwhile to investigate a transaction that received one signal (fraud probability 14.8%)?
(c) At what minimum prior probability is one signal enough to justify an investigation?

Problem 8 [B] For a transaction with posterior fraud probability $p = 0.148$, three actions are considered: (1) let it pass (expected loss: $8p$ million); (2) investigate (cost 0.4 million, no loss); (3) block (if the transaction is normal, the damage to the customer is 1 million; if fraudulent, no loss). Which action minimizes the expected loss?

Problem 9 [C]

(a) Derive Bayes' formula for event A_i from the definition of conditional probability and the law of total probability (the events $A_1, \dots, A_n$ form a partition).

(b) Prove the odds form: $\dfrac{P(A \mid B)}{P(\overline{A} \mid B)} = \dfrac{P(A)}{P(\overline{A})} \cdot \dfrac{P(B \mid A)}{P(B \mid \overline{A})}$.

Problem 10 [C] A family has two children.

(a) It is known that at least one child is a boy. What is the probability that both are boys?

(b) It is known that the older child is a boy. What is the probability that both are boys?

Problem 11 [C] The Monty Hall problem. Behind one of three doors is a car, behind the other two are goats. You have chosen door 1. The host, who knows where the car is, opens a door with a goat (from the two remaining; if both have goats, he chooses at random) and offers you the chance to switch. Find the probability of winning if you switch.

Problem 12 [C] Two fraud-alert systems (sensitivity 90%, false alarms 12%, fraud share 1%) raised signals. An analyst multiplied the likelihood ratios and obtained 36.2%. It then turned out that the second system simply copies the decision of the first. What should the estimate of the probability be?

STAT201 · STATISTICS · LECTURE 4

Random Variables and Discrete Distributions

How to describe an uncertain numerical outcome by a distribution, an expectation and a variance

Syllabus topic 4 · week 4 · 4 academic hours

LECTURE GOALS

- distinguish a discrete random variable from a continuous one and specify the probability mass function (PMF) and the cumulative distribution function (CDF);
- compute the expected value, variance and standard deviation of a discrete variable;
- apply the properties of $\mathrm{E}[aX+b]$ and $\mathrm{Var}(aX+b)$ and the addition rule for independent variables;
- recognize situations described by the binomial, geometric and Poisson distributions and compute their probabilities;
- use the expected value and risk (standard deviation) to compare decisions.

1 A random variable and its distribution

In Lectures 2–3 we worked with events. Most often an outcome is expressed by a number: the profit on a loan, the number of calls per hour, the number of defective parts in a batch. This is conveniently described by a function that assigns a number to each outcome.

RANDOM VARIABLE

A random variable X is a numerical function of the outcome of a random experiment. Values are denoted by lowercase letters: $\{X = x\}$ is the event "the variable X took the value x".

A random variable is *discrete* if it takes a finite or countable number of values (the number of customers, the number of defects). It is *continuous* if it takes any values on an interval (height, waiting time, an exchange rate). This lecture is devoted to discrete variables; continuous ones are covered in Lecture 5.

PROBABILITY MASS FUNCTION (PMF)

For a discrete variable the function $p(x) = P(X = x)$ is called the probability mass function. It has the properties

$$p(x) \geq 0, \qquad \sum_{x} p(x) = 1. \tag{1.1}$$

CUMULATIVE DISTRIBUTION FUNCTION (CDF)

$F(x) = P(X \le x) = \sum_{t \le x} p(t)$. This is a non-decreasing "step" function from 0 to 1. For integer values $P(a < X \le b) = F(b) - F(a)$.

EXAMPLE 1. CALLS TO A SUPPORT LINE

The number of contacts X in 10 minutes has the distribution

x	0	1	2	3	4
$P(X = x)$	0.10	0.25	0.35	0.20	0.10
$F(x)$	0.10	0.35	0.70	0.90	1.00

The probabilities sum to 1, so the distribution is valid. $P(X \ge 3) = 0.20 + 0.10 = 0.30$; $P(1 \le X \le 3) = F(3) - F(0) = 0.90 - 0.10 = 0.80$.

2 Expectation, variance, standard deviation

EXPECTED VALUE

$$\mu = \mathrm{E}[X] = \sum_x x\, p(x). \tag{2.1}$$

This is the average value we would get over a very large number of repetitions (an average weighted by probabilities).

VARIANCE AND STANDARD DEVIATION

$$\sigma^2 = \mathrm{Var}(X) = \mathrm{E}\big[(X-\mu)^2\big] = \sum_x (x-\mu)^2 p(x) = \mathrm{E}\big[X^2\big] - \mu^2, \qquad \sigma = \sqrt{\mathrm{Var}(X)}. \tag{2.2}$$

The second form in (2.2) (the computational formula) is usually more convenient: $\mathrm{E}\big[X^2\big] = \sum x^2 p(x)$.

EXAMPLE 2. CALLS: EXPECTATION AND SPREAD

$\mu = 0 \cdot 0.10 + 1 \cdot 0.25 + 2 \cdot 0.35 + 3 \cdot 0.20 + 4 \cdot 0.10 = 1.95$. $\mathrm{E}\big[X^2\big] = 0.25 + 4 \cdot 0.35 + 9 \cdot 0.20 + 16 \cdot 0.10 = 0.25 + 1.40 + 1.80 + 1.60 = 5.05$. Hence $\mathrm{Var}(X) = 5.05 - 1.95^2 = 1.2475$, $\sigma = 1.117$.

EXAMPLE 3. THE LOAN FROM LECTURE 3

The profit X (million UZS) takes the values $+1.2$, $+0.4$, -8.0 with probabilities 0.80, 0.15, 0.05. Then $\mathrm{E}[X] = 0.62$, $\mathrm{E}[X^2] = 1.44 \cdot 0.8 + 0.16 \cdot 0.15 + 64 \cdot 0.05 = 1.152 + 0.024 + 3.2 = 4.376$, $\mathrm{Var}(X) = 4.376 - 0.62^2 = 3.9916$, $\sigma = 1.998$.
The expected profit is positive (0.62), but the standard deviation (2.0) is more than three times larger. The reason is the rare but heavy default: the "average" result is never observed on an individual loan (neither 0.62 nor anything close to it is among the outcomes).

COMMON MISTAKE

The expected value need not be a possible value of the variable and does not guarantee the result of a single trial. It is a long-run average; on one loan you can lose 8.

3 Properties of expectation and variance

Linear transformations and sums

$$\mathrm{E}[aX+b] = a\mathrm{E}[X] + b, \qquad \mathrm{Var}(aX+b) = a^2\,\mathrm{Var}(X), \qquad \mathrm{SD}(aX+b) = |a|\,\mathrm{SD}(X),$$

$$\mathrm{E}[X+Y] = \mathrm{E}[X] + \mathrm{E}[Y] \text{ (always)},$$

$$\mathrm{Var}(X \pm Y) = \mathrm{Var}(X) + \mathrm{Var}(Y) \text{ (if } X, Y \text{ are independent)}.$$

A shift b changes the centre but not the spread; multiplication by a scales the spread by $|a|$ and the variance by a^2. The expectation of a sum equals the sum of expectations even for dependent variables, whereas for the variance independence is essential (the general case is in Lecture 6).

EXAMPLE 4. A PORTFOLIO OF 100 LOANS

Let the profits of 100 loans be independent and identically distributed, as above. The total profit is $T = X_1 + \cdots + X_{100}$:

$$\mathrm{E}[T] = 100 \cdot 0.62 = 62, \qquad \mathrm{Var}(T) = 100 \cdot 3.9916 = 399.16, \qquad \mathrm{SD}(T) = 19.98.$$

The average profit per loan $\bar{X} = T/100$ has $\mathrm{E}[\bar{X}] = 0.62$ and $\mathrm{SD}(\bar{X}) = 19.98/100 = 0.20$, i.e. the spread of the mean is $\sqrt{100} = 10$ times smaller than that of a single loan. This is the seed of the law of large numbers and the CLT (Lecture 7). Note: standard deviations do not add, variances do.

COMMON MISTAKE

$\mathrm{SD}(X+Y) \neq \mathrm{SD}(X) + \mathrm{SD}(Y)$ and $\mathrm{Var}(X-Y) = \mathrm{Var}(X) + \mathrm{Var}(Y)$ (not the difference): subtracting independent variables does not reduce the spread.

4 The binomial distribution

The Bernoulli scheme: n independent trials are performed, in each a "success" with probability p (the same each time). The number of successes X has the binomial distribution, $X \sim \text{Bin}(n, p)$.

Binomial distribution

$$P(X = k) = \binom{n}{k} p^k (1-p)^{n-k}, \quad k = 0, 1, \dots, n; \qquad \text{E}[X] = np, \quad \text{Var}(X) = np(1-p).$$

Conditions of applicability (always check them): (1) a fixed number of trials n; (2) two outcomes; (3) the same probability of success; (4) independence. The factor $\binom{n}{k}$ is the number of ways to choose in which k trials the success occurred (Lecture 2).

EXAMPLE 5. QUALITY CONTROL

The probability that a part is defective is 5%, a batch of 10 parts, $X \sim \text{Bin}(10, 0.05)$.

$$P(X = 0) = 0.95^{10} = 0.599,$$
$$P(X \geq 1) = 1 - 0.599 = 0.401,$$
$$P(X = 2) = \binom{10}{2} \cdot 0.05^2 \cdot 0.95^8 = 45 \cdot 0.0025 \cdot 0.6634 = 0.0746.$$

$\text{E}[X] = 0.5$, $\sigma = \sqrt{10 \cdot 0.05 \cdot 0.95} = 0.69$. In 40% of batches of 10 parts there is at least one defective, although "on average" there are only 0.5 defective parts.

EXAMPLE 6. CUSTOMERS WHO ACCEPTED AN OFFER

A bank calls 20 customers, each accepts with probability 0.3 independently, $X \sim \text{Bin}(20, 0.3)$: $\text{E}[X] = 6$, $\text{Var}(X) = 4.2$, $\sigma = 2.05$. $P(X = 6) = \binom{20}{6} 0.3^6\, 0.7^{14} = 0.1916$; $P(X \leq 4) = 0.2375$; $P(X \geq 8) = 0.2277$. Such calculations use a table or the Excel function `BINOM.DIST`.

5 The geometric distribution

If trials are repeated until the first success, the number of trials X has the geometric distribution:

$$P(X = k) = (1-p)^{k-1} p, \quad k = 1, 2, \dots; \qquad \text{E}[X] = \frac{1}{p}, \quad P(X \leq k) = 1 - (1-p)^k. \tag{5.1}$$

Example: the probability of closing a deal with one call is $p = 0.2$. Then the deal takes exactly 3 calls with probability $0.8^2 \cdot 0.2 = 0.128$, at most 3 calls with probability $1 - 0.8^3 = 0.488$, and on average $1/0.2 = 5$ calls are needed.

6 The Poisson distribution

If events occur independently of each other in continuous time (or space) with a constant average intensity, then the number of events X in a fixed interval, λ on average, has the Poisson distribution:

$$P(X=k)=\frac{e^{-\lambda}\lambda^{k}}{k!},\quad k=0,1,2,\dots;\qquad \mathrm{E}[X]=\mathrm{Var}(X)=\lambda. \tag{6.1}$$

EXAMPLE 7. CUSTOMERS AT A BRANCH

A branch receives on average $\lambda=3$ customers per 10 minutes. Then

$$P(X=0)=e^{-3}=0.0498,\quad P(X=2)=\frac{e^{-3}3^{2}}{2}=0.2240,$$
$$P(X\ge 4)=1-P(X\le 3)=0.3528.$$

The intensity is rescaled in proportion to the interval: in 20 minutes there are 6 customers on average, and, for example, $P(X\le 3)=0.1512$.

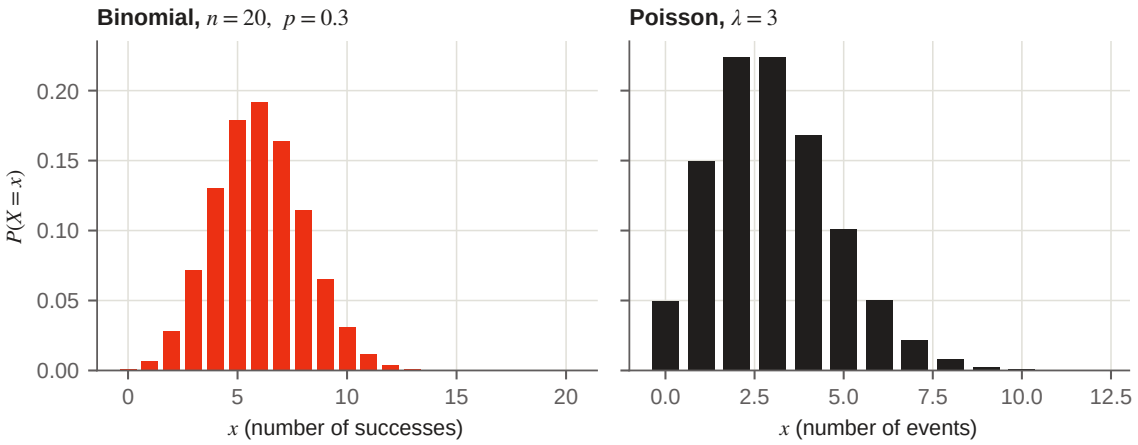


Fig. 1. Probability distributions: the binomial $\mathrm{Bin}(20,0.3)$ is almost symmetric and bell-shaped, centred at $\mu=6$; the Poisson with $\lambda=3$ is skewed to the right.

Poisson as an approximation to the binomial. For large n and small p the distribution $\mathrm{Bin}(n,p)$ is close to $\mathrm{Pois}(\lambda=np)$. For example, 200 operations, error probability 0.01: $P(X\ge 3)=0.3233$ by the binomial formula and 0.3233 by the Poisson with $\lambda=2$.

REMEMBER

How to choose a model: a fixed number n of trials with success probability p — binomial; the number of trials until the first success — geometric; the number of events in an interval with no upper bound — Poisson. For the Poisson the variance equals the expectation; if the spread of the data is noticeably larger than the mean, the model is inadequate.

7 Key terms

Term	*In Russian*	Meaning
Random variable	*Случайная величина*	a numerical function of the outcome
Probability mass function (PMF)	*Закон распределения*	$p(x) = P(X = x)$, $\sum p = 1$
Cumulative distribution function (CDF)	*Функция распределения*	$F(x) = P(X \le x)$
Expected value	*Математическое ожидание*	$\mu = \sum x\, p(x)$
Variance	*Дисперсия*	$\mathrm{E}[X^2] - \mu^2$
Binomial	*Биномиальное*	$\mathrm{Bin}(n, p)$: np, $np(1-p)$
Geometric	*Геометрическое*	number of trials until success, $1/p$
Poisson	*Пуассона*	events in an interval, λ

Formulas of Lecture 4

$$\mu = \sum x\, p(x), \qquad \sigma^2 = \sum x^2 p(x) - \mu^2,$$
$$\mathrm{E}[aX + b] = a\mu + b, \quad \mathrm{Var}(aX + b) = a^2\sigma^2, \quad \mathrm{Var}(X \pm Y) = \mathrm{Var}(X) + \mathrm{Var}(Y) \text{ (indep.)},$$
$$\mathrm{Bin}(n, p):\ P(k) = \tbinom{n}{k} p^k (1-p)^{n-k},\ \mu = np,\ \sigma^2 = np(1-p);$$
$$\mathrm{Geom}(p):\ P(k) = (1-p)^{k-1} p,\ \mu = 1/p; \qquad \mathrm{Pois}(\lambda):\ P(k) = e^{-\lambda}\lambda^k/k!,\ \mu = \sigma^2 = \lambda.$$

8 Practice problems

> Problem level is shown in square brackets: **[A]** — basic, tests understanding of definitions; **[B]** — standard computation; **[C]** — advanced, requires a derivation or an interpretation. Decimals are written with a point, as in a calculator, Excel and R. If a problem needs tables, use the appendix “Statistical Tables”.

Problem 1 [A] The distribution of the variable X: $P(1) = 0.2$, $P(2) = 0.3$, $P(3) = k$, $P(4) = 0.1$.

(a) Find k.
(b) Find $\mathrm{E}[X]$, $\mathrm{Var}(X)$, $\mathrm{SD}(X)$.
(c) Find $P(X \ge 3)$.

Problem 2 [A] For a variable X: $\mathrm{E}[X] = 40$, $\mathrm{SD}(X) = 6$. The variable X_2 is independent of X: $\mathrm{E}[X_2] = 25$, $\mathrm{SD}(X_2) = 8$.

(a) Find the expectation, variance and standard deviation of $Y = 3X - 10$.
(b) Find the expectation and standard deviation of $X + X_2$ and $X - X_2$.

Problem 3 [A] $X \sim \text{Bin}(5, 0.4)$.

(a) Build the table of $P(X = k)$, $k = 0, \dots, 5$.
(b) Find $\text{E}[X]$ and $\text{Var}(X)$ by the formulas.
(c) Find $P(X \geq 4)$.

Problem 4 [B] A borrower repays a loan with probability 0.9 independently of the others. The bank issued 12 loans.

(a) Find the expected number of repaid loans and the standard deviation.
(b) Find the probability that all 12 are repaid.
(c) Find the probability that at most one loan is not repaid.

Problem 5 [B] The average number of complaints to a support line is 1.5 per hour (a Poisson flow).

(a) Find the probability that there are no complaints in an hour.
(b) Find the probability of at least two complaints in an hour.
(c) Find the probability that at most one complaint arrives in two hours.

Problem 6 [B] Two projects are considered. Project A: profit $+10$ million with probability 0.6 and -5 million with probability 0.4. Project B: $+6$ million with probability 0.7 and -1 million with probability 0.3.

(a) Find the expected profit and standard deviation of each project.
(b) Which project would you choose if the goal is to maximize expected profit? And if stability matters? Justify.

Problem 7 [B] An insurance company sells a policy for 80 thousand UZS. Annual payouts on the policy: 0 with probability 0.94, 400 thousand with probability 0.05, 3000 thousand with probability 0.01.

(a) Find the expected payout and the standard deviation of the payout on one policy.
(b) Find the expected profit and the standard deviation of profit on one policy.
(c) The company sold 1000 independent policies. Find the expected total profit and its standard deviation.

Problem 8 [B] The profit of one loan has $\mu = 0.62$ million, $\sigma = 2.0$ million (Lecture 4, the loan example). The bank issues 25 independent loans. Find the expectation and standard deviation of the total profit and of the average profit per loan. How many times less variable is the average than one loan?

Problem 9 [B] A bank processes 300 transactions per minute, each erroneous with probability 0.006 independently.

(a) Find the probability of no errors by the exact binomial formula and by the Poisson approximation.
(b) Find the probability of at least three errors by both methods.

Problem 10 [B] The probability of closing a deal on one call is 0.25 (calls are independent). A manager calls until the first deal.

(a) What is the probability that the first deal comes on the fourth call?
(b) What is the probability that at most four calls suffice?

(c) How many calls are needed on average?

Problem 11 [C] Prove: (a) $\operatorname{Var}(X) = \mathrm{E}[X^2] - \mu^2$; (b) $\operatorname{Var}(aX + b) = a^2 \operatorname{Var}(X)$.

Problem 12 [C] Let $X \sim \operatorname{Bin}(n, p)$ be the number of successes, $X = I_1 + \cdots + I_n$, where I_i is the indicator of success in the i-th trial ($I_i = 1$ with probability p, otherwise 0, independent).

(a) Find $\mathrm{E}[I_i]$ and $\operatorname{Var}(I_i)$.
(b) Derive $\mathrm{E}[X] = np$ and $\operatorname{Var}(X) = np(1-p)$.
(c) At what p is the variance of a single trial maximal? What is it equal to?

STAT201 · STATISTICS · LECTURE 5

Continuous Distributions and the Normal Distribution

Density, area under a curve, standardization and working with the normal table

Syllabus topic 5 · week 5 · 4 academic hours

LECTURE GOALS

- explain why for a continuous variable probability is the area under the density and $P(X = x) = 0$;
- work with the uniform and exponential distributions;
- standardize a normal variable (z-score) and find probabilities and quantiles from the table;
- apply the 68–95–99.7 rule and find unknown parameters from a given probability;
- use the normality of a sum of independent normal variables and the normal approximation to the binomial distribution.

1 Probability density

For a continuous variable (time, mass, an amount of money) one cannot list the values and assign probabilities to them: there are infinitely many values, and the probability of any single value is zero. Instead a *density* $f(x)$ is specified.

PROBABILITY DENSITY

A function $f(x) \geq 0$ with total area under the curve equal to 1 is called a density. The probability of falling into an interval is the area under the curve above it:

$$P(a \leq X \leq b) = \int_a^b f(x)\,dx, \qquad P(X = x) = 0. \tag{1.1}$$

Therefore for a continuous variable it does not matter whether inequalities are strict or non-strict: $P(X < b) = P(X \leq b)$. The distribution function is defined as before: $F(x) = P(X \leq x)$ (the area to the left of x), and $P(a < X < b) = F(b) - F(a)$. The expectation and variance are given by the integrals $\mu = \int x f(x) dx$, $\sigma^2 = \int (x-\mu)^2 f(x) dx$, but for the standard families we shall use ready-made formulas.

COMMON MISTAKE

The value of the density $f(x)$ is not a probability and may exceed 1 (for example, the density of the uniform distribution on $[0, 0.5]$ equals 2). Probability is only area.

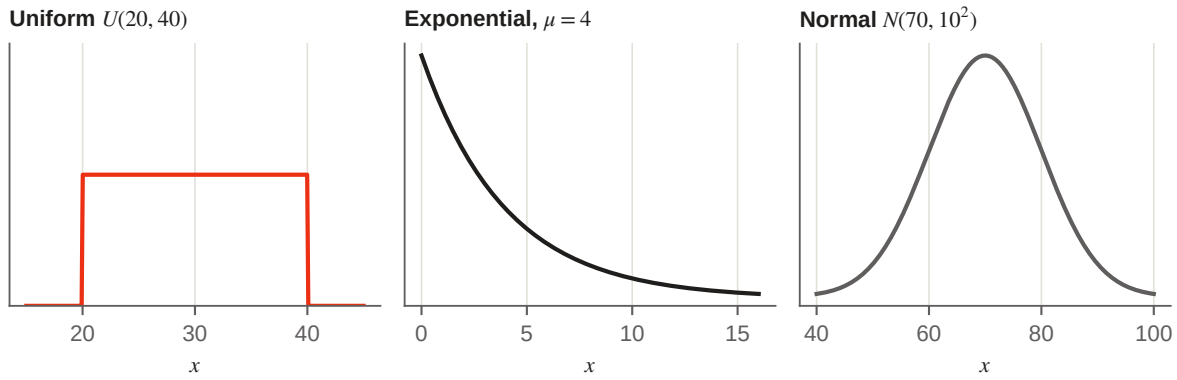


Fig. 1. Three densities: uniform (delivery time, min), exponential (waiting, min), normal (scores).

2 The uniform distribution

If all values on the interval $[a, b]$ are equally likely, $X \sim U(a, b)$:

$$f(x) = \frac{1}{b-a}\ (a \le x \le b), \qquad P(c \le X \le d) = \frac{d-c}{b-a}, \qquad \mu = \frac{a+b}{2}, \quad \sigma^2 = \frac{(b-a)^2}{12}. \tag{2.1}$$

EXAMPLE 1. DELIVERY TIME

A courier delivers an order in time $X \sim U(20, 40)$ minutes. $\mu = 30$, $\sigma = 20/\sqrt{12} = 5.77$. $P(25 \le X \le 35) = 10/20 = 0.5$; $P(X > 36) = 4/20 = 0.2$.

3 The exponential distribution

The waiting time until the next event of a Poisson flow (Lecture 4) with intensity λ is exponentially distributed:

$$f(x) = \lambda e^{-\lambda x}, \quad P(X > x) = e^{-\lambda x}, \quad F(x) = 1 - e^{-\lambda x}\ (x \ge 0); \qquad \mu = \sigma = \frac{1}{\lambda}. \tag{3.1}$$

EXAMPLE 2. WAITING IN A QUEUE

The average wait for service is 4 minutes, $\lambda = 1/4$. $P(X > 6) = e^{-1.5} = 0.2231$; $P(X < 2) = 1 - e^{-0.5} = 0.3935$; $P(3 < X < 5) = e^{-0.75} - e^{-1.25} = 0.4724 - 0.2865 = 0.1859$. The median $= 4 \ln 2 = 2.77$ minutes, less than the mean of 4: the distribution is right-skewed.

REMEMBER

The exponential distribution has no “memory”: $P(X > s + t \mid X > s) = P(X > t)$. If you have already waited 5 minutes, the remaining waiting time is distributed the same as for a customer who has just arrived.

4 The normal distribution

NORMAL DISTRIBUTION

$X \sim N(\mu, \sigma^2)$ if the density has the bell shape

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}}\, e^{-\frac{(x-\mu)^2}{2\sigma^2}}. \tag{4.1}$$

The curve is symmetric about μ; the parameter σ determines the width.

Variables that are formed as a sum of many small independent influences are (approximately) normal: measurement errors, height, scores on large tests, and, as we shall see in Lecture 7, sample means for any initial distribution.

4.1 Standardization

The area under $N(\mu, \sigma^2)$ cannot be taken "head-on", so all probabilities are reduced to the standard normal $Z \sim N(0, 1)$ using the z-score:

$$z = \frac{x - \mu}{\sigma}, \qquad P(X \le x) = \Phi(z), \tag{4.2}$$

where $\Phi(z)$ is the distribution function of Z (table in the appendix "Statistical Tables", Excel: `NORM.S.DIST(z,TRUE)`). The value z shows by how many standard deviations a value lies from the mean.

Useful properties: $\Phi(-z) = 1 - \Phi(z)$; $P(Z > z) = 1 - \Phi(z)$; $P(a < Z < b) = \Phi(b) - \Phi(a)$.

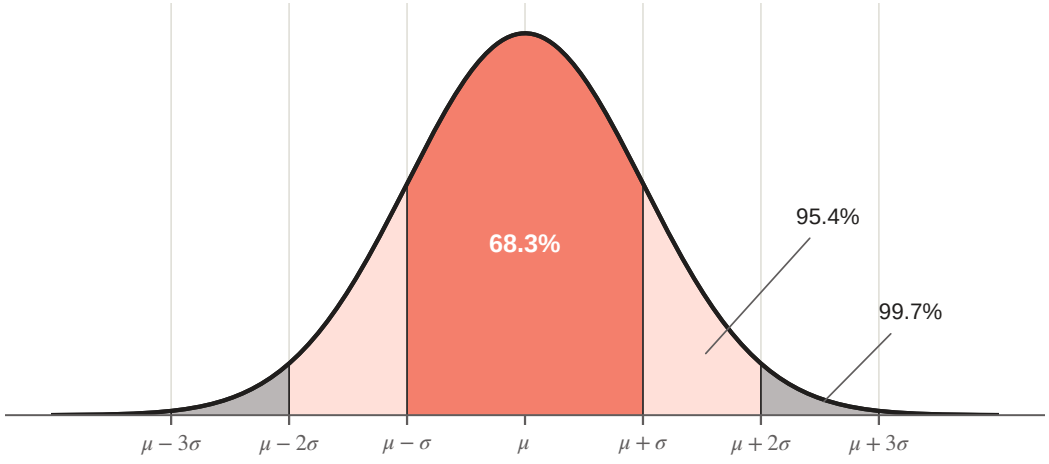


Fig. 2. The 68–95–99.7 rule: for a normal variable 68.3% of values lie within $\pm 1\sigma$, 95.4% within $\pm 2\sigma$, 99.7% within $\pm 3\sigma$.

EXAMPLE 3. EXAM SCORES

Scores $X \sim N(70, 10^2)$.

- $P(X > 85)$: $z = 1.5$, $P = 1 - \Phi(1.5) = 1 - 0.9332 = 0.0668$.
- $P(60 < X < 80)$: $z = \pm 1$, $P = 0.8413 - 0.1587 = 0.6827$.
- $P(65 < X < 90)$: $z = -0.5$ and 2: $P = 0.9772 - 0.3085 = 0.6687$.

4.2 The inverse problem: quantiles

To find the value x above (below) which a given share lies, we find z from the table "in reverse" and go back: $x = \mu + z\sigma$. Frequently used values: $z_{0.90} = 1.2816$, $z_{0.95} = 1.6449$, $z_{0.975} = 1.9600$, $z_{0.99} = 2.3263$, $z_{0.995} = 2.5758$.

EXAMPLE 4. THE TOP 10% AND THE QUARTILES

The cut-off for the top 10% in the exam: $x = 70 + 1.2816 \cdot 10 = 82.8$. The lower quartile is $Q_1 = 70 - 0.6745 \cdot 10 = 63.3$, the upper $Q_3 = 76.7$; hence $IQR = 13.5 = 1.349\,\sigma$ (a useful guide from Lecture 1: for normal data $IQR \approx 1.35\sigma$).

EXAMPLE 5. INCOMES

Monthly income $X \sim N(5.5;\ 1.2^2)$ million UZS. $P(X > 7) = P(Z > 1.25) = 0.1056$; $P(X < 4) = 0.1056$ (symmetry: $z = -1.25$); $P(4 < X < 7) = 0.7887$. The central 95% of values: $5.5 \pm 1.96 \cdot 1.2$, i.e. from 3.15 to 7.85 million.

4.3 Unknown parameters

If the probability is known but a parameter is not, we write down the standardization and solve the equation. For example, $X \sim N(90, \sigma^2)$ and $P(X > 100) = 0.10$. Then $z = 1.2816 = \dfrac{100 - 90}{\sigma}$, $\sigma = 10/1.2816 = 7.80$. Another example: $X \sim N(\mu, 8^2)$ and $P(X < 50) = 0.20$: $z_{0.20} = -0.8416$, $50 = \mu - 0.8416 \cdot 8$, $\mu = 56.73$.

4.4 Comparison through z-scores

The z-score allows results on different scales to be compared. A student scored 78 on a test $N(70, 8^2)$ and 640 on a test $N(580, 60^2)$: $z_1 = 1.0$, $z_2 = 1.0$. The relative results are the same.

4.5 Sums of normal variables

A linear combination of independent normal variables is normal, and the parameters are computed by the rules of Lecture 4. Let $X \sim N(40, 6^2)$, $Y \sim N(25, 8^2)$ be independent. Then $X + Y \sim N(65, 10^2)$ and $X - Y \sim N(15, 10^2)$; $P(X + Y > 80) = P(Z > 1.5) = 0.0668$, $P(X < Y) = P(X - Y < 0) = P(Z < -1.5) = 0.0668$.

5 Normal approximation to the binomial

For large n the distribution $\mathrm{Bin}(n, p)$ is well approximated by the normal $N(np,\ np(1-p))$ (condition: $np \geq 10$ and $n(1-p) \geq 10$). Since the binomial distribution is discrete, a *continuity correction* is introduced: $P(X \leq k) \approx \Phi\left(\dfrac{k + 0.5 - np}{\sqrt{np(1-p)}}\right)$.

EXAMPLE 6. 100 CUSTOMERS

$X \sim \text{Bin}(100, 0.3)$: $\mu = 30$, $\sigma = \sqrt{21} = 4.58$. Exactly $P(X \le 35) = 0.8839$. Without the correction $\Phi(1.09) = 0.8624$, with the correction $\Phi\left(\frac{35.5-30}{4.58}\right) = \Phi(1.20) = 0.8849$. The correction noticeably improves the accuracy of the approximation.

6 Key terms

Term	*In Russian*	Meaning
Probability density function (pdf)	*Плотность*	$P(a \le X \le b) = \int_a^b f$
Uniform	*Равномерное*	$\mu = \frac{a+b}{2}$, $\sigma^2 = \frac{(b-a)^2}{12}$
Exponential	*Экспоненциальное*	$P(X > x) = e^{-\lambda x}$, $\mu = 1/\lambda$
Normal	*Нормальное*	bell-shaped, $N(\mu, \sigma^2)$
z-score	*z-оценка*	$z = (x - \mu)/\sigma$
Standard normal	*Стандартное нормальное*	$Z \sim N(0, 1)$, function Φ
Quantile	*Квантиль*	the value to the left of which lies a given share
Continuity correction	*Поправка на непрерывность*	± 0.5 when approximating a discrete variable

Formulas of Lecture 5

$$U(a,b):\ \mu = \tfrac{a+b}{2},\ \sigma^2 = \tfrac{(b-a)^2}{12}; \qquad \text{Exp}(\lambda):\ P(X > x) = e^{-\lambda x},\ \mu = \sigma = \tfrac{1}{\lambda};$$

$$X \sim N(\mu, \sigma^2):\ z = \frac{x - \mu}{\sigma},\ \ P(X \le x) = \Phi(z),\ \ x = \mu + z\sigma;$$

$$\text{Bin}(n,p) \approx N(np,\ np(1-p)),\ \ P(X \le k) \approx \Phi\Big(\tfrac{k+0.5-np}{\sqrt{np(1-p)}}\Big).$$

7 Practice problems

Problem level is shown in square brackets: **[A]** — basic, tests understanding of definitions; **[B]** — standard computation; **[C]** — advanced, requires a derivation or an interpretation. Decimals are written with a point, as in a calculator, Excel and R. If a problem needs tables, use the appendix “Statistical Tables”.

Problem 1 [A] A bus runs at 15-minute intervals. A passenger arrives at a random moment, the waiting time is $X \sim U(0, 15)$.

(a) Find μ and σ.
(b) Find $P(X > 10)$ and $P(5 < X < 8)$.

Problem 2 [A] The service time of a customer at a bank is exponential with a mean of 5 minutes.

(a) What is the parameter λ?
(b) Find the probability that service lasts more than 3 minutes, and the probability that it lasts less than 10 minutes.
(c) Find the median.

Problem 3 [A] $X \sim N(200, 20^2)$. Find $P(X < 230)$, $P(X > 170)$, $P(190 < X < 215)$.

Problem 4 [B] A machine packs coffee, the mass of a pack is $X \sim N(500, 4^2)$ g.

(a) Find the share of packs weighing less than 495 g.
(b) Find the share of packs weighing more than 510 g.
(c) To what mean should the machine be set so that only 1% of packs weigh less than 495 g (the standard deviation stays the same)?

Problem 5 [B] The size of a consumer loan is $X \sim N(40, 12^2)$ million UZS.

(a) Find the share of loans larger than 60 million.
(b) Find the share of loans smaller than 30 million.
(c) Find the cut-off for the largest 5% of loans and the interval containing the central 90% of loans.

Problem 6 [B] For $X \sim N(90, \sigma^2)$ it is known that $P(X > 100) = 0.10$. For $Y \sim N(\mu, 8^2)$ it is known that $P(Y < 50) = 0.20$. Find σ and μ.

Problem 7 [B] Student A scored 82 on an exam $N(70, 8^2)$, student B scored 640 on an exam $N(580, 60^2)$. Who showed the better relative result? What share of test-takers scored lower than each?

Problem 8 [B] The returns of two independent assets (%): $R_1 \sim N(8, 4^2)$, $R_2 \sim N(5, 3^2)$. The portfolio is the sum $R_1 + R_2$.

(a) Find the distribution of $R_1 + R_2$.
(b) Find the probability of a negative total return.
(c) Find the probability that $R_1 < R_2$.

Problem 9 [B] 40% of a shop's visitors make a purchase, independently. 200 visitors came in a day.

(a) Check that the normal approximation is applicable and find μ and σ of the number of buyers.
(b) Find the probability that there are at least 90 buyers (with a continuity correction).

Problem 10 [B] The time between customer arrivals is exponential with a mean of 2 minutes.

(a) Find the probability of waiting longer than 3 minutes for the next customer.
(b) Two minutes have already passed with no customers. What is the probability that we must wait at least 3 more minutes?
(c) What is the distribution of the number of customers in 10 minutes? Find the probability that nobody arrives in 10 minutes.

Problem 11 [C] Prove the memorylessness of the exponential distribution: $P(X > s + t \mid X > s) = P(X > t)$.

Problem 12 [C] For $X \sim U(a, b)$ derive $\mathrm{E}[X]$ and $\mathrm{Var}(X)$ by integration.

Problem 13 [C] Let $X \sim N(\mu, \sigma^2)$. Show that the z-score $Z = (X - \mu)/\sigma$ has $\mathrm{E}[Z] = 0$ and $\mathrm{Var}(Z) = 1$. Explain why $IQR \approx 1.349\sigma$ implies that for $IQR = 27$ in normal data $\sigma \approx 20$.

STAT201 · STATISTICS · LECTURE 6

Joint Distributions, Covariance and Correlation

How to describe two random variables at once, and why the relationship between assets determines portfolio risk

Syllabus topic 6 · week 6 · 4 academic hours

LECTURE GOALS

- read and build a joint distribution table and find marginal and conditional distributions;
- check the independence of two discrete variables;
- compute covariance and the correlation coefficient (from a distribution and from a sample) and interpret them;
- find the variance of a sum and a difference of dependent variables;
- compute the risk of a two-asset portfolio and explain the effect of diversification.

1 The joint distribution

So far we have studied a single variable. In economics pairs are interesting: price and demand, the returns of two assets, the sales of two shops. A *joint distribution* specifies the probabilities $p(x, y) = P(X = x,\ Y = y)$ of all pairs of values.

EXAMPLE 1. SALES OF TWO SHOPS

X is the number of units of a product sold per hour in shop A, Y in shop B (values $0, 1, 2$).

	$X = 0$	$X = 1$	$X = 2$	$P(Y = y)$
$Y = 0$	0.10	0.10	0.05	**0.25**
$Y = 1$	0.10	0.30	0.10	**0.50**
$Y = 2$	0.05	0.05	0.15	**0.25**
$P(X = x)$	**0.25**	**0.45**	**0.30**	1

The sum of all nine probabilities is 1. The row and column sums (the margins of the table) give the *marginal* distributions of X and Y.

The marginal distribution is obtained by summation: $p_X(x) = \sum_y p(x, y)$. It gives the usual characteristics: $\mathrm{E}[X] = 0 \cdot 0.25 + 1 \cdot 0.45 + 2 \cdot 0.30 = 1.05$, $\mathrm{Var}(X) = 1.65 - 1.05^2 = 0.5475$; for Y: $\mathrm{E}[Y] = 1$, $\mathrm{Var}(Y) = 0.5$.

1.1 Conditional distributions and independence

The conditional distribution is obtained as in Lecture 3: $p(y \mid x) = p(x,y)/p_X(x)$. For $X = 2$ (the column is divided by 0.30): $P(Y=0 \mid X=2) = 0.1\overline{6}$, $P(Y=1 \mid X=2) = 0.\overline{3}$, $P(Y=2 \mid X=2) = 0.5$. The conditional expectation is $\mathrm{E}[Y \mid X=2] = 0.333 + 1.0 = 1.33$, whereas $\mathrm{E}[Y] = 1$: when shop A sells a lot, shop B on average does too.

INDEPENDENCE

Discrete variables are independent if $p(x,y) = p_X(x)\,p_Y(y)$ for *all* pairs (x,y).

In our example $p(0,0) = 0.10 \neq 0.25 \cdot 0.25 = 0.0625$, so X and Y are dependent. One violation is enough.

2 Covariance and correlation

COVARIANCE

$$\mathrm{Cov}(X,Y) = \mathrm{E}[(X-\mu_X)(Y-\mu_Y)] = \mathrm{E}[XY] - \mu_X\mu_Y, \qquad \mathrm{E}[XY] = \sum_{x,y} xy\,p(x,y). \quad (2.1)$$

A positive covariance means that large values of X are on average accompanied by large values of Y; a negative one means the opposite. Its value depends on the units of measurement, so a dimensionless coefficient is used to assess the strength of the relationship.

CORRELATION

$$\rho = \mathrm{Corr}(X,Y) = \frac{\mathrm{Cov}(X,Y)}{\sigma_X\sigma_Y}, \qquad -1 \le \rho \le 1. \quad (2.2)$$

EXAMPLE 2. CONTINUATION: SALES OF TWO SHOPS

Nonzero terms in $\mathrm{E}[XY]$ come only from $x, y \ge 1$: $1\cdot 1\cdot 0.30 + 2\cdot 1\cdot 0.10 + 1\cdot 2\cdot 0.05 + 2\cdot 2\cdot 0.15 = 0.30 + 0.20 + 0.10 + 0.60 = 1.20$. Therefore $\mathrm{Cov}(X,Y) = 1.20 - 1.05\cdot 1 = 0.15$, $\rho = \dfrac{0.15}{\sqrt{0.5475\cdot 0.5}} = 0.287$ — the relationship is positive and weak.

Properties: $\mathrm{Cov}(X,X) = \mathrm{Var}(X)$; $\mathrm{Cov}(aX+b,\, cY+d) = ac\ \mathrm{Cov}(X,Y)$; $\mathrm{Corr}(aX+b,\, cY+d) = \mathrm{sign}(ac)\ \mathrm{Corr}(X,Y)$ (correlation does not depend on units). If X and Y are independent, then $\mathrm{Cov} = 0$ and $\rho = 0$.

COMMON MISTAKE

The converse is false: $\rho = 0$ does not mean independence. Correlation measures only a *linear* relationship. Example: X takes the values $-2, -1, 0, 1, 2$ with equal probability, and $Y = X^2$; Y is completely determined by X, but $\mathrm{Cov}(X,Y) = 0$.

2.1 Sample covariance and correlation

From observations (x_i, y_i), $i = 1..n$:

$$s_{xy} = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{n-1}, \qquad r = \frac{s_{xy}}{s_x s_y} = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum (x_i - \bar{x})^2 \sum (y_i - \bar{y})^2}}. \tag{2.3}$$

EXAMPLE 3. ADVERTISING AND SALES

Six months: advertising spend x (million UZS) $= 2, 3, 5, 6, 8, 10$; sales y (million UZS) $= 15, 17, 22, 31, 33, 40$. $\bar{x} = 5.667$, $\bar{y} = 26.333$. $\sum (x_i - \bar{x})^2 = 45.33$, $\sum (y_i - \bar{y})^2 = 487.33$, $\sum (x_i - \bar{x})(y_i - \bar{y}) = 145.67$ (terms $41.56;\ 24.89;\ 2.89;\ 1.56;\ 15.56;\ 59.22$). Hence $s_{xy} = 145.67/5 = 29.13$, $s_x = 3.01$, $s_y = 9.87$, $r = \dfrac{145.67}{\sqrt{45.33 \cdot 487.33}} = 0.980$.

REMEMBER

Correlation is not causation. A high r may arise because of a third factor, because of reverse causation, or by chance. We discuss this in detail in Lectures 12–13.

3 The sum and difference of dependent variables

Variance of a sum

$$\operatorname{Var}(X \pm Y) = \operatorname{Var}(X) + \operatorname{Var}(Y) \pm 2 \operatorname{Cov}(X, Y).$$

For independent variables the earlier rule remains (Lecture 4). In the shops example: $\operatorname{Var}(X + Y) = 0.5475 + 0.5 + 2 \cdot 0.15 = 1.3475$, and $\operatorname{Var}(X - Y) = 1.0475 - 0.30 = 0.7475$. The expectation is always $\mathrm{E}[X + Y] = 2.05$. A positive relationship makes the sum more variable and the difference less so.

4 A portfolio of two assets

The asset returns are R_1, R_2 with expectations μ_1, μ_2, standard deviations σ_1, σ_2 and correlation ρ. A share w of capital is invested in the first asset, $1 - w$ in the second. The portfolio return is $R_p = wR_1 + (1-w)R_2$:

$$\mathrm{E}\big[R_p\big] = w\mu_1 + (1-w)\mu_2, \qquad \sigma_p^2 = w^2\sigma_1^2 + (1-w)^2\sigma_2^2 + 2w(1-w)\rho\,\sigma_1\sigma_2. \tag{4.1}$$

EXAMPLE 4. DIVERSIFICATION

$\sigma_1 = 10\%$, $\sigma_2 = 15\%$, $w = 0.5$. The standard deviation of the portfolio:

ρ	1	0.5	0	−0.5	−1
σ_p, %	12.50	10.90	9.01	6.61	2.50

Only at $\rho = 1$ does the portfolio risk equal the average risk of the assets (12.5%). The lower the correlation, the stronger the diversification effect; the expected return does not depend on ρ.

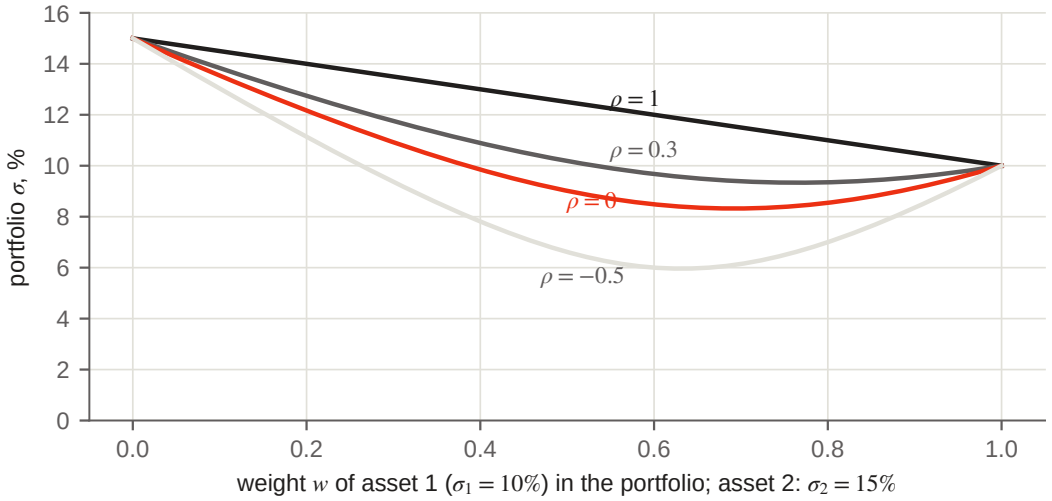


Fig. 1. Portfolio risk as a function of the weight of asset 1 for different correlations. The lower ρ, the deeper the "dip" of the curve.

EXAMPLE 5. MINIMUM RISK

For $\rho = 0.3$ the weight of asset 1 that minimizes the variance is $w^* = \dfrac{\sigma_2^2 - \rho\sigma_1\sigma_2}{\sigma_1^2 + \sigma_2^2 - 2\rho\sigma_1\sigma_2} = \dfrac{0.0225 - 0.0045}{0.01 + 0.0225 - 0.009} = 0.766$, $\sigma_p = 9.33\%$ — less than either asset alone (10% and 15%).

The same reasoning explains why the risk of a portfolio of n equally risky loans with pairwise correlation ρ does not fall to zero: $\sigma_{\bar{X}} = \sigma\sqrt{\frac{1+(n-1)\rho}{n}}$. For $\sigma = 2.0$, $n = 100$: at $\rho = 0$ we get 0.20, at $\rho = 0.1$ already 0.66. Systematic (common) risk is not eliminated by diversification.

5 Key terms

Term	*In Russian*	Meaning
Joint distribution	*Совместное распределение*	$p(x,y) = P(X=x, Y=y)$
Marginal distribution	*Маргинальное распределение*	$p_X(x) = \sum_y p(x,y)$
Conditional distribution	*Условное распределение*	$p(y \mid x) = p(x,y)/p_X(x)$
Covariance	*Ковариация*	$\mathrm{E}[XY] - \mu_X\mu_Y$
Correlation	*Корреляция*	$\rho = \mathrm{Cov}/(\sigma_X\sigma_Y)$
Diversification	*Диверсификация*	risk reduction thanks to imperfect correlation

Formulas of Lecture 6

$$\mathrm{Cov}(X,Y) = \mathrm{E}[XY] - \mu_X\mu_Y, \qquad \rho = \frac{\mathrm{Cov}(X,Y)}{\sigma_X\sigma_Y},$$

$$r = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum (x_i - \bar{x})^2 \sum (y_i - \bar{y})^2}},$$

$$\mathrm{Var}(X \pm Y) = \mathrm{Var}(X) + \mathrm{Var}(Y) \pm 2\,\mathrm{Cov}(X,Y),$$

$$\mathrm{Cov}(aX + b, cY + d) = ac\,\mathrm{Cov}(X,Y),$$

$$\sigma_p^2 = w^2\sigma_1^2 + (1-w)^2\sigma_2^2 + 2w(1-w)\rho\sigma_1\sigma_2.$$

6 Practice problems

Problem level is shown in square brackets: **[A]** — basic, tests understanding of definitions; **[B]** — standard computation; **[C]** — advanced, requires a derivation or an interpretation. Decimals are written with a point, as in a calculator, Excel and R. If a problem needs tables, use the appendix “Statistical Tables”.

Problem 1 [A] The joint distribution of two indicators: $P(X=0, Y=0) = 0.3$, $P(X=0, Y=1) = 0.2$, $P(X=1, Y=0) = 0.1$, $P(X=1, Y=1) = 0.4$.

(a) Find the marginal distributions of X and Y.
(b) Find $\mathrm{Cov}(X,Y)$ and ρ.
(c) Find $P(Y=1 \mid X=1)$. Are X and Y independent?

Problem 2 [A] $\mathrm{Var}(X) = 9$, $\mathrm{Var}(Y) = 16$, $\mathrm{Cov}(X,Y) = 6$. Find ρ, $\mathrm{Var}(X+Y)$, $\mathrm{Var}(X-Y)$, $\mathrm{Var}(2X - 3Y)$.

Problem 3 [A] Check independence in two tables (rows $Y = 0, 1$; columns $X = 0, 1$):

(1) $p(0,0)=0.2$, $p(1,0)=0.3$, $p(0,1)=0.2$, $p(1,1)=0.3$;
(2) $p(0,0)=0.1$, $p(1,0)=0.4$, $p(0,1)=0.3$, $p(1,1)=0.2$.
(The first index is the value of X, the second the value of Y.)

Problem 4 [B] Five students: study time x (h) $=1,2,3,4,5$ and score $y=50,55,65,70,85$.

(a) Find $\bar{x}$, $\bar{y}$ and $\sum(x_i-\bar{x})(y_i-\bar{y})$.
(b) Find the sample covariance and correlation.

Problem 5 [B] Assets: $\mu_1=9\%$, $\sigma_1=12\%$; $\mu_2=14\%$, $\sigma_2=20\%$; $\rho=0.2$. Portfolio: 40% in asset 1 and 60% in asset 2.

(a) Find the expected return and standard deviation of the portfolio.
(b) Compare σ_p with the weighted average of the standard deviations. What does the difference show?

Problem 6 [B] $\sigma_1=10\%$, $\sigma_2=15\%$, the assets are uncorrelated ($\rho=0$). At what weight w of asset 1 is the portfolio variance minimal? What is σ_p?

Problem 7 [B] For the data of Problem 4, study time was measured not in hours but in minutes ($x'=60x$), and scores as a fraction of the maximum (suppose the maximum is 100, $y'=y/100$). How do s_{xy} and r change? Compute.

Problem 8 [B] $X\in\{0,1,2\}$ is the number of complaints per week, $Y\in\{1,2\}$ the shift. Joint probabilities $p(x,y)$: $p(0,1)=0.1$, $p(1,1)=0.2$, $p(2,1)=0.1$, $p(0,2)=0.2$, $p(1,2)=0.3$, $p(2,2)=0.1$.

(a) Find the marginal distributions and $\mathrm{E}[X]$, $\mathrm{E}[Y]$.
(b) Find $\mathrm{Cov}(X,Y)$ and ρ.
(c) Find the conditional distribution of X given $Y=2$ and $\mathrm{E}[X\mid Y=2]$.

Problem 9 [B] The costs of two components: $\sigma_1=4$, $\sigma_2=6$. Find the standard deviation of the total cost for $\rho=0.5$, $\rho=0$ and $\rho=-0.5$.

Problem 10 [B] A bank issued 50 loans, each with $\sigma=2.0$ million. Find the standard deviation of the average profit for pairwise correlation $\rho=0$ and $\rho=0.2$; find the limit as $n\to\infty$ for $\rho=0.2$.

Problem 11 [C] Prove that (a) $\mathrm{Cov}(X,Y)=\mathrm{E}[XY]-\mu_X\mu_Y$; (b) $\mathrm{Var}(X+Y)=\mathrm{Var}(X)+\mathrm{Var}(Y)+2\,\mathrm{Cov}(X,Y)$.

Problem 12 [C] Let $Y=aX+b$, $a\neq 0$. Show that $\rho_{XY}=+1$ for $a>0$ and -1 for $a<0$.

Problem 13 [C] X takes the values $-1,0,1$ with probabilities $\frac14,\frac12,\frac14$, and $Y=X^2$. Find $\mathrm{Cov}(X,Y)$ and prove that X and Y are dependent.

STAT201 · STATISTICS · LECTURE 7

Sampling Distributions and the Central Limit Theorem

Why the sample mean fluctuates, how to measure this fluctuation, and why it is almost always normal

Syllabus topic 7 · week 7 · 4 academic hours

LECTURE GOALS

- distinguish a population from a sample, a parameter from a statistic;
- explain that the sample mean is a random variable and describe its distribution (expectation and standard error);
- state the central limit theorem (CLT) and the conditions for its use;
- find probabilities for $\bar{X}$ and $\hat{p}$ using the normal distribution;
- determine the sample size from a required precision and understand why $n-1$ is used in the sample variance.

1 Population, sample, statistic

The *population* is all the objects we want to learn about (all bank customers, all parts of a batch). The *sample* is its part that we actually observe. A *parameter* is a number characterizing the population (μ, σ, p); it is fixed and usually unknown. A *statistic* is a number computed from a sample ($\bar{x}$, s, $\hat{p}$); it is known but *changes from sample to sample*.

For conclusions to be reliable the sample must be random (each object has a known nonzero chance of entering it). A simple random sample of n independent observations is denoted $X_1, \dots, X_n$. A non-representative sample (for example, a phone survey during working hours only) gives biased estimates that cannot be fixed by increasing n.

SAMPLING DISTRIBUTION

The sampling distribution of a statistic is the distribution of the values it takes over all possible samples of size n.

2 The sample mean

Let $X_1, \dots, X_n$ be independent, $\mathrm{E}[X_i] = \mu$, $\mathrm{Var}(X_i) = \sigma^2$. The mean is $\bar{X} = \frac{1}{n}\sum X_i$. From the properties of Lecture 4:

$$\mathrm{E}\big[\bar{X}\big] = \mu, \qquad \mathrm{Var}(\bar{X}) = \frac{\sigma^2}{n}, \qquad \mathrm{SE}(\bar{X}) = \sigma_{\bar{X}} = \frac{\sigma}{\sqrt{n}}. \tag{2.1}$$

The first equality means that $\bar{X}$ is an *unbiased* estimator of μ: averaged over all samples it hits the target. The second: the spread decreases as n grows, but slowly — to halve the error one needs a sample four times larger. The standard deviation of a statistic is called the *standard error* (SE).

COMMON MISTAKE

σ is the spread of individual observations, $\text{SE} = \sigma/\sqrt{n}$ is the spread of the mean. These are different numbers: in problems about "the sample mean" divide by $\sqrt{n}$, in problems about "a single observation" do not.

EXAMPLE 1. A SAMPLING DISTRIBUTION IN A SMALL EXAMPLE

The population is $\{1, 2, 3\}$, $\mu = 2$, $\sigma^2 = 2/3$. Take samples of 2 elements with replacement (9 equally likely pairs). The mean $\bar{x}$ takes the values $1; 1.5; 2; 2.5; 3$ with probabilities $\frac{1}{9}, \frac{2}{9}, \frac{3}{9}, \frac{2}{9}, \frac{1}{9}$. The expectation is $\mathrm{E}\big[\bar{X}\big] = 2 = \mu$, the variance $\mathrm{Var}(\bar{X}) = \frac{1}{3} = \frac{\sigma^2}{n} = \frac{2/3}{2}$ — formula (2.1) is confirmed by direct enumeration.

If the sample is drawn without replacement from a finite population of size N and makes up a noticeable share ($n/N > 5\%$), the standard error is multiplied by the correction $\sqrt{\frac{N-n}{N-1}}$. For example, $N = 500$, $n = 100$, $\sigma = 4$: $\text{SE} = 0.4 \cdot 0.895 = 0.358$ instead of 0.4.

3 The central limit theorem

CLT

Let $X_1, \dots, X_n$ be independent identically distributed variables with finite μ and σ^2. Then for large n the distribution of $\bar{X}$ is approximately normal:

$$\bar{X} \;\approx\; N\!\left(\mu,\, \frac{\sigma^2}{n}\right), \qquad Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \approx N(0,1), \tag{3.1}$$

whatever the original distribution. Similarly for the sum: $\sum X_i \approx N(n\mu,\, n\sigma^2)$.

Rule of thumb: $n \geq 30$ is enough for moderately skewed distributions; for strongly skewed ones or ones with outliers more is needed; if the original distribution is normal, $\bar{X}$ is normal for any n.

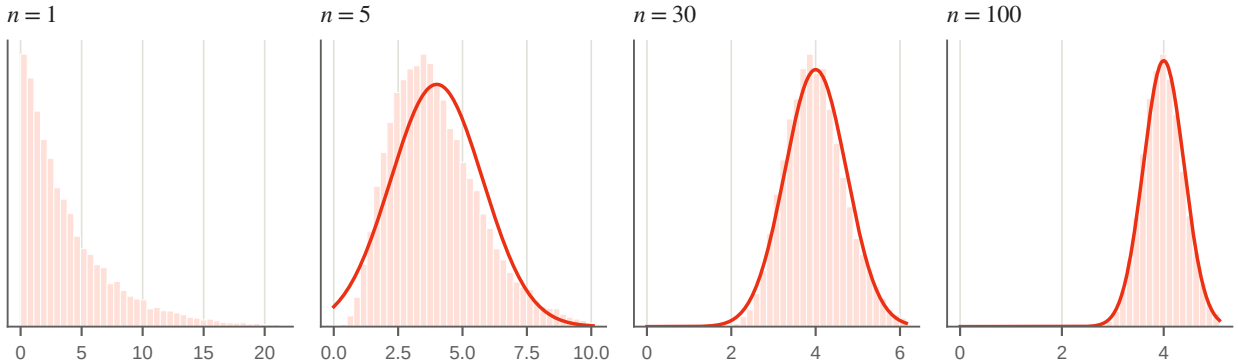


Fig. 1. Simulation: the distribution of the sample mean from an exponential population ($\mu = \sigma = 4$) over 20 000 samples. For $n = 1$ strongly skewed; for $n = 30$ almost bell-shaped; the red line is the normal approximation $N(4, (4/\sqrt{n})^2)$.

EXAMPLE 2. PROCESSING TIME OF AN APPLICATION

The processing time of a loan application: $\mu = 12$ days, $\sigma = 4$. A sample of $n = 36$ applications. SE $= 4/6 = 0.667$. $P(\bar{X} > 13) = P(Z > 1.5) = 0.0668$. $P(11 < \bar{X} < 13) = P(-1.5 < Z < 1.5) = 0.8664$. The mean of 36 applications lies in 95% of cases within $12 \pm 1.96 \cdot 0.667$, i.e. from 10.69 to 13.31 days. Note: for a single application $P(X > 13)$ is substantially larger (about 40% for a normal X).

EXAMPLE 3. A SKEWED POPULATION

Waiting time is exponential with a mean of 4 minutes. $n = 40$, SE $= 4/\sqrt{40} = 0.632$. $P(\bar{X} > 5) \approx P(Z > 1.58) = 0.0569$. The exact value (a sum of exponential variables has a gamma distribution) is 0.0646; the discrepancy remains because of the noticeable skewness, but the order of magnitude is right.

4 The sample proportion

For a proportion $\hat{p} = X/n$, where $X \sim \text{Bin}(n, p)$ is the number of "successes" in the sample, by the CLT (normal approximation to the binomial, Lecture 5):

$$\text{E}[\hat{p}] = p, \qquad \text{SE}(\hat{p}) = \sqrt{\frac{p(1-p)}{n}}, \qquad \hat{p} \approx N\left(p,\ \frac{p(1-p)}{n}\right) \text{ when } np \geq 10,\ n(1-p) \geq 10. \tag{4.1}$$

EXAMPLE 4. CUSTOMER SATISFACTION

Among bank customers 40% are satisfied with the service. $n = 100$ were surveyed. SE $= \sqrt{0.4 \cdot 0.6/100} = 0.049$. $P(\hat{p} > 0.50) = P(Z > 2.04) = 0.0206$. $P(0.35 < \hat{p} < 0.45) = P(|Z| < 1.02) = 0.6926$: two thirds of samples give a proportion within ± 5 percentage points of the true one.

5 Sample size and precision

Suppose we need the sample mean to differ from μ by no more than E with probability 95%. The condition $1.96\,\sigma/\sqrt{n} \leq E$ gives

$$n \geq \left(\frac{1.96\,\sigma}{E}\right)^2. \tag{5.1}$$

For $\sigma = 4$ days and $E = 1$ day: $n \geq (7.84)^2 = 61.5$, i.e. $n = 62$. For proportions one uses $n \geq \dfrac{1.96^2\,p(1-p)}{E^2}$; when p is unknown the worst case $p = 0.5$ is taken: for $E = 0.03$ (3 percentage points) we get $n \geq 1067.1$, i.e. 1068 respondents. This number explains why opinion polls usually contain about a thousand people.

6 The sample variance and division by $n-1$

The sample variance $s^2 = \dfrac{\sum(X_i - \bar{X})^2}{n-1}$ is an unbiased estimator of σ^2: $\mathrm{E}[s^2] = \sigma^2$. If we divide by n, the estimate is systematically too low, because deviations are computed from $\bar{X}$ rather than from the unknown μ: the sample is "closer to its own mean" than to μ.

EXAMPLE 5. A CHECK ON A SMALL POPULATION

In the example above ($\{1, 2, 3\}$, samples of 2 with replacement) $\sigma^2 = 0.667$. The average value of s^2 with the divisor $n-1=1$ is $0.667 = \sigma^2$, and with the divisor $n=2$ it is 0.333, half as much.

REMEMBER

The sampling distribution of s^2 (more precisely, of $(n-1)s^2/\sigma^2$) for a normal population is chi-squared with $n-1$ degrees of freedom. It will be needed in Lecture 11.

7 Key terms

Term	*In Russian*	Meaning
Population	*Генеральная совокупность*	all objects of the study
Sample	*Выборка*	the observed part of the population
Parameter / statistic	*Параметр / статистика*	μ, σ, p / $\bar{x}, s, \hat{p}$
Sampling distribution	*Выборочное распределение*	the distribution of a statistic over all samples
Standard error	*Стандартная ошибка*	$\sigma/\sqrt{n}$; $\sqrt{p(1-p)/n}$
Unbiasedness	*Несмещённость*	$\mathrm{E}[\text{estimator}] = \text{parameter}$
Central limit theorem (CLT)	*ЦПТ*	$\bar{X} \approx N(\mu, \sigma^2/n)$

Formulas of Lecture 7

$$\mathrm{E}[\bar{X}] = \mu, \quad \mathrm{SE}(\bar{X}) = \frac{\sigma}{\sqrt{n}}, \quad Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \approx N(0,1);$$

$$\mathrm{E}[\hat{p}] = p, \quad \mathrm{SE}(\hat{p}) = \sqrt{\frac{p(1-p)}{n}};$$

$$n \geq \left(\frac{1.96\sigma}{E}\right)^2, \qquad n \geq \frac{1.96^2 p(1-p)}{E^2}; \qquad s^2 = \frac{\sum(x_i - \bar{x})^2}{n-1}.$$

8 Practice problems

> Problem level is shown in square brackets: **[A]** — basic, tests understanding of definitions; **[B]** — standard computation; **[C]** — advanced, requires a derivation or an interpretation. Decimals are written with a point, as in a calculator, Excel and R. If a problem needs tables, use the appendix “Statistical Tables”.

Problem 1 [A] The average bill in a chain of cafés is $\mu = 50$ thousand UZS, $\sigma = 12$. A sample of $n = 36$ bills is taken.

(a) Find the expectation and standard error of $\bar{X}$.
(b) Find $P(\bar{X} > 53)$ and $P(48 < \bar{X} < 52)$.

Problem 2 [A] $\sigma = 20$. Find the standard error of the mean for $n = 25$ and $n = 100$. What sample size is needed for SE $= 1$?

Problem 3 [A] The share of customers with the mobile app is $p = 0.30$. $n = 200$ were surveyed. Find SE$(\hat{p})$, $P(\hat{p} > 0.35)$ and $P(0.28 < \hat{p} < 0.32)$.

Problem 4 [B] The size of a deposit is strongly right-skewed: $\mu = 8$ million, $\sigma = 5$ million. A sample of $n = 50$ deposits.

(a) May the normal distribution be used for $\bar{X}$? Justify.
(b) Find $P(\bar{X} < 7)$ and $P(\bar{X} > 9.5)$.
(c) Can $P(X < 7)$ be found in the same way for a single deposit?

Problem 5 [B] Each of 100 transactions has $\mu = 2$ million UZS and $\sigma = 1.5$ million, the transactions are independent. Find the approximate probability that the total volume exceeds 230 million.

Problem 6 [B] We need to estimate the average time to resolve a request ($\sigma \approx 15$ minutes) to within $E = 2$ minutes at 95% confidence; then to estimate the share of satisfied customers to within ± 0.04.

(a) Find the required n for the mean.
(b) How does n change if the required precision is $E = 1$ minute?
(c) Find n for the proportion with unknown p and with $p \approx 0.2$.

Problem 7 [B] The defect rate on a line is $p = 0.05$. A batch of $n = 400$ parts is inspected.

(a) Check the conditions of the normal approximation and find SE$(\hat{p})$.
(b) Find the probability that the sample defect rate exceeds 7%.
(c) Find the probability that the sample defect rate turns out below 3%.

Problem 8 [B] The population is $\{2, 4, 6\}$; samples of 2 with replacement are taken (9 equally likely).

(a) Find μ and σ^2.
(b) Write the distribution of $\bar{X}$ and check $\mathrm{E}[\bar{X}] = \mu$, $\mathrm{Var}(\bar{X}) = \sigma^2/2$.
(c) Find the mean value of s^2 (with divisor $n - 1$).

Problem 9 [B] From a population of $N = 400$ customers with $\sigma = 20$ a sample of $n = 100$ is drawn without replacement. Find the standard error of the mean with and without the finite population correction.

Problem 10 [C] Derive $\mathrm{Var}(\bar{X}) = \sigma^2/n$ and show that $\mathrm{E}[s^2] = \sigma^2$ if $s^2 = \frac{1}{n-1}\sum (X_i - \bar{X})^2$.

Problem 11 [C] The law of large numbers. For $\sigma = 4$ and $\varepsilon = 1$ Chebyshev's inequality gives $P(|\bar{X} - \mu| \geq \varepsilon) \leq \sigma^2/(n\varepsilon^2)$. Estimate the probability for $n = 100$ and compare with the normal approximation. What does the probability tend to as $n \to \infty$?

Problem 12 [C] An online poll in the bank's Telegram channel showed 80% satisfied customers with $n = 5000$. Suppose $\mathrm{SE} \approx 0.006$. Does this mean that the share of satisfied customers among all customers lies within $0.80 \pm 2 \cdot 0.006$? Explain.

STAT201 · STATISTICS · LECTURE 8

The CLT in Practice and Midterm Review

Applications of the central limit theorem, a map of the course (weeks 1–7), common mistakes and a practice exam

Syllabus topic 8 · week 8 · 4 academic hours

LECTURE GOALS

- apply the CLT to sums, means and proportions in practical problems (portfolio risk, inventory, quality control);
- choose the right model and formula from the wording of a problem;
- find and correct common mistakes in calculations;
- solve a practice midterm in exam format, finishing within the time limit.

1 Applications of the CLT

Let us recall the main result of Lecture 7: for independent identically distributed variables with μ and σ

$$\bar{X} \approx N\Big(\mu, \frac{\sigma^2}{n}\Big), \qquad \sum_{i=1}^{n} X_i \approx N(n\mu,\ n\sigma^2), \qquad \hat{p} \approx N\Big(p, \frac{p(1-p)}{n}\Big).$$

Three steps for solving any such problem: (1) write out μ, σ, n and decide what the question is about (a sum, a mean or a proportion); (2) find the parameters of the approximating normal distribution (for a sum $n\mu$ and $\sigma\sqrt{n}$; for a mean μ and $\sigma/\sqrt{n}$); (3) standardize and find the probability from the table.

EXAMPLE 1. CASH LIMIT OF AN ATM

Daily cash withdrawal by one customer: $\mu = 300$ thousand UZS, $\sigma = 200$ thousand. $n = 100$ independent customers come to the ATM. The total withdrawal is $T \approx N(30\,000,\ 2000^2)$ (since $\sigma\sqrt{n} = 200 \cdot 10 = 2000$). The ATM holds 33 000 thousand. The probability of a shortfall: $P(T > 33\,000) = P(Z > 1.5) = 0.0668$. For the probability of a shortfall to be at most 1%, a stock of $30\,000 + 2.326 \cdot 2000 = 34\,653$ thousand UZS is needed.

EXAMPLE 2. QUALITY CONTROL BY PROPORTION

The defect rate in a production process is 8% when operating normally. 200 items are taken from a batch and the line is stopped if the sample defect rate exceeds 12%. SE $= \sqrt{0.08 \cdot 0.92/200} = 0.0192$. A false stop (under normal operation): $P(\hat{p} > 0.12) = P(Z > 2.09) = 0.0183$.

COMMON MISTAKE

For a sum the standard deviation is $\sigma\sqrt{n}$, for a mean it is $\sigma/\sqrt{n}$. The most common mistake in the exam is to mix them up. A common-sense check: a mean fluctuates *less* than an individual value, a sum *more*.

2 Course map: weeks 1–7

Week	Task	Tool
1	describe data	$\bar{x},\ s,$ Me, $Q_1, Q_3,\ IQR$; outlier fences $Q_1 - 1.5IQR, Q_3 + 1.5IQR$; $CV = s/\bar{x}$; r
2	probability of an event	$P(A \cup B)$, independence, $\binom{n}{k}$, $P(n,k)$, trees
3	updating from data	$P(A \mid B)$, total probability, Bayes, odds
4	discrete random variable	$\mathrm{E}[X]$, Var, Bin, Geom, Pois
5	continuous variable	$z = (x-\mu)/\sigma$, Φ, U, Exp, normal approximation to Bin
6	two variables at once	Cov, ρ, $\mathrm{Var}(X \pm Y)$, portfolio
7	sample → population	$\mathrm{SE} = \sigma/\sqrt{n}$, CLT, $\hat{p}$, sample size

2.1 How to choose a model

- "How many successes in n independent trials" — Bin(n,p); for $np, n(1-p) \geq 10$ — the normal approximation.
- "How many events per hour/day with no upper limit" — Poisson; "how long to wait for the next one" — exponential.
- "Measured on a continuous scale, symmetric, bell-shaped" — normal; a question about the *mean* of a large sample — always normal by the CLT.
- "Something is known, find the probability of the cause" — Bayes.
- "Two variables, find the risk of the sum" — $\mathrm{Var}(X+Y)$ with the covariance.

2.2 Common mistakes

1. Confusing $P(A \mid B)$ and $P(B \mid A)$ (Lecture 3).
2. Using the binomial model without independence or without a constant p.
3. Adding standard deviations instead of variances; forgetting that $\mathrm{Var}(X-Y) = \mathrm{Var}(X) + \mathrm{Var}(Y)$ for independent variables.
4. Using σ instead of $\sigma/\sqrt{n}$ for a mean.
5. Dividing by n instead of $n-1$ when computing s^2 from a sample.
6. Thinking that $\rho = 0$ means independence, and that a high correlation means causation.
7. Getting the sign wrong with z-scores: $P(Z > z) = 1 - \Phi(z)$.
8. Not checking the conditions of applicability (normal approximation, independence of observations, representativeness).

3 Format of the practice exam

At the end of the section there is a practice midterm of 12 problems of varying difficulty. Recommendations: solve against the clock (90 minutes), using only a calculator and the tables from the appendix; first write out “given” and “find”, then the formula and the substitution, and only then the number; write the answer with units and an interpretation (in words, what it means). Check the solutions (in the combined instructor’s file) after your attempt, not before.

What to remember for the midterm

$$s^2 = \tfrac{1}{n-1}\sum (x_i - \bar{x})^2; \quad IQR = Q_3 - Q_1; \quad P(A \cup B) = P(A) + P(B) - P(A \cap B);$$

$$P(A \mid B) = \tfrac{P(A \cap B)}{P(B)}; \quad P(A_i \mid B) = \tfrac{P(B \mid A_i)P(A_i)}{\sum_j P(B \mid A_j)P(A_j)};$$

$$\mathrm{E}[X] = \sum x p(x); \quad \mathrm{Var}(X) = \mathrm{E}\left[X^2\right] - \mu^2;$$

$$\text{Bin}: \binom{n}{k} p^k (1-p)^{n-k}; \quad \text{Pois}: e^{-\lambda}\lambda^k / k!;$$

$$z = \tfrac{x-\mu}{\sigma}; \quad \mathrm{Var}(X \pm Y) = \mathrm{Var}\, X + \mathrm{Var}\, Y \pm 2\,\mathrm{Cov};$$

$$\mathrm{SE}(\bar{X}) = \tfrac{\sigma}{\sqrt{n}}; \quad \mathrm{SE}(\hat{p}) = \sqrt{\tfrac{p(1-p)}{n}}.$$

4 Practice problems

> Problem level is shown in square brackets: **[A]** — basic, tests understanding of definitions; **[B]** — standard computation; **[C]** — advanced, requires a derivation or an interpretation. Decimals are written with a point, as in a calculator, Excel and R. If a problem needs tables, use the appendix “Statistical Tables”.

Problem 1 [B] The time (minutes) spent by ten customers on arranging a service: $12,\ 15,\ 15,\ 18,\ 20,\ 22,\ 25,\ 28,\ 30,\ 45$.

(a) Find the mean, median, variance and standard deviation.
(b) Find Q_1, Q_3, IQR and the outlier fences. Are there outliers?
(c) Find the skewness coefficient $Sk = 3(\bar{x} - \mathrm{Me})/s$ and explain the sign. How do the mean and median change without the value 45?

Problem 2 [A] $P(A) = 0.5$, $P(B) = 0.4$, $P(A \cap B) = 0.2$. Find $P(A \cup B)$, $P(\overline{A} \cap \overline{B})$, $P(A \mid B)$ and check independence.

Problem 3 [B] An online shop: 60% of orders come through the website (5% are returned), 40% through the app (10% are returned).

(a) What is the probability that an order will be returned?
(b) An order was returned. What is the probability that it was placed through the app?

Problem 4 [A] A group has 10 students (5 boys and 5 girls). A committee of 3 is chosen at random. What is the probability that it contains at least one girl?

Problem 5 [B] A customer takes out a subscription with probability 0.2, independently. The offer was made to 15 customers. Find $\mathrm{E}[X]$, $\mathrm{Var}(X)$, the probability of no subscriptions and the probability of at least four subscriptions.

Problem 6 [B] There are on average 2.5 requests per minute (Poisson). Find the probability of at most one request per minute and of at least three.

Problem 7 [B] A test score is $X \sim N(500, 50^2)$.

(a) Find the share scoring above 580.
(b) Find the share scoring between 450 and 550.
(c) Find the 90th percentile.

Problem 8 [B] A game: a win of $+50$ with probability 0.3 and a loss of -10 with probability 0.7. Find the expected result and the standard deviation. Is it worth playing repeatedly?

Problem 9 [B] Two returns with correlation $\rho = -0.2$ have $\sigma_1 = 8\%$, $\sigma_2 = 12\%$. Find the risk of a 50/50 portfolio.

Problem 10 [B] The average number of purchases per customer is $\mu = 100$, $\sigma = 24$; a sample of $n = 64$. Find SE, $P(\bar{X} < 95)$ and $P(97 < \bar{X} < 103)$.

Problem 11 [B] The share of customers who cancelled the service during a year is $p = 0.25$. A sample of $n = 300$. Find $\mathrm{SE}(\hat{p})$ and $P(\hat{p} > 0.30)$.

Problem 12 [C] For the conditions of Problem 10 ($\sigma = 24$) find the sample size for which, with probability 95%, the sample mean differs from μ by no more than 4. Then show how n changes if we want to halve the permissible error, and explain this in terms of SE.

STAT201 · STATISTICS · LECTURE 9

Estimation and Confidence Intervals

How to report, instead of a single number, a range of plausible values of a parameter and honestly measure uncertainty

Syllabus topic 9 · week 9 · 4 academic hours

LECTURE GOALS

- distinguish point and interval estimates and explain the properties of a good estimator;
- construct confidence intervals for a mean (with known and unknown σ) and for a proportion;
- use the t-distribution, find the degrees of freedom and critical values;
- interpret the confidence level correctly and avoid typical wording errors;
- assess how the width of an interval depends on n, σ and the confidence level; determine the required sample size.

1 Point and interval estimates

A *point estimate* is a single number computed from a sample: $\bar{x}$ for μ, $\hat{p}$ for p, s^2 for σ^2. A good estimator is:

- *unbiased* — its average over all samples equals the parameter (Lecture 7);
- *consistent* — it approaches the parameter as n grows;
- *efficient* — it has the smallest variance among unbiased estimators.

But a single number says nothing about precision. A sample mean of 35.75 may be close to μ or far from it — that depends on the SE. A *confidence interval* (CI) gives a range of plausible values of the parameter together with a measure of the reliability of the method.

CONFIDENCE INTERVAL

An interval with confidence level $1-\alpha$ is built from a sample so that in the long run (under repeated sampling) the share of intervals covering the true parameter equals $1-\alpha$. The general form:

$$\text{estimate} \;\pm\; \underbrace{\text{critical value} \times \text{SE}}_{\text{margin of error } E}. \tag{1.1}$$

2 CI for the mean when σ is known

If σ is known (rare, but a convenient starting point) and either the population is normal or $n \geq 30$, then by the CLT

$$\bar{x} \pm z_{\alpha/2}\frac{\sigma}{\sqrt{n}}, \qquad z_{0.05} = 1.645\ (90\%), \quad z_{0.025} = 1.960\ (95\%), \quad z_{0.005} = 2.576\ (99\%). \tag{2.1}$$

EXAMPLE 1. APPLICATION PROCESSING TIME

A sample of $n = 36$ applications, $\bar{x} = 12.4$ days, $\sigma = 4$. SE $= 4/6 = 0.667$. 95% CI: $12.4 \pm 1.96 \cdot 0.667 = 12.4 \pm 1.31$, i.e. $(11.09;\ 13.71)$. 99% CI: $12.4 \pm 2.576 \cdot 0.667 = (10.68;\ 14.12)$ — wider.

3 How to understand the confidence level correctly

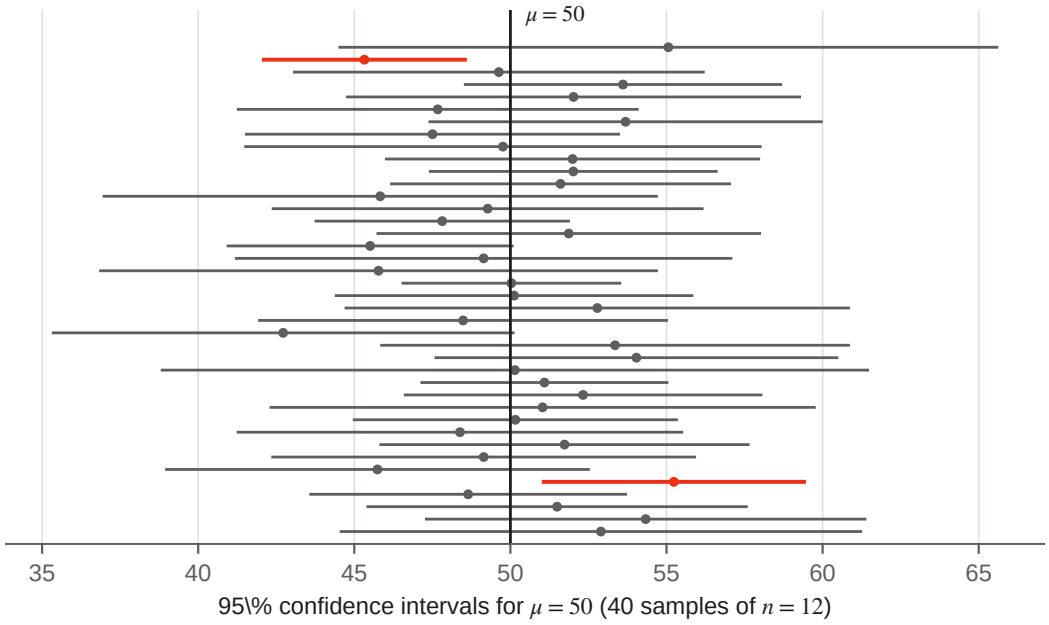


Fig. 1. 40 independent samples of 12 observations from $N(50, 10^2)$; each gives its own 95% interval. Two of the forty (red) did not cover $\mu = 50$ — exactly what 95% confidence allows.

The parameter μ is fixed, the interval's endpoints are random. "95% confidence" is a property of the *method*: 95% of the intervals built this way cover μ. For a specific computed interval $(11.09; 13.71)$ it is incorrect to say "with probability 95% μ lies in this interval" (it either contains μ or it does not), but it is acceptable to say: "we are 95% confident that the interval covers μ" or "values of μ from 11.1 to 13.7 are consistent with the data".

COMMON MISTAKE

Wrong: "95% of the observations lie in the confidence interval". A CI is for the *mean*, not for individual observations; the limits for 95% of observations (a prediction interval) are much wider.

4 Unknown σ: the t-distribution

Usually σ is unknown and is replaced by the sample s. But s is itself random, especially for small n. The statistic

$$t = \frac{\bar{X} - \mu}{s/\sqrt{n}} \tag{4.1}$$

has not a normal but a *Student distribution* with $\nu = n - 1$ degrees of freedom (if the population is normal). It is symmetric like the normal but has "heavier tails"; as ν grows it approaches $N(0, 1)$.

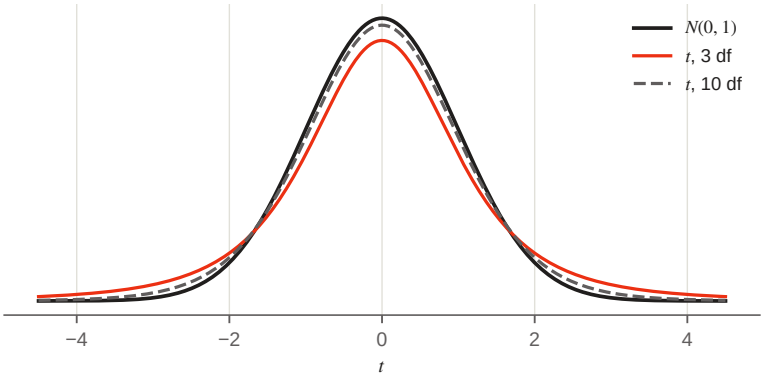


Fig. 2. The Student distribution with few degrees of freedom is wider than the normal; with 10 degrees of freedom it is already almost indistinguishable.

CI for the mean when σ is unknown

$$\bar{x} \pm t_{\alpha/2,\,n-1}\,\frac{s}{\sqrt{n}}.$$

Critical values $t_{0.025,\nu}$ (for 95%): $\nu = 4$: 2.776; 9: 2.262; 11: 2.201; 14: 2.145; 24: 2.064; 29: 2.045; 59: 2.001; 99: 1.984; ∞: 1.960 (the full table is in the appendix).

EXAMPLE 2. COFFEE SHOPS: MONTHLY REVENUE

From Lecture 1: 12 branches of a coffee-shop chain, $\bar{x} = 35.75$, $s = 20.33$ (monthly revenue of a branch, million UZS). SE $= 20.33/\sqrt{12} = 5.869$, $t_{0.025,11} = 2.201$, $E = 2.201 \cdot 5.869 = 12.92$. 95% CI: $35.75 \pm 12.92 = (22.83;\ 48.67)$. 99% CI ($t_{0.005,11} = 3.106$): $35.75 \pm 18.23 = (17.52;\ 53.98)$. If $z = 1.96$ had been used by mistake, the margin of error would have been 11.50 instead of 12.92 — the interval would have been too narrow and would "promise" more than there really is.

REMEMBER

Conditions: (1) the sample is random, observations are independent; (2) the population is approximately normal *or* n is large enough ($\gtrsim 30$). For $n < 30$ with clear skewness or outliers the t-interval is unreliable: in the coffee-shop example the outlier (branch 12) strongly affects $\bar{x}$ and s, and the result should be supplemented by an interval without it.

5 What determines the width of an interval

The margin of error $E = t \cdot s/\sqrt{n}$ decreases when: n grows (by a factor of $\sqrt{n}$ — four times as much data for an interval half as wide); the spread s decreases; the confidence level decreases (90% is narrower, 99% wider). The choice of confidence level is a compromise between certainty and precision; by default 95%.

6 CI for a proportion

CI for a proportion

$$\hat{p} \pm z_{\alpha/2}\sqrt{\frac{\hat{p}(1-\hat{p})}{n}} \qquad (\text{for } n\hat{p} \geq 10,\ n(1-\hat{p}) \geq 10).$$

EXAMPLE 3. SHARE OF SATISFIED CUSTOMERS

Of 400 respondents 240 are satisfied: $\hat{p} = 0.60$, SE $= \sqrt{0.6 \cdot 0.4/400} = 0.0245$. 95% CI: $0.60 \pm 1.96 \cdot 0.0245 = 0.60 \pm 0.048 = (0.552;\ 0.648)$. 90% CI: $(0.560;\ 0.640)$.

For small samples and extreme proportions the simple interval is unreliable. For $n = 20$, $x = 2$ ($\hat{p} = 0.10$) it gives $(-0.031;\ 0.231)$ — with a negative lower limit; the Wilson interval $(0.028;\ 0.301)$ is more correct. Rule: if the conditions $n\hat{p} \geq 10$ and $n(1-\hat{p}) \geq 10$ are not met, use the Wilson or an exact interval.

7 Sample size for a given precision

$$n \geq \left(\frac{z_{\alpha/2}\,\sigma}{E}\right)^2, \qquad n \geq \frac{z_{\alpha/2}^2\,\hat{p}(1-\hat{p})}{E^2} \ \text{(in the worst case } \hat{p} = 0.5). \tag{7.1}$$

For a mean with $\sigma \approx 20$, $E = 3$ and 99% confidence: $n \geq (2.576 \cdot 20/3)^2 = 294.9$, i.e. 295. For a proportion with $E = 0.03$ and a preliminary estimate $p \approx 0.6$: $n \geq 1.96^2 \cdot 0.24/0.0009 = 1024.4$, i.e. 1025; without a preliminary estimate (worst case) 1068. The value of σ is taken from a pilot sample or past data; with a t-interval the final n is slightly larger.

8 CI for a variance (for information)

If the population is normal, then $(n-1)s^2/\sigma^2$ has a χ^2 distribution with $n-1$ degrees of freedom, and

$$\frac{(n-1)s^2}{\chi^2_{\alpha/2,\,n-1}} \ \leq\ \sigma^2 \ \leq\ \frac{(n-1)s^2}{\chi^2_{1-\alpha/2,\,n-1}}. \tag{8.1}$$

For the coffee shops: $s^2 = 413.3$, $\chi^2_{0.025,11} = 21.92$, $\chi^2_{0.975,11} = 3.816$: $\sigma^2 \in (207.4;\ 1191.5)$, i.e. $\sigma \in (14.4;\ 34.5)$. The interval is very wide: with $n = 12$ the spread is estimated poorly. This interval is sensitive to departures from normality.

9 Key terms

Term	*In Russian*	Meaning
Point estimate	*Точечная оценка*	a single number from a sample
Confidence interval (CI)	*Доверительный интервал*	estimate $\pm$ critical value $\times$ SE
Confidence level	*Уровень доверия*	$1-\alpha$
Margin of error	*Предельная ошибка*	E — half the width of the CI
Degrees of freedom	*Степени свободы*	$\nu = n-1$ for a mean
Student's t distribution	*Распределение Стьюдента*	heavy tails, $\to N(0,1)$

Formulas of Lecture 9

$$\bar{x} \pm z_{\alpha/2}\frac{\sigma}{\sqrt{n}}, \qquad \bar{x} \pm t_{\alpha/2,\,n-1}\frac{s}{\sqrt{n}}, \qquad \hat{p} \pm z_{\alpha/2}\sqrt{\frac{\hat{p}(1-\hat{p})}{n}};$$

$$n \geq \left(\frac{z\sigma}{E}\right)^2, \qquad n \geq \frac{z^2\hat{p}(1-\hat{p})}{E^2}.$$

10 Practice problems

> Problem level is shown in square brackets: **[A]** — basic, tests understanding of definitions; **[B]** — standard computation; **[C]** — advanced, requires a derivation or an interpretation. Decimals are written with a point, as in a calculator, Excel and R. If a problem needs tables, use the appendix "Statistical Tables".

Problem 1 [A] The mean score of a sample of 49 students is 82, $\sigma = 14$. Construct a 95% CI for the mean.

Problem 2 [A] Find from the table the critical value t for: (a) a 95% CI, $n = 10$; (b) a 99% CI, $n = 25$; (c) a 90% CI, $n = 16$.

Problem 3 [A] A sample of 25 customers: $\bar{x} = 82$ thousand UZS, $s = 10$. Construct a 95% CI for the average bill. Why is t used here rather than z?

Problem 4 [B] The cost of 16 randomly selected orders: $\bar{x} = 250$ thousand, $s = 40$ thousand. Construct a 99% CI for the average order cost and state what happens to the width when moving to 95%.

Problem 5 [B] Of 500 transactions checked, 85 are flagged as suspicious. Construct 95% and 90% CIs for the share of suspicious transactions.

Problem 6 [B] Delivery times of 8 orders (minutes): $18, 22, 25, 21, 30, 24, 27, 23$.

(a) Find $\bar{x}$ and s.
(b) Construct a 95% CI for the mean delivery time.
(c) What assumption is needed, and how can it be checked on such data?

Problem 7 [B] To estimate the average response time ($\sigma \approx 20$ s) a margin of error $E = 3$ s is required. Find n for 95% and 99% confidence. Then find n for estimating the share of satisfied customers with $E = 0.03$ at 95% confidence if (a) p is unknown; (b) preliminarily $p \approx 0.6$.

Problem 8 [B] An analyst writes: "With probability 95% the average processing time lies in the interval $(22.8;\ 48.7)$" and "95% of branches have revenue in this interval". Find and correct the errors in both statements.

Problem 9 [B] How does the width of a CI for a mean (σ known) change if (a) n grows fourfold; (b) the confidence level changes from 95% to 99%; (c) σ decreases by 20%?

Problem 10 [B] In a sample of 80 borrowers, 6 were overdue. Check the conditions, construct the simple 95% CI for the proportion and the Wilson interval. Compare and comment.

Problem 11 [C] Derive the CI for μ with known σ from the condition $P(-z_{\alpha/2} < Z < z_{\alpha/2}) = 1 - \alpha$, where $Z = (\bar{X} - \mu)/(\sigma/\sqrt{n})$.

Problem 12 [C] For a sample of $n = 2$ observations $t_{0.025,1} = 12.706$. Explain why a 95% CI with $n = 2$ is almost useless, while for $n = 100$ ($t = 1.984$) it practically coincides with the z-interval. What role does the estimate s play here?

STAT201 · STATISTICS · LECTURE 10

Hypothesis Testing I: Logic, the p-value and Errors

How to decide formally whether the data agree with a statement about a parameter, and what the researcher risks

Syllabus topic 10 · week 10 · 4 academic hours

LECTURE GOALS

- formulate the null and alternative hypotheses and choose a one-sided or two-sided test;
- carry out a z- and t-test for a mean and a z-test for a proportion in five steps;
- compute and correctly interpret the p-value;
- explain Type I and Type II errors, the significance level and power;
- relate hypothesis testing to confidence intervals and distinguish statistical significance from practical significance.

1 The logic of hypothesis testing

Hypothesis testing is a formalized answer to the question “do the data agree with a statement about the population?”. The scheme resembles the legal presumption of innocence: the null hypothesis is not rejected until the data provide sufficiently strong evidence against it.

HYPOTHESES

The *null hypothesis* H_0 is a statement of “no effect” (an equality sign): $\mu = \mu_0$, $p = p_0$. The *alternative hypothesis* H_1 (or H_a) is what we want to prove: two-sided $\mu \neq \mu_0$; one-sided $\mu > \mu_0$ or $\mu < \mu_0$. The direction is chosen *before* looking at the data, based on the research question.

EXAMPLE 1. FORMULATIONS

A bank promises to process an application in 12 days on average. Customers complain that it takes longer. $H_0 : \mu = 12$, $H_1 : \mu > 12$ (one-sided). If the manager wants to know whether the time has changed at all: $H_1 : \mu \neq 12$.

1.1 Five steps

1. Formulate H_0, H_1 and choose the significance level α (usually 0.05).
2. Check the conditions (randomness and independence, normality or large n).
3. Compute the test statistic.
4. Find the p-value (or the critical value).

5. Make a decision and state the conclusion in the language of the problem.

THE p-VALUE

The p-value is the probability (given that H_0 is true) of obtaining a value of the statistic as extreme as, or more extreme than, the observed one. A small p means: the observed result is hard to explain by chance under H_0. Decision: if $p \le \alpha$, H_0 is rejected; otherwise there is no basis to reject it.

COMMON MISTAKE

The p-value is *not* the probability that H_0 is true, and not the probability of an erroneous decision. It is $P(\text{data} \mid H_0)$, not $P(H_0 \mid \text{data})$ — the same transposition of conditions as in Lecture 3. Moreover, "we do not reject H_0" does not mean "we have proved H_0": perhaps there are too few data.

2 Tests for a mean

2.1 The z-test (σ known)

$$z = \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}}. \tag{2.1}$$

When H_0 is true the statistic is approximately $N(0,1)$. The p-value: two-sided $2P(Z > |z|)$; right-sided $P(Z > z)$; left-sided $P(Z < z)$. Critical values at $\alpha = 0.05$: ± 1.96 (two-sided), 1.645 (one-sided).

EXAMPLE 2. APPLICATION PROCESSING TIME, $\sigma = 4$, $n = 36$

$H_0 : \mu = 12$, $H_1 : \mu \neq 12$, $\alpha = 0.05$. (a) $\bar{x} = 12.4$: $z = \dfrac{0.4}{0.667} = 0.60$, $p = 2 \cdot 0.274 = 0.549 > 0.05$ — no grounds to reject H_0. (b) $\bar{x} = 13.6$: $z = \dfrac{1.6}{0.667} = 2.40$, $p = 2 \cdot 0.0082 = 0.0164 < 0.05$ — H_0 is rejected: the average time differs from 12 days.

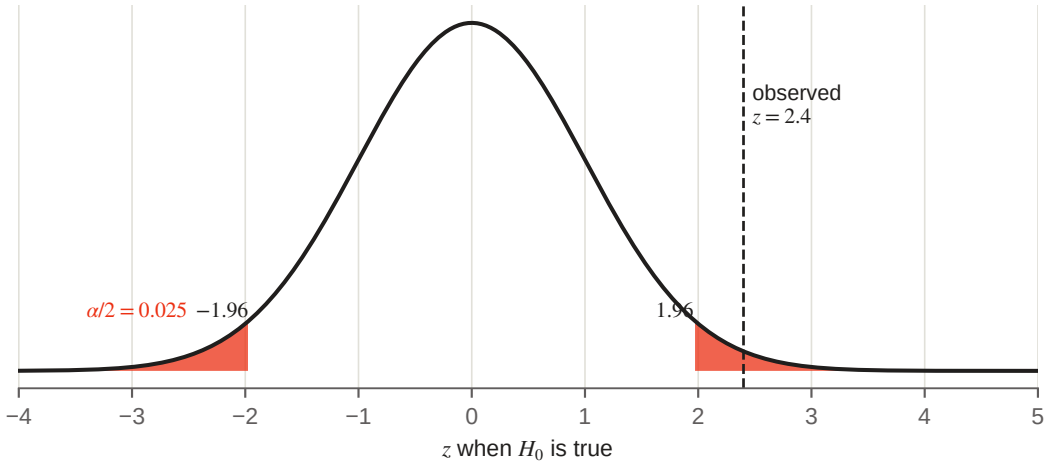


Fig. 1. A two-sided test at $\alpha = 0.05$: the rejection regions are 2.5% in each tail. The observed $z = 2.4$ falls into the right rejection region; the p-value is the area beyond $|z| = 2.4$ in both tails.

2.2 The t-test (σ unknown)

We replace σ by s and use the t-distribution with $n-1$ degrees of freedom:

$$t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}, \qquad \nu = n - 1. \tag{2.2}$$

EXAMPLE 3. BRANCH REVENUE

The chain considers a monthly revenue of 30 million UZS per branch to be the norm. For 12 branches $\bar{x} = 35.75$, $s = 20.33$. $H_0 : \mu = 30$, $H_1 : \mu \neq 30$. $t = \dfrac{5.75}{20.33/\sqrt{12}} = \dfrac{5.75}{5.869} = 0.98$, $\nu = 11$, $p = 0.348$. The critical value is ± 2.201. H_0 cannot be rejected: the difference of 5.75 million is easily explained by chance with such a spread. Note that the 95% CI $(22.83;\ 48.67)$ contains 30.

EXAMPLE 4. A ONE-SIDED TEST

A sample $n = 25$, $\bar{x} = 48.2$, $s = 5$. $H_0 : \mu = 50$, $H_1 : \mu < 50$. $t = -1.8/1 = -1.80$, $\nu = 24$; $p = P(t_{24} < -1.80) = 0.042 < 0.05$ — we reject (critical value -1.711). A two-sided test would have given $p = 0.084$ and would not have rejected H_0: the direction must be chosen in advance and justified, not fitted to the result.

3 The test for a proportion

For $H_0 : p = p_0$ the standard error is computed *from* p_0 (we assume H_0 is true):

$$z = \frac{\hat{p} - p_0}{\sqrt{p_0(1-p_0)/n}}, \qquad \text{condition: } np_0 \geq 10,\ n(1-p_0) \geq 10. \tag{3.1}$$

EXAMPLE 5. DEFECTS ON A LINE

The permissible defect rate is $p_0 = 0.10$. In a sample of 200 items 30 are defective, $\hat{p} = 0.15$. $H_1 : p > 0.10$. $z = \dfrac{0.05}{\sqrt{0.09/200}} = \dfrac{0.05}{0.0212} = 2.36$, $p = 0.0092$. We reject H_0: the defect rate exceeds the permissible level.

EXAMPLE 6. SATISFACTION

The hypothesis “exactly 50% are satisfied” against $\neq$; 240 of 400 are satisfied. $z = \dfrac{0.10}{\sqrt{0.25/400}} = 4.00$, $p \approx 6 \cdot 10^{-5}$. We reject H_0. (In a CI for a proportion the standard error is computed from $\hat{p}$, in a test from p_0.)

4 Type I and Type II errors, significance level and power

	H_0 is true	H_0 is false
Do not reject H_0	correct $(1-\alpha)$	Type II error (β)
Reject H_0	Type I error (α)	correct: **power** $1-\beta$

The *significance level* $\alpha = P(\text{reject } H_0 \mid H_0 \text{ true})$ is the risk of a false alarm; we set it ourselves. *Power* $1-\beta$ is the probability of detecting an effect when there is one. For a two-sided z-test at a true μ and $\delta = (\mu - \mu_0)/(\sigma/\sqrt{n})$ the power equals $P(Z > 1.96 - \delta) + P(Z < -1.96 - \delta)$.

EXAMPLE 7. POWER

$H_0 : \mu = 12$, $\sigma = 4$, $n = 36$, $\alpha = 0.05$. If the true $\mu = 13.5$: $\delta = 2.25$, power $= 0.614$ (Type II error $\beta = 0.386$). At $\mu = 14$ the power is 0.851; at $\mu = 12.5$ only 0.117.

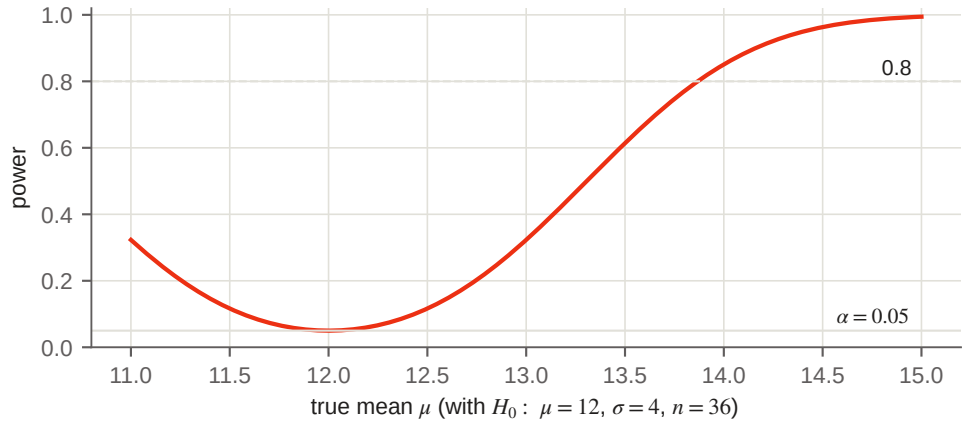


Fig. 2. The power of a two-sided z-test as a function of the true μ. At $\mu = 12$ (true H_0) it equals $\alpha = 0.05$; the larger the effect, the higher the power.

Power increases with n, with the size of the effect and with α, and with a decrease in σ. For 80% power and an effect $\Delta = \mu - \mu_0$ one needs $n \approx \left(\dfrac{(z_{\alpha/2} + z_\beta)\sigma}{\Delta} \right)^2$, where $z_\beta = 0.8416$. For $\Delta = 1.5$, $\sigma = 4$: $n = \left(\dfrac{2.80 \cdot 4}{1.5} \right)^2 = 55.8 \approx 56$.

REMEMBER

Lowering α reduces the risk of a false alarm but lowers power. To reduce both errors at once n must be increased. The choice of α depends on the cost of errors: for testing drug safety a false "approval" costs more than for a marketing hypothesis.

5 The link with confidence intervals

A two-sided test at level α rejects $H_0 : \mu = \mu_0$ if and only if μ_0 lies outside the $(1-\alpha)$ CI. In example (b): the 95% CI at $\bar{x} = 13.6$ is $(12.29;\ 14.91)$, 12 lies outside — the test rejects; for (a) the CI

$(11.09;\ 13.71)$ contains 12 — it does not. The CI is more informative: it shows not only "yes/no" but also the size of the effect.

6 Significance and practical importance

With a huge sample a negligible effect becomes "statistically significant": $\bar{x} = 30.3$ against $\mu_0 = 30$ with $\sigma = 4$ and $n = 10\,000$ gives $z = 7.5$, $p < 10^{-12}$, but a difference of 0.3 (1%) may have no practical meaning. And conversely: a large effect in a small sample may be "insignificant" (low power). Always report the effect size and CI along with p.

If many hypotheses are tested, then at $\alpha = 0.05$ about 5% of true nulls will be "significant" by chance. 20 independent tests give a chance of at least one false alarm of $1 - 0.95^{20} = 64\%$. For such cases corrections are applied (for example, Bonferroni: α/m).

7 Key terms

Term	*In Russian*	Meaning
Null hypothesis	*Нулевая гипотеза*	H_0 — "no effect"
Alternative hypothesis	*Альтернативная*	H_1 — what we want to prove
Test statistic	*Статистика критерия*	z or t
p-value	*p-значение*	$P(\text{more extreme} \mid H_0)$
Significance level	*Уровень значимости*	$\alpha = P(\text{Type I error})$
Type II error	*Ошибка II рода*	β
Power	*Мощность*	$1 - \beta$

Formulas of Lecture 10

$$z = \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}}, \qquad t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}\ (\nu = n - 1), \qquad z = \frac{\hat{p} - p_0}{\sqrt{p_0(1-p_0)/n}};$$

p-value: $2P(Z > |z|)$ or $P(Z > z)$ or $P(Z < z)$;

decision: $p \le \alpha \Rightarrow$ reject H_0;

$$\text{power} = 1 - \beta; \qquad n \approx \left(\frac{(z_{\alpha/2} + z_\beta)\sigma}{\Delta}\right)^2.$$

8 Practice problems

> Problem level is shown in square brackets: **[A]** — basic, tests understanding of definitions; **[B]** — standard computation; **[C]** — advanced, requires a derivation or an interpretation. Decimals are written with a point, as in a calculator, Excel and R. If a problem needs tables, use the appendix “Statistical Tables”.

Problem 1 [A] Formulate H_0 and H_1 (stating the type of test): (a) a manufacturer claims the average pack mass is 500 g, an inspector suspects underweight; (b) a bank wants to check whether the share of overdue payments has changed compared with the previous 6%; (c) a manager believes that a new application form has shortened the average processing time from 15 minutes.

Problem 2 [A] $H_0 : \mu = 50$, $H_1 : \mu \neq 50$, $\sigma = 10$, sample $n = 100$, $\bar{x} = 52.5$, $\alpha = 0.05$. Carry out the test.

Problem 3 [A] A study obtained $p = 0.03$. Which statements are true? (1) The probability that H_0 is true is 3%. (2) If H_0 were true, a result at least as extreme as the observed one would occur in 3% of samples. (3) The effect is large. (4) At $\alpha = 0.05$ H_0 is rejected; at $\alpha = 0.01$ it is not.

Problem 4 [B] $H_0 : \mu = 50$, $H_1 : \mu \neq 50$. A sample $n = 9$, $\bar{x} = 52$, $s = 6$. Carry out the t-test ($\alpha = 0.05$) and find the p-value.

Problem 5 [B] The average delivery time should be 250 minutes. For 16 orders $\bar{x} = 245$, $s = 20$. Test $H_0 : \mu = 250$ against $H_1 : \mu < 250$.

Problem 6 [B] A company claims that 30% of customers use the app. In a random sample of 500 customers, 170 use the app.

(a) Test $H_0 : p = 0.30$ against $H_1 : p \neq 0.30$ at $\alpha = 0.05$.
(b) How does the conclusion change for $H_1 : p > 0.30$?
(c) Construct a 95% CI and compare with the result of (a).

Problem 7 [B] Describe Type I and Type II errors in the language of the problem: (a) a bank’s monitoring system tests H_0: “the transaction is normal”; (b) testing a new drug, H_0: “the drug has no effect”. Which error is more costly in each case and how does this affect the choice of α?

Problem 8 [B] $H_0 : \mu = 100$, $\sigma = 15$, $n = 36$, $\alpha = 0.05$ (two-sided).

(a) Find the power of the test at a true $\mu = 105$.
(b) What n is needed for 80% power at $\mu = 105$?

Problem 9 [B] $n = 64$, $\bar{x} = 103$, $\sigma = 16$. (a) Construct a 95% CI for μ. (b) Without conducting a test, determine the result of a two-sided test at the 0.05 level for $H_0 : \mu = 100$ and for $H_0 : \mu = 98$. (c) Verify this by computing z and p.

Problem 10 [B] An analyst tested 10 independent hypotheses at the level $\alpha = 0.05$ and found one “significant”. (a) What is the probability of at least one false alarm if all 10 null hypotheses are true? (b) What significance level for each test does the Bonferroni correction give?

Problem 11 [C] Show that when H_0 is true and the statistic is continuous, the p-value (for a right-sided test) is uniformly distributed on $[0,1]$, i.e. $P(p \le \alpha) = \alpha$ for any α.

Problem 12 [C] Show that a two-sided z-test at level α rejects $H_0 : \mu = \mu_0$ if and only if μ_0 does not belong to the $(1-\alpha)$ CI $\bar{x} \pm z_{\alpha/2}\sigma/\sqrt{n}$. Then, for $n = 10\,000$, $\bar{x} = 30.3$, $\sigma = 4$, $\mu_0 = 30$, find the p-value and the CI and comment on the practical significance.

STAT201 · STATISTICS · LECTURE 11

Hypothesis Testing II: Comparing Groups, χ^2 and ANOVA

Comparing two means and proportions, paired data, contingency tables and comparing three or more groups

Syllabus topic 11 · week 11 · 4 academic hours

LECTURE GOALS

- choose between independent and paired samples and carry out the corresponding t-test;
- compare two proportions with a z-test and construct a CI for the difference;
- test the independence of two categorical attributes and goodness of fit to a distribution using the χ^2 test;
- carry out a one-way analysis of variance (ANOVA) and interpret the F-statistic;
- judge the practical significance of differences and avoid typical mistakes (multiple comparisons, the wrong choice of test).

1 How to choose a test

The logic of hypothesis testing (Lecture 10) remains the same; the statistic and its distribution change. The choice depends on the type of data and the plan of collection:

Question	Test
Two means, independent groups	two-sample t-test (Welch)
Two measurements on the same objects (before/after)	paired t-test on the differences
Two proportions	two-sample z-test for proportions
Relationship between two categorical attributes	χ^2 test of independence
Does a distribution match a given one	χ^2 goodness-of-fit test
Three or more means	one-way ANOVA (F-test)

2 Two means: independent samples

Let $\bar{x}_1, s_1, n_1$ and $\bar{x}_2, s_2, n_2$ be obtained from two independent populations. $H_0 : \mu_1 - \mu_2 = 0$. The standard error of the difference (variances of independent variables add):

$$\mathrm{SE}(\bar{x}_1 - \bar{x}_2) = \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}, \qquad t = \frac{\bar{x}_1 - \bar{x}_2}{\mathrm{SE}}, \tag{2.1}$$

and the degrees of freedom (the Welch–Satterthwaite formula)

$$\nu \approx \frac{\left(s_1^2/n_1 + s_2^2/n_2\right)^2}{\dfrac{(s_1^2/n_1)^2}{n_1 - 1} + \dfrac{(s_2^2/n_2)^2}{n_2 - 1}}. \tag{2.2}$$

This is Welch's test: it does not require equal variances and is the default setting in modern software. (If the variances are assumed equal, one can use the pooled estimate $s_p^2 = \frac{(n_1-1)s_1^2 + (n_2-1)s_2^2}{n_1+n_2-2}$ and $\nu = n_1 + n_2 - 2$.) The CI for the difference: $(\bar{x}_1 - \bar{x}_2) \pm t_{\alpha/2,\nu}\,\mathrm{SE}$.

EXAMPLE 1. TRANSACTION TIME AT TWO BRANCHES

Branch 1: $n_1 = 12$, $\bar{x}_1 = 14.2$ min, $s_1 = 3.1$. Branch 2: $n_2 = 15$, $\bar{x}_2 = 12.1$, $s_2 = 2.6$. $H_0 : \mu_1 = \mu_2$, $H_1 : \mu_1 \neq \mu_2$, $\alpha = 0.05$. $\mathrm{SE} = \sqrt{3.1^2/12 + 2.6^2/15} = \sqrt{0.801 + 0.451} = 1.119$; $t = \dfrac{2.1}{1.119} = 1.88$; $\nu = 21.5$. $p = 0.074 > 0.05$ — at the 5% level the difference is not proven. The 95% CI of the difference: $2.1 \pm 2.077 \cdot 1.119 = (-0.22;\ 4.42)$ — contains 0, consistent with the test. With a pooled variance we would get almost the same ($t = 1.92$, $p = 0.067$). A one-sided test ($H_1 : \mu_1 > \mu_2$) would give $p = 0.037$, but, as said, the direction is chosen before the data.

3 Paired data

If the measurements relate to the same objects (before and after training; two methods on the same customers), the samples are dependent. Then we compute the *differences* $d_i = x_i^{\text{after}} - x_i^{\text{before}}$ and apply a one-sample t-test to them:

$$t = \frac{\bar{d}}{s_d/\sqrt{n}}, \qquad \nu = n - 1. \tag{3.1}$$

EXAMPLE 2. STAFF TRAINING

Scores of 8 employees before training: $62, 70, 55, 68, 74, 60, 66, 71$; after: $66, 73, 58, 69, 79, 62, 70, 72$. Differences d: $4, 3, 3, 1, 5, 2, 4, 1$; $\bar{d} = 2.875$, $s_d = 1.458$, $\mathrm{SE} = 0.515$. $t = 5.58$, $\nu = 7$, $p = 0.0008$. We reject $H_0 : \mu_d = 0$: training raised the score by 2.9 on average (95% CI: $(1.66;\ 4.09)$).

COMMON MISTAKE

Paired data must not be analysed as independent. For the same data Welch's test gives $t = 0.89$, $p = 0.39$ — the effect "gets lost" because of the large spread between employees (which the paired design eliminates). The converse also holds: independent samples must not be "paired" arbitrarily.

4 Two proportions

$H_0 : p_1 = p_2$. Under H_0 both proportions estimate the same p, so the test uses the pooled proportion $\hat{p} = \frac{x_1+x_2}{n_1+n_2}$:

$$z = \frac{\hat{p}_1 - \hat{p}_2}{\sqrt{\hat{p}(1-\hat{p})\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}}. \tag{4.1}$$

For a CI of the difference the separate error $\sqrt{\frac{\hat{p}_1(1-\hat{p}_1)}{n_1} + \frac{\hat{p}_2(1-\hat{p}_2)}{n_2}}$ is used.

EXAMPLE 3. TWO MAILING VARIANTS

Offer A: 120 responses out of 400 ($\hat{p}_1 = 0.300$); offer B: 150 out of 420 ($\hat{p}_2 = 0.357$). $\hat{p} = 270/820 = 0.329$, SE $= \sqrt{0.329 \cdot 0.671 \cdot (1/400 + 1/420)} = 0.0328$. $z = \dfrac{-0.0571}{0.0328} = -1.74$, $p = 0.082$ (two-sided). The difference is not proven at the 5% level. 95% CI of the difference $p_1 - p_2$: $-0.057 \pm 1.96 \cdot 0.0327 = (-0.121;\ 0.007)$.

5 The χ^2 test of independence

For two categorical attributes we build an $r \times c$ contingency table. Under H_0 (the attributes are independent) the expected count of a cell is $E_{ij} = \dfrac{(\text{row } i \text{ total})(\text{column } j \text{ total})}{N}$.

$$\chi^2 = \sum \frac{(O-E)^2}{E}, \qquad \nu = (r-1)(c-1). \tag{5.1}$$

A large value means that the observed counts are far from the expected ones. The test is always right-sided. Condition: all $E_{ij} \geq 5$.

EXAMPLE 4. SERVICE CHANNEL AND SATISFACTION

500 customers:

	Satisfied	Not satisfied	Total
Website	90	60	150
App	150	50	200
Branch	85	65	150
Total	325	175	500

Expected: website $97.5/52.5$; app $130/70$; branch $97.5/52.5$ (for example, $E_{11} = 150 \cdot 325/500 = 97.5$). The terms $\frac{(O-E)^2}{E}$: $0.58,\ 1.07;\ 3.08,\ 5.71;\ 1.60,\ 2.98$. $\chi^2 = 15.02$, $\nu = 2$, critical value 5.991, $p = 0.0005$. We reject independence: the share of satisfied customers differs by channel (60%, 75%, 56.7%). The cell contributions show that the main difference is the app customers. The strength of association by Cramér's $V = \sqrt{\chi^2/(N \min(r-1, c-1))} = 0.17$ — weak.

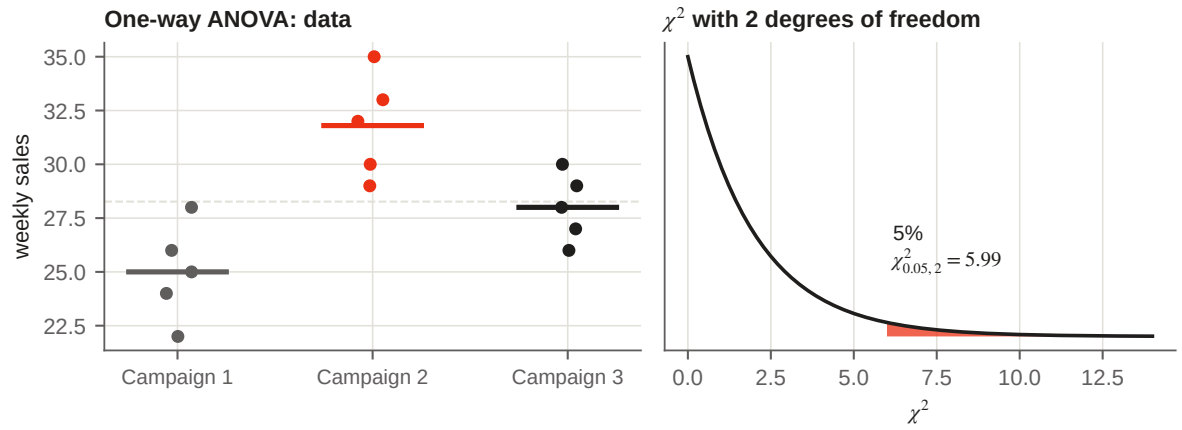


Fig. 1. Left: data of three advertising campaigns (dots) and their means; the dashed line is the grand mean. Right: the χ^2 distribution with 2 degrees of freedom and the rejection region at $\alpha = 0.05$.

5.1 The χ^2 goodness-of-fit test

It tests whether the distribution of a categorical variable matches a given one: $\chi^2 = \sum \frac{(O_i - E_i)^2}{E_i}$, $\nu = k - 1$ (minus the number of estimated parameters).

EXAMPLE 5. WORKLOAD BY DAY

A manager assumes that requests are uniformly distributed over the five working days. In a week there are 120 requests: $15, 20, 24, 22, 39$. Expected $120/5 = 24$ per day. The terms: $3.375;\ 0.667;\ 0;\ 0.167;\ 9.375$; $\chi^2 = 13.58$, $\nu = 4$, critical value 9.488, $p = 0.0088$. The hypothesis of uniformity is rejected — the main contribution comes from Friday.

6 Comparing several means: ANOVA

For three or more groups, multiple t-tests lead to an inflated Type I error (Lecture 10). Analysis of variance tests $H_0 : \mu_1 = \mu_2 = \cdots = \mu_k$ with one test. The idea: compare the variability *between* group means with the variability *within* groups.

$$SSB = \sum n_j(\bar{x}_j - \bar{x})^2, \quad SSW = \sum\sum (x_{ij} - \bar{x}_j)^2, \quad SST = SSB + SSW, \tag{6.1}$$

$$F = \frac{MSB}{MSW} = \frac{SSB/(k-1)}{SSW/(N-k)}, \qquad \nu_1 = k - 1,\ \nu_2 = N - k. \tag{6.2}$$

If H_0 is true, both variance estimates are close and $F \approx 1$; a large F means that the group means differ more than chance allows. Conditions: independence, approximate normality within groups, similar variances (the ratio of the largest s to the smallest no more than 2).

EXAMPLE 6. THREE ADVERTISING CAMPAIGNS

Sales over five weeks: C1: $22, 25, 24, 28, 26$ ($\bar{x} = 25.0$); C2: $30, 33, 29, 35, 32$ (31.8); C3: $27, 26, 30, 29, 28$ (28.0). Grand mean 28.27.

Source	SS	ν	MS	F
Between groups	116.13	2	58.07	**13.20**
Within groups	52.80	12	4.40	
Total	168.93	14		

Critical value $F_{0.05;2,12} = 3.885$; $p = 0.0009$. We reject H_0: mean sales differ. The share of explained variability is $\eta^2 = SSB/SST = 0.69$.

ANOVA says only that at least one pair of means differs. To find out which one, pairwise comparisons with a multiplicity correction (Bonferroni, Tukey) are used. In the example with $MSW = 4.4$ ($\nu = 12$), pairwise t-tests with the Bonferroni correction (multiplying p by 3): C1 vs C2: $p = 0.0008$; C2 vs C3: $p = 0.043$; C1 vs C3: $p = 0.129$ — the difference between C1 and C3 is not proven.

7 Practical remarks

1. Significance $\neq$ importance: report the difference of means and the CI.
2. Check the conditions: outliers and strong skewness with small n make t-tests unreliable; for χ^2 — expected counts ≥ 5.
3. An experiment with random assignment to groups permits causal conclusions; an observational comparison does not (Lecture 13).
4. A null result ($p > \alpha$) is "no evidence of a difference", not "evidence of no difference"; look at the width of the CI and the power.

8 Key terms

Term	*In Russian*	Meaning
Independent samples	*Независимые выборки*	different objects in the groups
Paired samples	*Парные выборки*	the same objects, two measurements
Pooled proportion	*Объединённая доля*	$\hat{p} = (x_1 + x_2)/(n_1 + n_2)$
Contingency table	*Таблица сопряжённости*	counts for two attributes
Expected count	*Ожидаемая частота*	$E = \text{row} \times \text{column}/N$
Analysis of variance (ANOVA)	*Дисперсионный анализ*	comparing $k \geq 3$ means
F-statistic	*F-статистика*	MSB/MSW

Formulas of Lecture 11

$$t = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{s_1^2/n_1 + s_2^2/n_2}}, \qquad t = \frac{\bar{d}}{s_d/\sqrt{n}}, \qquad z = \frac{\hat{p}_1 - \hat{p}_2}{\sqrt{\hat{p}(1-\hat{p})(1/n_1 + 1/n_2)}};$$

$$\chi^2 = \sum \frac{(O-E)^2}{E}, \quad \nu = (r-1)(c-1); \qquad F = \frac{SSB/(k-1)}{SSW/(N-k)}.$$

9 Practice problems

> Problem level is shown in square brackets: **[A]** — basic, tests understanding of definitions; **[B]** — standard computation; **[C]** — advanced, requires a derivation or an interpretation. Decimals are written with a point, as in a calculator, Excel and R. If a problem needs tables, use the appendix “Statistical Tables”.

Problem 1 [A] Choose the test (two-sample t, paired t, z for two proportions, χ^2 independence, χ^2 goodness of fit, ANOVA): (a) the average bill in Tashkent, Samarkand and Bukhara; (b) the weight of 20 customers before and after a programme; (c) the share of satisfied customers among men and among women; (d) the relationship between education level (3 categories) and the type of tariff chosen (4 types); (e) the average response time of two groups of operators; (f) whether the distribution of orders over the days of the week is uniform.

Problem 2 [A] Group 1: $n = 40$, $\bar{x} = 50$, $s = 8$; group 2: $n = 50$, $\bar{x} = 46$, $s = 10$. Test $H_0 : \mu_1 = \mu_2$ (H_1 two-sided, $\alpha = 0.05$) and construct a 95% CI for the difference.

Problem 3 [A] The balance of 6 customers (thousand UZS) before and after an advertising campaign: before $40, 52, 38, 45, 60, 48$; after $44, 55, 37, 50, 66, 49$. Test whether the balance increased (two-sided test, $\alpha = 0.05$).

Problem 4 [B] Branch A: $n = 15$, $\bar{x} = 78$, $s = 12$; branch B: $n = 18$, $\bar{x} = 70$, $s = 9$ (customer-service scores). Carry out Welch’s test for $H_0 : \mu_A = \mu_B$ and construct a 95% CI for the difference. Compare with the conclusion from the CI.

Problem 5 [B] Defect rate on line A: 18 out of 300; on line B: 33 out of 400. Test $H_0 : p_A = p_B$ and construct a 95% CI for the difference.

Problem 6 [B] Of 200 borrowers who received a loan under scheme A, 45 became overdue; of 200 under scheme B — 30. Carry out a χ^2 test of independence (scheme and overdue), $\alpha = 0.05$, and compare with the z-test for two proportions.

Problem 7 [B] A survey of 350 customers: region (North, Centre, South) and preferred product (loan, deposit, card):

	Loan	Deposit	Card
North	40	30	30
Centre	50	60	40
South	30	20	50

Test the independence of region and product ($\alpha = 0.05$). In which cell is the largest deviation?

Problem 8 [B] A die was rolled 60 times; faces $1, \dots, 6$ came up $8, 12, 9, 11, 6, 14$ times. Test the fairness of the die ($\alpha = 0.05$).

Problem 9 [B] One-way ANOVA: 3 groups of $n = 6$ ($N = 18$); means $20, 24, 28$; $MSW = 16$.

(a) Find SSB, MSB and F.
(b) Find the critical value $F_{0.05}$ and draw a conclusion.

Problem 10 [B] The time to perform an operation in three crews (4 observations each): A: $10, 12, 11, 13$; B: $14, 15, 13, 16$; C: $12, 11, 13, 12$. Build the ANOVA table and draw a conclusion ($\alpha = 0.05$).

Problem 11 [C] For two groups ANOVA is equivalent to a two-sample t-test with a pooled variance: $F = t^2$. Verify on the data of crews A and B from the previous problem.

Problem 12 [C] In the staff-training data (Lecture 11) $s_{\text{before}} = 6.34$, $s_{\text{after}} = 6.59$, and the correlation between scores before and after is $r = 0.975$. Explain, via the variance of the difference, why the paired test is much more powerful than the independent one. Find the SE for both approaches.

STAT201 · STATISTICS · LECTURE 12

Simple Linear Regression

How to describe the relationship between two quantitative variables by a straight line, estimate it from data and assess its quality

Syllabus topic 12 · week 12 · 4 academic hours

LECTURE GOALS

- draw a scatter plot and write the simple linear regression model $y = \beta_0 + \beta_1 x + \varepsilon$;
- compute the OLS estimates b_0, b_1 by hand and through the correlation, interpret the slope and intercept;
- compute residuals, SSE, SSR, SST, R^2 and the residual standard error s_e;
- make predictions and understand the danger of extrapolation;
- analyse a residual plot and find violations of the model's conditions.

1 The model

Correlation (Lecture 6) measures the strength of a linear relationship but does not say *by how much* y changes on average when x grows by one unit. Regression does that. Suppose for n objects an *explanatory* variable x (predictor) and a *dependent* variable y (response) are measured.

THE SIMPLE LINEAR REGRESSION MODEL

$$y_i = \beta_0 + \beta_1 x_i + \varepsilon_i, \qquad \varepsilon_i \sim \text{independent, } \mathrm{E}[\varepsilon_i] = 0,\ \mathrm{Var}(\varepsilon_i) = \sigma^2. \tag{1.1}$$

β_1 is the change in the mean of y when x grows by one unit; β_0 is the mean of y at $x = 0$; ε is the random error (everything not explained by x). The parameters β_0, β_1 are unknown and are estimated from a sample.

We are modelling the *conditional mean* $\mathrm{E}[y \mid x] = \beta_0 + \beta_1 x$. This does not mean that x is the cause of y (Lecture 13).

2 The method of least squares

The line $\hat{y} = b_0 + b_1 x$ is chosen so that the sum of squared vertical deviations (the residuals $e_i = y_i - \hat{y}_i$) is minimal:

$$\min_{b_0, b_1} \sum_{i=1}^{n} (y_i - b_0 - b_1 x_i)^2. \tag{2.1}$$

The solution (setting the derivatives to zero):

$$b_1 = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sum (x_i - \bar{x})^2} = \frac{S_{xy}}{S_{xx}} = r\,\frac{s_y}{s_x}, \qquad b_0 = \bar{y} - b_1 \bar{x}. \tag{2.2}$$

From the second equality it follows that the line always passes through the point $(\bar{x}, \bar{y})$. From the formula $b_1 = r\, s_y / s_x$ it follows that the sign of the slope coincides with the sign of the correlation.

EXAMPLE 1. ADVERTISING AND SALES

Data from Lecture 6: advertising spend $x = 2, 3, 5, 6, 8, 10$ and sales $y = 15, 17, 22, 31, 33, 40$ (million UZS). $\bar{x} = 5.667$, $\bar{y} = 26.333$, $S_{xx} = 45.33$, $S_{yy} = 487.33$, $S_{xy} = 145.67$.

$$b_1 = \frac{145.67}{45.33} = 3.213, \qquad b_0 = 26.333 - 3.213 \cdot 5.667 = 8.125, \qquad \hat{y} = 8.125 + 3.213\,x.$$

Interpretation: each additional million of advertising is associated with an increase in sales of 3.21 million UZS on average. The intercept 8.1 is sales at zero spend (here $x = 0$ lies outside the data range 2–10, so it has no literal interpretation).

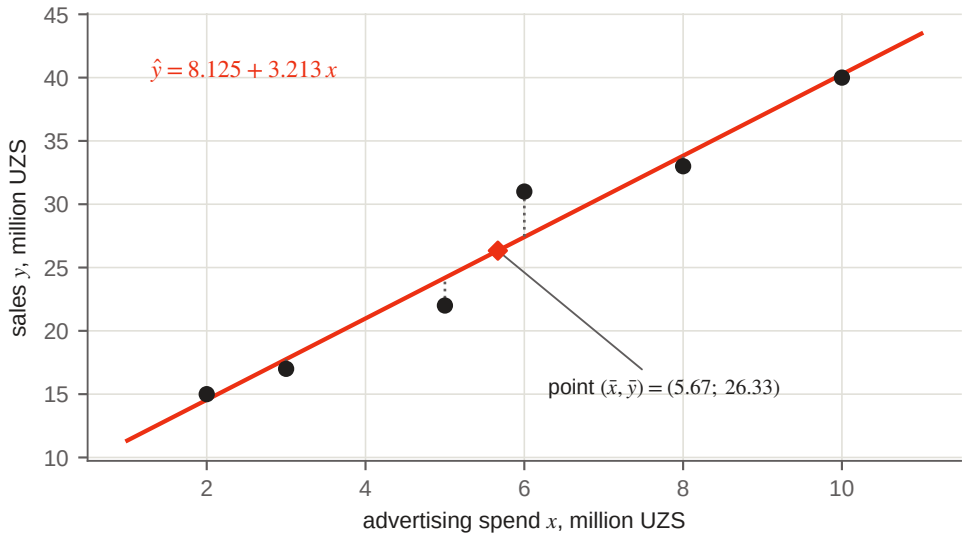


Fig. 1. The scatter plot and the OLS line. The dashed segments are the residuals; the line minimizes the sum of their squares. The diamond is the point $(\bar{x}, \bar{y})$ through which the line passes.

3 Residuals and goodness of fit

The predictions $\hat{y}_i = b_0 + b_1 x_i$: $14.55;\ 17.77;\ 24.19;\ 27.40;\ 33.83;\ 40.26$. The residuals $e_i = y_i - \hat{y}_i$: $0.45;\ -0.77;\ -2.19;\ 3.60;\ -0.83;\ -0.26$. Properties of OLS residuals: $\sum e_i = 0$ and $\sum x_i e_i = 0$ (check this!).

The total variability of y is decomposed into explained and unexplained:

$$\underbrace{\sum (y_i - \bar{y})^2}_{SST} = \underbrace{\sum (\hat{y}_i - \bar{y})^2}_{SSR} + \underbrace{\sum (y_i - \hat{y}_i)^2}_{SSE}, \qquad R^2 = \frac{SSR}{SST} = 1 - \frac{SSE}{SST}. \tag{3.1}$$

COEFFICIENT OF DETERMINATION

R^2 is the share of the variation in y explained by the linear dependence on x. In simple regression $R^2 = r^2$.

In the example $SST = 487.33$, $SSE = 19.27$, $SSR = 468.06$; $R^2 = 0.9605$ ($= 0.980^2$): advertising explains 96% of the variation in sales. The residual standard error

$$s_e = \sqrt{\frac{SSE}{n-2}} = \sqrt{\frac{19.27}{4}} = 2.195 \tag{3.2}$$

shows the typical size of a prediction error in the units of y (we divide by $n-2$ because two parameters were estimated).

COMMON MISTAKE

A high R^2 proves neither causation nor the correctness of the model; a low R^2 does not mean there is no relationship (there may be a nonlinear relationship or a significant but small effect). Always look at the residual plot.

4 Prediction and extrapolation

The prediction of the mean at $x = 7$: $\hat{y} = 8.125 + 3.213 \cdot 7 = 30.62$ million UZS. This is acceptable, since 7 lies inside the range $[2; 10]$ (*interpolation*). The prediction at $x = 12$ gives 46.7, but this is *extrapolation*: beyond the data linearity is not guaranteed (market saturation, diminishing returns), and such a prediction cannot be trusted.

REMEMBER

The regression of y on x and the regression of x on y are different lines (except when $|r| = 1$). If you need to predict y, the explanatory variable is x; there is no symmetry in OLS. Units: b_1 is measured in units of y per unit of x; when the units of x are changed (million $\to$ thousand) b_1 changes, r and R^2 do not.

5 Diagnostics from the residuals

Conditions of the model: (1) linearity; (2) independence of errors; (3) constant variance of errors (homoscedasticity); (4) approximately normal errors (important for small n for inference, Lecture 13). The residual plot of e_i against $\hat{y}_i$ is examined: with a good fit the points are randomly scattered around zero with no structure.

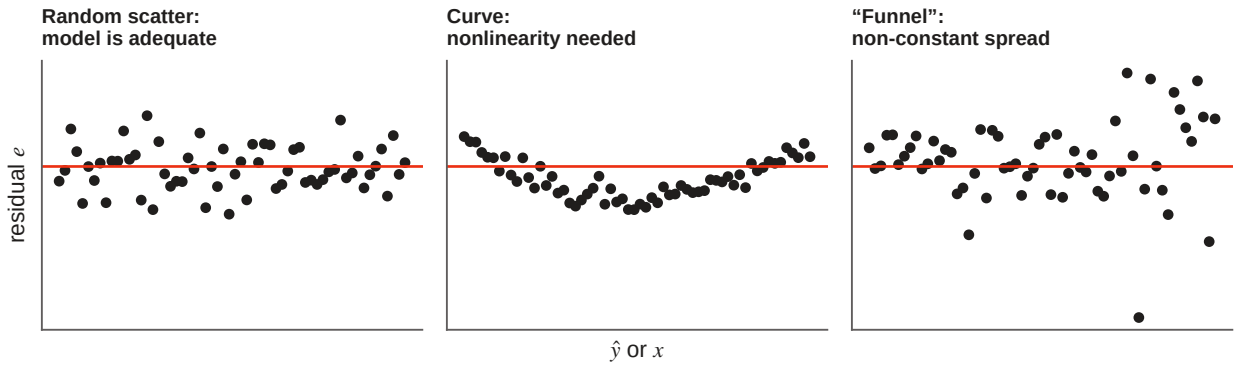


Fig. 2. Three typical residual patterns. A curve means that the relationship is nonlinear, a “funnel” that the spread grows with x (heteroscedasticity).

5.1 Outliers and influential observations

A point with a large residual is an outlier in y. A point with an unusual value of x (far from $\bar{x}$) has high leverage and may “pull” the line. Always find out the reason (a typing error, a special case) and compare results with and without the point (as in Lecture 1 for branch 12), rather than deleting data for the sake of a nice line.

6 Key terms

Term	*In Russian*	Meaning
Response (dependent) variable, y	*Зависимая переменная*	the variable being explained
Predictor (explanatory) variable, x	*Объясняющая переменная*	the factor
Ordinary least squares (OLS)	*МНК*	minimum of $\sum e_i^2$
Residual	*Остаток*	$e_i = y_i - \hat{y}_i$
Coefficient of determination, R^2	*Коэффициент детерминации*	$SSR/SST = r^2$
Residual standard error	*Стандартная ошибка регрессии*	$s_e = \sqrt{SSE/(n-2)}$
Extrapolation	*Экстраполяция*	a prediction outside the range of x
Homoscedasticity	*Гомоскедастичность*	constant variance of errors

Formulas of Lecture 12

$$b_1 = \frac{S_{xy}}{S_{xx}} = r\frac{s_y}{s_x}, \qquad b_0 = \bar{y} - b_1\bar{x}, \qquad e_i = y_i - \hat{y}_i;$$

$$SST = SSR + SSE, \qquad R^2 = \frac{SSR}{SST} = r^2, \qquad s_e = \sqrt{\frac{SSE}{n-2}}.$$

7 Practice problems

> Problem level is shown in square brackets: **[A]** — basic, tests understanding of definitions; **[B]** — standard computation; **[C]** — advanced, requires a derivation or an interpretation. Decimals are written with a point, as in a calculator, Excel and R. If a problem needs tables, use the appendix “Statistical Tables”.

Problem 1 [A] The regression equation of delivery cost y (thousand UZS) on distance x (km): $\hat{y} = 10 + 2.5x$.

(a) Interpret b_1 and b_0.
(b) Find the prediction at $x = 8$ km and the residual for the observation $(8;\ 32)$.

Problem 2 [A] Given: $\bar{x} = 40$, $\bar{y} = 100$, $s_x = 10$, $s_y = 30$, $r = 0.6$. Find the regression equation of y on x and R^2.

Problem 3 [A] Study time x (h): $1, 2, 3, 4, 5$; scores y: $50, 55, 65, 70, 85$. Find the regression equation, the prediction at $x = 4$ and the residual for this observation.

Problem 4 [B] For the data of the previous problem find all predictions and residuals, check $\sum e_i = 0$, find SSE, SST, R^2 and s_e.

Problem 5 [B] Apartment area x (m^2): $40, 50, 60, 80, 100$; price y (thousand USD): $38, 45, 50, 68, 84$.

(a) Find the regression equation.
(b) Find R^2 and s_e.
(c) Predict the price of a 70 m^2 apartment. Can the model be used for a 200 m^2 apartment?

Problem 6 [B] For a regression on 22 observations $SST = 500$, $SSE = 120$.

(a) Find SSR and R^2.
(b) Find $|r|$ and s_e.
(c) What can be said about the sign of r?

Problem 7 [B] In the regression of Lecture 12 ($\hat{y} = 8.125 + 3.213x$, x and y in million UZS) express x in thousands of UZS ($x' = 1000x$). How do b_0, b_1, r, R^2 and s_e change?

Problem 8 [B] Using the regression of Lecture 12 an analyst predicts sales for advertising of 25 million UZS and gets $\hat{y} = 88.5$. What is wrong with this prediction? What should he do?

Problem 9 [B] For the data of Lecture 12 the regression of y on x gives $b_1 = 3.213$. (a) Find the slope of the regression of x on y. (b) Check that the product of the two slopes equals r^2. Can x be found from a predicted y simply by solving the equation $y = b_0 + b_1 x$ for x?

Problem 10 [C] Derive the OLS estimates: find b_0 and b_1 minimizing $Q = \sum (y_i - b_0 - b_1 x_i)^2$.

Problem 11 [C] Show that (a) $\sum e_i = 0$; (b) $R^2 = r^2$ in simple regression.

STAT201 · STATISTICS · LECTURE 13

Regression Inference, Causation and Multiple Regression

Is the relationship found significant, how accurate are the estimates and predictions, and when regression does not answer the question “why”

Syllabus topic 13 · week 13 · 4 academic hours

LECTURE GOALS

- test the hypothesis $H_0 : \beta_1 = 0$ (t- and F-test) and construct a CI for the slope;
- construct a confidence interval for the mean response and a prediction interval for a new observation;
- interpret regression software output;
- distinguish correlation from causation; explain the role of omitted variables, reverse causation and Simpson’s paradox;
- understand the meaning of “all else equal” in multiple regression, R^2_{adj} and multicollinearity.

1 From the sample to the population

The coefficients b_0, b_1 (Lecture 12) are computed from a sample and fluctuate from sample to sample. As for a mean, we need to know their standard errors. When the conditions of the model hold (linearity, independence, constant variance, approximately normal errors):

$$\mathrm{SE}(b_1) = \frac{s_e}{\sqrt{S_{xx}}}, \qquad \mathrm{SE}(b_0) = s_e\sqrt{\frac{1}{n} + \frac{\bar{x}^2}{S_{xx}}}, \qquad \frac{b_1 - \beta_1}{\mathrm{SE}(b_1)} \sim t_{n-2}. \tag{1.1}$$

The standard error of the slope decreases when n grows, when s_e decreases and when the values of x are more spread out (S_{xx} larger).

1.1 Testing the significance of the slope

$H_0 : \beta_1 = 0$ (no linear relationship) against $H_1 : \beta_1 \neq 0$:

$$t = \frac{b_1}{\mathrm{SE}(b_1)}, \qquad \nu = n - 2; \qquad \text{CI: } b_1 \pm t_{\alpha/2,\,n-2}\,\mathrm{SE}(b_1). \tag{1.2}$$

EXAMPLE 1. ADVERTISING AND SALES

From Lecture 12: $b_1 = 3.213$, $s_e = 2.195$, $S_{xx} = 45.33$. $\mathrm{SE}(b_1) = 2.195/\sqrt{45.33} = 0.326$. $t = 3.213/0.326 = 9.86$, $\nu = 4$, $p = 0.0006$. We reject H_0: the relationship is significant. 95% CI: $3.213 \pm 2.776 \cdot 0.326 = (2.31;\ 4.12)$ — a 1 million increase in advertising is associated with a sales increase of 2.3–4.1 million. For the intercept: $\mathrm{SE}(b_0) = 2.053$, $t = 3.96$, $p = 0.017$, CI $(2.42;\ 13.83)$.
Testing $H_0 : \beta_1 = 2$ (what if the return on advertising is 2): $t = (3.213 - 2)/0.326 = 3.72$, $p = 0.020$ — we reject: the return is significantly greater than 2.

1.2 The regression ANOVA table and the F-test

Source	SS	ν	MS	F
Regression	$SSR = 468.06$	1	468.06	**97.15**
Residual	$SSE = 19.27$	$n - 2 = 4$	$4.82 = s_e^2$	
Total	$SST = 487.33$	5		

$F = MSR/MSE$ tests the same hypothesis $\beta_1 = 0$; in simple regression $F = t^2$ ($9.86^2 = 97.15$) and the p-values coincide. The F-test becomes a test in its own right in multiple regression (significance of the model as a whole).

2 Intervals for prediction

At $x = x_0$ two questions are distinguished. (1) What is the *average* value of y for all objects with x_0? — the confidence interval for the mean response. (2) What will y be for a *new* object with x_0? — the prediction interval (it is wider because the random error ε is added):

$$\hat{y}_0 \pm t_{\alpha/2,n-2}\, s_e \sqrt{\frac{1}{n} + \frac{(x_0 - \bar{x})^2}{S_{xx}}} \quad \text{(mean)}, \tag{2.1}$$

$$\hat{y}_0 \pm t_{\alpha/2,n-2}\, s_e \sqrt{1 + \frac{1}{n} + \frac{(x_0 - \bar{x})^2}{S_{xx}}} \quad \text{(individual)}. \tag{2.2}$$

EXAMPLE 2. PREDICTION AT $x_0 = 7$

$\hat{y}_0 = 30.62$. $\sqrt{1/6 + (7 - 5.667)^2/45.33} = 0.4538$, $s_e \cdot 0.4538 = 0.996$. The interval for the mean: $30.62 \pm 2.776 \cdot 0.996 = (27.85;\ 33.38)$. The prediction interval: $\sqrt{1 + 0.2059} = 1.098$, $2.195 \cdot 1.098 = 2.410$; $30.62 \pm 2.776 \cdot 2.410 = (23.93;\ 37.31)$. The interval for the mean is narrowest at $\bar{x}$ and grows toward the edges: at $x_0 = 10$ the half-width for the mean is ± 4.64 against ± 2.48 at $x_0 = \bar{x} = 5.67$.

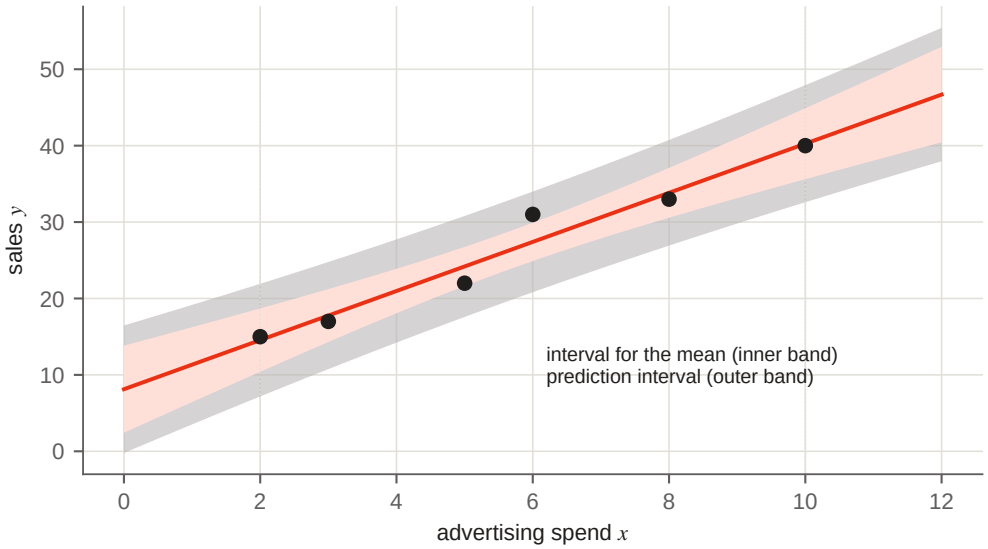


Fig. 1. The regression line, the 95% CI for the mean response (inner band, narrowing near $\bar{x}$) and the 95% prediction interval for a new observation (outer band).

COMMON MISTAKE

A prediction interval is not reducible to a CI for the mean: even with an infinite sample, individual values remain scattered around the line by $\pm 1.96\,\sigma$. Do not confuse "the average for a group" with "the value for a particular customer".

3 Reading software output

Typical output (Excel, R, Python) contains: coefficients, their standard errors, t-statistics and p-values, R^2, s_e, F and the p-value of F. For our example: Intercept 8.125 (SE 2.053, $t = 3.96$, $p = 0.017$); Slope 3.213 (SE 0.326, $t = 9.86$, $p = 0.0006$); $R^2 = 0.960$; $s_e = 2.195$; $F = 97.15$ (Significance $F = 0.0006$), $n = 6$. The ability to "read" such output in a minute is a standard exam requirement.

4 Correlation and causation

The significance of the slope says only that the linear relationship in the data is unlikely to be a coincidence. It does not mean that a change in x causes a change in y. Reasons for false conclusions:

- **An omitted (lurking) variable** Z affects both quantities. Example: the more firefighters at a fire, the greater the damage — but both quantities are determined by the scale of the fire. Ice-cream sales and the number of drownings rise together because of hot weather.
- **Reverse causation**: banks with higher profits spend more on advertising — and not only does advertising raise profit.
- **Coincidence** (especially when searching through many variables).
- **Simpson's paradox**: a relationship in the pooled data reverses sign within groups.

EXAMPLE 3. SIMPSON'S PARADOX

A call centre tests two conversation scripts with customers (success = the customer took out a product).

	Simple customers	Difficult customers	Total
Script A	81/87 (93.1%)	192/263 (73.0%)	273/350 (78.0%)
Script B	234/270 (86.7%)	55/80 (68.8%)	289/350 (82.6%)

In each group script A is better, but in total B looks better: script A was used more often with difficult customers (263 of 350), and B with simple ones (270 of 350). The customer type is a confounding variable: it is related both to the choice of script and to the result. The correct conclusion is drawn taking the groups into account.

REMEMBER

Causal conclusions are reliable when the values of x are assigned *randomly* (a randomized experiment, an A/B test): then, by construction, x is unrelated to omitted factors. With observational data one needs (a) control of confounding variables, (b) a justification of the direction of causation and (c) cautious wording: "is associated with", not "leads to".

5 Multiple regression (an introduction)

A model with several explanatory variables: $y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \cdots + \beta_k x_k + \varepsilon$. The coefficient β_j shows the change in the mean of y when x_j grows by one unit *with the other variables held constant* (ceteris paribus).

EXAMPLE 4. ADVERTISING, PRICE AND SALES

10 weeks: x_1 — advertising spend, x_2 — price, y — sales.

Week	1	2	3	4	5	6	7	8	9	10
x_1	2	3	5	6	8	10	4	7	9	1
x_2	12	11	11	10	9	9	12	10	9	13
y	20	24	31	38	47	52	26	41	48	15

Regression on x_1 alone: $\hat{y} = 11.0 + 4.22x_1$, $R^2 = 0.989$. Multiple regression: $\hat{y} = 40.86 + 3.19\,x_1 - 2.29\,x_2$, $R^2 = 0.996$, $R^2_{\text{adj}} = 0.995$, $s_e = 0.944$. Interpretation: at a constant price an additional million of advertising gives on average $+3.19$ in sales (not 4.22: the simple regression "attributed" to advertising part of the price effect, because in weeks with more advertising the price was lowered, $r(x_1, x_2) = -0.95$); at constant advertising a price increase of one unit reduces sales by 2.29. Both coefficients are significant ($p = 0.00003$ and $p = 0.014$). The prediction for $x_1 = 6$, $x_2 = 10$: 37.2.

REMEMBER

R^2 does not decrease when variables are added, so $R^2_{\text{adj}} = 1 - \dfrac{SSE/(n-k-1)}{SST/(n-1)}$ is used to compare models. A strong correlation among the explanatory variables (*multicollinearity*) makes the estimates unstable: large SE and "jumping" coefficients. Categorical factors are entered through indicators (0/1): the coefficient on an indicator is the difference in means between the category and the base one, all else equal.

6 Key terms

Term	*In Russian*	Meaning
Standard error of the slope	*Стандартная ошибка наклона*	$s_e/\sqrt{S_{xx}}$
Slope t-test	*t-тест наклона*	$t = b_1/\operatorname{SE}(b_1)$, $n-2$ df
Prediction interval	*Интервал предсказания*	for a new observation
CI for the mean response	*Доверительный интервал отклика*	for the mean of y at x_0
Confounder	*Искажающая переменная*	related to both x and y
Simpson's paradox	*Парадокс Симпсона*	the relationship reverses sign within groups
Multiple regression	*Множественная регрессия*	several predictors
Multicollinearity	*Мультиколлинеарность*	strong relationship among predictors

Formulas of Lecture 13

$$\operatorname{SE}(b_1) = \frac{s_e}{\sqrt{S_{xx}}}, \quad t = \frac{b_1}{\operatorname{SE}(b_1)}\ (\nu = n-2), \quad b_1 \pm t_{\alpha/2}\operatorname{SE}(b_1); \quad F = \frac{SSR/1}{SSE/(n-2)} = t^2;$$

$$\hat{y}_0 \pm t_{\alpha/2} s_e \sqrt{\tfrac{1}{n} + \tfrac{(x_0-\bar{x})^2}{S_{xx}}}\ \text{(mean)}; \qquad \hat{y}_0 \pm t_{\alpha/2} s_e \sqrt{1 + \tfrac{1}{n} + \tfrac{(x_0-\bar{x})^2}{S_{xx}}}\ \text{(individual)}.$$

7 Practice problems

Problem level is shown in square brackets: **[A]** — basic, tests understanding of definitions; **[B]** — standard computation; **[C]** — advanced, requires a derivation or an interpretation. Decimals are written with a point, as in a calculator, Excel and R. If a problem needs tables, use the appendix "Statistical Tables".

Problem 1 [A] In a regression on $n = 27$ observations $b_1 = 2.4$, $\operatorname{SE}(b_1) = 0.6$. Test $H_0 : \beta_1 = 0$ ($\alpha = 0.05$) and construct a 95% CI for β_1.

Problem 2 [A] The sample correlation is $r = 0.45$ with $n = 30$. Test $H_0 : \rho = 0$ using $t = r\sqrt{n-2}/\sqrt{1-r^2}$ ($\alpha = 0.05$).

Problem 3 [A] For each statement name a probable lurking variable or an alternative explanation: (a) cities with more churches have more crime; (b) employees who have had more training earn more; (c) firms with high advertising have larger profits.

Problem 4 [B] Study-time data: $x = 1, 2, 3, 4, 5$, $y = 50, 55, 65, 70, 85$ (Lecture 12): $b_1 = 8.5$, $s_e = 3.03$, $S_{xx} = 10$.

(a) Find $\mathrm{SE}(b_1)$ and test the significance of the slope.
(b) Construct a 95% CI for β_1.
(c) Test $H_0 : \beta_1 = 5$ against $H_1 : \beta_1 \neq 5$.

Problem 5 [B] For the data of the previous problem, at $x_0 = 4.5$ hours find the 95% CI for the mean score and the 95% prediction interval for a new student. Explain the difference.

Problem 6 [B] Apartment data (Lecture 12): $\hat{y} = 5.79 + 0.776x$, $s_e = 1.576$, $n = 5$, $S_{xx} = 2320$, $\bar{x} = 66$.

(a) Test the significance of the slope and construct a 95% CI.
(b) Find the 95% CI for the mean price of 70 m^2 apartments and the prediction interval for the price of a particular 70 m^2 apartment.

Problem 7 [B] In the output of a regression on $n = 27$ observations it is known that: $SSR = 160$, $SSE = 90$, $b_1 = 2.4$.

(a) Fill in the ANOVA table (ν, MS, F).
(b) Find R^2 and s_e.
(c) Find the t-statistic of the slope and $\mathrm{SE}(b_1)$.

Problem 8 [B] Loan applications are submitted at two branches. Approved: branch X — small loans $18/20$, large loans $24/80$; branch Y — small $68/80$, large $5/20$. Find the approval rates by group and overall. Which branch is "better"? Explain the paradox.

Problem 9 [B] An online shop introduced free delivery for some customers, choosing those who order more often. A month later their average bill rose by 12% relative to the rest. Can we say that free delivery increased the bill? How should a study be organized to draw a causal conclusion?

Problem 10 [B] A model of weekly sales: $\hat{y} = 20 + 3x_1 - 1.5x_2$ (x_1 — advertising, x_2 — price); $n = 20$, $R^2 = 0.80$.

(a) Interpret the coefficients.
(b) Find the prediction at $x_1 = 8$, $x_2 = 10$.
(c) Find R^2_{adj}.

Problem 11 [C] Derive $\mathrm{Var}(b_1) = \sigma^2/S_{xx}$ using the representation $b_1 = \sum c_i y_i$, where $c_i = (x_i - \bar{x})/S_{xx}$, and the y_i are independent with variance σ^2.

Problem 12 [C] Show that $\mathrm{Var}(\hat{y}_0) = \sigma^2 \left(\dfrac{1}{n} + \dfrac{(x_0 - \bar{x})^2}{S_{xx}} \right)$ and explain where the "1+" in the prediction interval comes from.

STAT201 · STATISTICS · LECTURE 14

Project Lab: From Question to Conclusion

How to carry out a small statistical study, choose a method, present the result and make the work reproducible

Topic 14 of the syllabus · week 14 · 4 academic hours

LECTURE GOALS

- organize a study in stages: question → data → description → method → checking conditions → conclusion → report;
- choose a method according to the type of question and data, and plan the sample size;
- present results: a table, a graph, an effect estimate with a confidence interval, and a worded conclusion;
- ensure reproducibility (data, formulas, code) and avoid "tuning" results (p-hacking);
- use AI tools responsibly in data analysis: verify, do not take on trust.

1 Stages of a statistical study

1. **Question.** Formulated before the analysis and must be testable: "Do the monthly expenses of mobile-app customers and branch customers differ?" For each question fix the unit of observation, the population and the variables.
2. **Data.** Where they come from, how they were collected (random sample, experiment, observation), how many values are missing, which units are used. The way the data were collected determines which conclusions are allowed (lectures 7 and 13).
3. **Cleaning and description.** Check for entry errors, duplicates and outliers; descriptive statistics and graphs (lecture 1). Outliers are not removed automatically: find the reason and show the results with and without them.
4. **Method.** Chosen by the type of question (section 2).
5. **Checking conditions.** Independence, normality or large n, equal variances, linearity, etc.
6. **Conclusion.** Effect estimate, CI, p-value, practical significance, limitations.
7. **Report.** A short, precise text with a table/graph and honest caveats.

2 Choosing a method

Question	Data	Method
What is the mean / proportion?	1 quantitative / 1 categorical	CI (t or z), one-sample test
Do 2 groups differ?	quantitative, 2 independent groups	Welch test + CI for the difference + effect size
Did it change after an intervention?	paired measurements	paired t-test
Do proportions differ?	categorical, 2 groups	two-proportion z-test
Are two categories related?	two categorical	χ^2 test of independence, Cramér's V
Do ≥ 3 groups differ?	quantitative, k groups	ANOVA and pairwise comparisons with a correction
How does y depend on x?	two quantitative	regression: b_1, CI, R^2, residuals
Need a prediction for a new object?	regression	prediction interval

3 Worked mini-project

Question. Does the monthly spending of a customer (million UZS) differ between mobile-app users and customers served in branches? Data: a random sample of customers of one bank over one month, 12 app customers and 10 branch customers. For app users the customer tenure (years) is known.

App	1	2	3	4	5	6	7	8	9	10	11	12
Spending	4.2	5.1	3.8	6.0	4.9	5.5	4.4	5.8	6.3	4.7	5.2	5.0
Tenure	1	2	1	4	3	3	2	4	5	2	3	3

Branch	1	2	3	4	5	6	7	8	9	10
Spending	3.9	4.1	3.5	4.8	5.2	3.7	4.0	4.6	3.3	4.4

Description. App: $n = 12$, $\bar{x} = 5.075$, median 5.05, $s = 0.745$, range 3.8–6.3. Branch: $n = 10$, $\bar{x} = 4.150$, median 4.05, $s = 0.599$, range 3.3–5.2. There are no outliers, the distributions are symmetric, and the spreads are similar ($s_1/s_2 = 1.24 < 2$). The small sample sizes ($n < 30$) require approximate normality, which is acceptable in the absence of outliers and skewness.

Method and result. Two-sample Welch test ($H_0 : \mu_1 = \mu_2$). Difference 0.925 million, SE $= 0.286$, $t = 3.23$, $\nu \approx 20.0$, $p = 0.0042$. 95% CI for the difference: $(0.33;\ 1.52)$ million. Effect size (Cohen's d) $= \dfrac{0.925}{s_p} = \dfrac{0.925}{0.683} = 1.35$ — large. 95% CIs for the mean spending: app $(4.60;\ 5.55)$, branch $(3.72;\ 4.58)$.

Regression within the app group. Spending y on tenure x: $\hat{y} = 3.48 + 0.580\,x$, SE$(b_1) = 0.0626$, $t = 9.27$, $p < 0.001$, $R^2 = 0.896$, $s_e = 0.252$. The 95% CI for the slope is $(0.44;\ 0.72)$: each year of tenure is associated with an increase in mean spending of 0.44–0.72 million.

Wording the conclusion (an example of a good paragraph).

REMEMBER

"Among 22 bank customers, the mean monthly spending of app users (5.08 million UZS, $s = 0.74$, $n = 12$) is higher than that of branch customers (4.15, $s = 0.60$, $n = 10$). The difference was 0.93 million UZS (95% CI from 0.33 to 1.52; Welch test, $t(20.0) = 3.23$, $p = 0.004$). The effect is large ($d = 1.35$). Because customers were not randomly assigned to channels, the difference cannot be interpreted as the effect of the app itself: it may reflect differences in the customer mix (for example, in tenure and income). The sample is small, so the result should be confirmed on a larger data set."

4 How to report results

- First the answer to the question, then the number with units and CI, then the caveats. Do not make the reader search for the conclusion.
- Give the *effect size* and CI, not only the p-value; round p sensibly ($p = 0.004$; $p < 0.001$), not "$p = 0.0000$".
- Distinguish *statistical* from *practical* significance; interpret in the units of the problem.
- Do not write "proved" or "H_0 is accepted": say "the data are consistent with…" / "there is insufficient evidence…".
- Graphs: labeled axes with units, a zero baseline for bar charts, no needless effects; for small samples show the individual points.
- Tables: no more than 3 significant digits, with an explanation of the notation.
- State the number of observations, the sampling method, missing values and limitations (causality, representativeness).

4.1 Typical violations

Choosing the test after looking at the results; repeating tests until $p < 0.05$ (p-hacking); deleting "inconvenient" observations without justification; ignoring multiple comparisons (20 tests give a 64% chance of a false discovery); interpreting correlation as causation; reporting only "significant" results (publication bias); applying a method without checking its conditions.

5 Planning: sample size for comparing two proportions

For $\alpha = 0.05$ (two-sided) and power 80%, the number of observations *in each group* is

$$n \approx \frac{(z_{\alpha/2} + z_{\beta})^2 \left[p_1(1-p_1) + p_2(1-p_2)\right]}{(p_1 - p_2)^2}, \qquad (z_{\alpha/2} + z_{\beta})^2 = (1.96 + 0.8416)^2 = 7.85. \quad (5.1)$$

To detect a rise in conversion from 10% to 12%: $n \approx 7.85 \cdot (0.09 + 0.1056)/0.0004 = 3838$ per group. Small effects require very large samples, which is why it is important to define the minimum practically significant effect in advance.

6 Reproducibility

A result is reproducible if another person gets the same numbers from your data and description. Practice: keep the raw data unchanged and the cleaned data separately; record all transformations as formulas/scripts rather than manual edits; keep file versions; state the random seed for random procedures; comment on what each step does.

Task	Excel	Python (scipy)
Mean, s	`AVERAGE`, `STDEV.S`	`x.mean()`, `x.std(ddof=1)`
$P(Z < z)$	`NORM.S.DIST(z,TRUE)`	`norm.cdf(z)`
Critical t	`T.INV.2T(0.05,df)`	`t.ppf(0.975,df)`
Welch test	`T.TEST(A,B,2,3)`	`ttest_ind(a,b,equal_var=False)`
Paired test	`T.TEST(A,B,2,1)`	`ttest_rel(a,b)`
χ^2 test	`CHISQ.TEST(obs,exp)`	`chi2_contingency(obs)`
ANOVA	Data Analysis → ANOVA	`f_oneway(g1,g2,g3)`
Regression	`LINEST`, Data Analysis	`linregress(x,y)`

7 Responsible use of AI

Language models and other AI tools can explain methods, help write code and formulas, and suggest a report structure. But:

- **Check the numbers.** A model may make arithmetic errors and invent table values; recompute key results independently (calculator, Excel, code) and compare with the tables.
- **Check the method.** Make sure the test fits the data and the conditions hold; AI often picks the "standard" method without looking at the data structure (paired samples treated as independent!).
- **Do not invent data or sources.** Any data must be real or explicitly marked as teaching/simulated.
- **Protect personal and commercially sensitive data:** do not upload it to external services without permission; aggregate and anonymize.
- **Disclose the use of AI** in accordance with your organization's rules; you are responsible for the conclusion.
- **Do not replace understanding.** If you cannot explain why the method was chosen and what the result means, the report is not ready.

8 Key terms

Term	*In Russian*	Meaning
Effect size	*Размер эффекта*	difference / Cohen's $d = \Delta / s_p$
Reproducibility	*Воспроизводимость*	the same data and code give the same numbers
p-hacking	*p-hacking*	tuning the analysis until p is significant
Multiple comparisons	*Множественные сравнения*	growth of Type I error with many tests
Representativeness	*Репрезентативность*	the sample reflects the population

Formulas of lecture 14

$$d = \frac{\bar{x}_1 - \bar{x}_2}{s_p}, \quad s_p = \sqrt{\frac{(n_1-1)s_1^2 + (n_2-1)s_2^2}{n_1+n_2-2}};$$

$$n_{\text{group}} \approx \frac{(z_{\alpha/2} + z_{\beta})^2 [p_1(1-p_1) + p_2(1-p_2)]}{(p_1 - p_2)^2};$$

Bonferroni: $\alpha^* = \alpha / m$; $\quad P(\text{at least one false alarm}) = 1 - (1-\alpha)^m$.

9 Practice problems

> Problem level is shown in square brackets: **[A]** — basic, tests understanding of definitions; **[B]** — standard computation; **[C]** — advanced, requires a derivation or an interpretation. Decimals are written with a point, as in a calculator, Excel and R. If a problem needs tables, use the appendix "Statistical Tables".

Problem 1 [A] For the project "What influences the loan amount?", classify the variables and suggest one graph for each: (a) loan amount; (b) loan type (consumer / mortgage / auto); (c) borrower's years of employment; (d) region; (e) loan amount versus years of employment.

Problem 2 [A] Choose a method: (a) has the mean application processing time changed after a new system was introduced (for the same 30 employees); (b) are the customer's gender and the chosen card type related; (c) do the mean service ratings differ across 4 branches; (d) by how much does spending grow when income grows by 1 million; (e) do the default rates differ between two lending schemes.

Problem 3 [A] Using the data of the worked mini-project (branch: $n = 10$, $\bar{x} = 4.15$, $s = 0.599$), construct a 95% CI for the mean spending of branch customers.

Problem 4 [B] Two samples of customers: group 1: $n = 40$, $\bar{x} = 52$, $s = 10$; group 2: $n = 45$, $\bar{x} = 47$, $s = 12$.

(a) Find Cohen's d.
(b) Carry out the Welch test and construct a 95% CI for the difference.

(c) Do “statistical” and “practical” significance agree? How would you word the conclusion?

Problem 5 [B] Find the flaws in this report: “We surveyed 15 customers chosen by the manager from those who agreed to answer. The mean rating is 4.5 out of 5. The test gave $p = 0.00$, so our service is unquestionably the best in the industry, and the new loyalty program **proved** an increase in satisfaction. Outliers (two ratings of 1) were removed.”

Problem 6 [B] An analyst split customers into 20 subgroups (by age, region, channel, etc.) and for each tested whether the mean purchase differs from the bank-wide mean ($\alpha = 0.05$). In 2 subgroups $p < 0.05$, and he published only those.

(a) What is the probability of at least one “significant” subgroup if there are in fact no differences (tests independent)?
(b) What is the p threshold under the Bonferroni correction? How should the results be reported correctly?

Problem 7 [B] The spending data are in columns A (app, A2:A13) and B (branch, B2:B11). Write Excel formulas for: (a) the mean and s of column A; (b) the p-value of the Welch test; (c) the critical t for a 95% CI for the mean of A; (d) $P(Z < 1.5)$. What is the result of (b)?

Problem 8 [B] For a pilot sample $s = 0.745$. What sample size is needed so that the 95% CI for the mean spending has a margin of error $E = 0.2$ million UZS?

Problem 9 [B] Regression of spending on tenure (app): $\hat{y} = 3.48 + 0.580x$, $\mathrm{SE}(b_1) = 0.0626$, $n = 12$, tenure observed from 1 to 5 years.

(a) Construct a 95% CI for β_1.
(b) Predict the mean spending at a tenure of 3 years. Can a prediction be made for a tenure of 10 years?
(c) Does $b_1 > 0$ mean that tenure increases spending?

Problem 10 [B] In the app data, customer 9 had spending 6.3. Suppose 14.5 was mistakenly entered. Recompute the mean, s and the Welch test (branch unchanged). Compare with the original result and state what should be done with such a value.

Problem 11 [C] An AI assistant reported for the mini-project data: “$t = 2.90$, $\nu = 22$, $p = 0.008$, the difference of means is significant.” How can you check this claim? What is wrong?

Problem 12 [C] For an A/B test of conversion the baseline proportion is 30%; you need to detect a rise to 35% ($\alpha = 0.05$, power 80%). (a) Find n per group. (b) What happens if you need to detect a rise to 32%? (c) Why is it important to plan in advance?

STAT201 · STATISTICS · LECTURE 15

Final Review and Exam Preparation

Course summary, a master table of methods, a formula sheet and a practice exam paper

Topic 15 of the syllabus · week 15 · 4 academic hours

LECTURE GOALS

- link the topics of the course into one chain: data → probability → distributions → sampling → estimation → hypothesis testing → regression;
- quickly choose a method and formula from the statement of a problem;
- use the z, t, χ^2, F tables (appendix “Statistical Tables”);
- solve a practice exam in the format of the final assessment and evaluate your own preparation.

1 The running logic of the course

The whole course answers one question: *how to make well-founded decisions under uncertainty*. The main links:

1. **Describing data** (lecture 1): center, spread, shape, outliers, the relationship between two variables.
2. **Probability** (2–3): events, addition and multiplication rules, conditional probability, Bayes’ theorem.
3. **Random variables** (4–6): expectation and variance, binomial, Poisson, normal and other distributions, covariance and correlation.
4. **Linking sample and population** (7–8): sampling distribution, standard error, CLT.
5. **Statistical inference** (9–11): CIs and hypothesis tests for a mean, proportion, difference, categories and several groups.
6. **Models** (12–13): regression, inference, causality.
7. **Practice** (14): study design, report, reproducibility, responsible use of AI.

The general template of inference is the same everywhere: $\dfrac{\text{estimate} - \text{value under } H_0}{\text{SE}}$ is compared with the normal or t distribution. The ability to find the right SE and the right number of degrees of freedom is the core of the course.

2 Master table of standard errors and tests

Parameter	Standard error	Distribution	Degrees of freedom
μ (σ known)	$\sigma/\sqrt{n}$	$N(0,1)$	—
μ (σ unknown)	$s/\sqrt{n}$	t	$n-1$
p	$\sqrt{p_0(1-p_0)/n}$ (test); $\sqrt{\hat{p}(1-\hat{p})/n}$ (CI)	$N(0,1)$	—
$\mu_1-\mu_2$ (indep.)	$\sqrt{s_1^2/n_1+s_2^2/n_2}$	t (Welch)	Welch formula
μ_d (pairs)	$s_d/\sqrt{n}$	t	$n-1$
p_1-p_2	$\sqrt{\hat{p}(1-\hat{p})(\frac{1}{n_1}+\frac{1}{n_2})}$ (test)	$N(0,1)$	—
β_1	$s_e/\sqrt{S_{xx}}$	t	$n-2$
$r\times c$ table	$\sum(O-E)^2/E$	χ^2	$(r-1)(c-1)$
k means	$F=MSB/MSW$	F	$k-1$; $N-k$

3 Formula sheet of the course

Data description and probability

$$\bar{x}=\tfrac{1}{n}\sum x_i,\quad s^2=\tfrac{1}{n-1}\sum(x_i-\bar{x})^2,\quad CV=\tfrac{s}{\bar{x}},\quad IQR=Q_3-Q_1;$$

outlier fences: $Q_1-1.5\,IQR$ and $Q_3+1.5\,IQR$;

$$P(A\cup B)=P(A)+P(B)-P(A\cap B);\quad P(A\cap B)=P(A)P(B\mid A);$$

$$P(A\mid B)=\tfrac{P(A\cap B)}{P(B)};\quad C_n^k=\tfrac{n!}{k!(n-k)!};\quad P(A_i\mid B)=\tfrac{P(B|A_i)P(A_i)}{\sum_j P(B|A_j)P(A_j)};$$

odds: $\text{odds}_{\text{post}}=\text{odds}_{\text{prior}}\cdot LR$.

Random variables and distributions

$$\mu=\sum xp(x),\quad \sigma^2=\sum x^2p(x)-\mu^2;\quad \mathrm{E}[aX+b]=a\mu+b,\quad \mathrm{Var}(aX+b)=a^2\sigma^2;$$

$$\mathrm{Bin}(n,p):\ \mu=np,\ \sigma^2=np(1-p);\qquad \mathrm{Pois}(\lambda):\ \mu=\sigma^2=\lambda;$$

$$\mathrm{Exp}(\lambda):\ \mu=\tfrac{1}{\lambda};\qquad U(a,b):\ \mu=\tfrac{a+b}{2},\ \sigma^2=\tfrac{(b-a)^2}{12};$$

$$z=\tfrac{x-\mu}{\sigma};\quad \mathrm{Cov}(X,Y)=\mathrm{E}[XY]-\mu_X\mu_Y;\quad \rho=\tfrac{\mathrm{Cov}}{\sigma_X\sigma_Y};$$

$$\mathrm{Var}(X\pm Y)=\mathrm{Var}\,X+\mathrm{Var}\,Y\pm 2\,\mathrm{Cov}\,.$$

Inference and regression

$$\mathrm{SE}(\bar X)=\tfrac{\sigma}{\sqrt{n}};\quad \bar x\pm t_{\alpha/2,n-1}\tfrac{s}{\sqrt{n}};\quad \hat p\pm z_{\alpha/2}\sqrt{\tfrac{\hat p(1-\hat p)}{n}};$$

$$n\ge\left(\tfrac{z\sigma}{E}\right)^2;\quad n\ge\tfrac{z^2p(1-p)}{E^2};$$

$$t=\tfrac{\bar x-\mu_0}{s/\sqrt{n}};\quad z=\tfrac{\hat p-p_0}{\sqrt{p_0(1-p_0)/n}};\quad t=\tfrac{\bar x_1-\bar x_2}{\sqrt{s_1^2/n_1+s_2^2/n_2}};\quad t=\tfrac{\bar d}{s_d/\sqrt{n}};$$

$$\chi^2=\sum\tfrac{(O-E)^2}{E};\quad F=\tfrac{SSB/(k-1)}{SSW/(N-k)};$$

$$b_1=\tfrac{S_{xy}}{S_{xx}}=r\tfrac{s_y}{s_x};\quad b_0=\bar y-b_1\bar x;\quad R^2=\tfrac{SSR}{SST};\quad s_e=\sqrt{\tfrac{SSE}{n-2}};\quad \mathrm{SE}(b_1)=\tfrac{s_e}{\sqrt{S_{xx}}};$$

$$\hat y_0\pm t\,s_e\sqrt{1+\tfrac1n+\tfrac{(x_0-\bar x)^2}{S_{xx}}}.$$

4 Typical exam mistakes

- Mixing up $\sigma/\sqrt{n}$ (mean), $\sigma\sqrt{n}$ (sum) and σ (a single observation).
- Not checking the conditions of applicability (normal approximation, independence, $n\ge30$, expected counts ≥5).
- z instead of t when σ is unknown and n is small; the wrong number of degrees of freedom.
- The wrong direction of the alternative and a wrong p-value (doubling for a two-sided test).
- Interpreting a CI or p-value as the "probability of the hypothesis"; concluding that "H_0 was proved".
- Analyzing paired data as independent, and vice versa.
- Extrapolating a regression; drawing causal conclusions from correlation.
- Missing units and a verbal interpretation in the answer.

5 How to prepare and how to take the exam

1. Solve the practice paper (below) in 120 minutes without hints, then check against the solutions and analyze each mistake: which topic does it belong to.
2. Make your own one-page "formula sheet" (even if tables are allowed); this is the best way to memorize formulas.
3. In the exam: read all the problems; start with those you know how to do; in each one write "given / find / method / computation / conclusion in words".
4. Check reasonableness: probability in $[0,1]$, variance ≥0, $|r|\le1$, a CI symmetric about the estimate, a p-value consistent with the decision and the CI.
5. Round only at the end; keep intermediate values to 4–5 digits.

6 Key terms of the course

Term	*In Russian*	Meaning
Sampling distribution	*Выборочное распределение*	distribution of a statistic
Standard error	*Стандартная ошибка*	SD of a statistic
Confidence interval	*Доверительный интервал*	estimate $\pm$ critical$\times$SE
p**-value**	*p-значение*	$P(\text{at least as extreme} \mid H_0)$
Power	*Мощность*	$1-\beta$
Residual	*Остаток*	$y-\hat{y}$
Confounder	*Искажающая переменная*	related to x and y

7 Practice problems

> Problem level is shown in square brackets: **[A]** — basic, tests understanding of definitions; **[B]** — standard computation; **[C]** — advanced, requires a derivation or an interpretation. Decimals are written with a point, as in a calculator, Excel and R. If a problem needs tables, use the appendix “Statistical Tables”.

Problem 1 [A] Monthly sales (million UZS) of eight managers: $14,\ 18,\ 20,\ 22,\ 25,\ 27,\ 31,\ 63$.

(a) Find the mean, median and standard deviation.
(b) Find Q_1, Q_3, IQR and the outlier fences. Are there outliers?
(c) Which measure of center is preferable and why? How does the mean change without the largest value?

Problem 2 [B] The default rate among borrowers is 4%. A scoring model flags 80% of future defaulters and 10% of reliable borrowers as “risky”. A borrower is flagged. What is the probability that he will default? What does the probability become if the borrower is flagged by two independent scoring models with the same characteristics?

Problem 3 [B] A customer accepts an offer with probability 0.25, independently of others; the offer is made to 12 customers.

(a) Find the expected number of acceptances and the standard deviation.
(b) Find the probability that at least 5 accept.
(c) Find the probability that exactly 3 accept.

Problem 4 [A] Test scores $X \sim N(65, 8^2)$. Find $P(X > 75)$, $P(60 < X < 70)$ and the 85th percentile.

Problem 5 [B] Number of purchases per customer: $\mu = 2.5$, $\sigma = 1.2$. A sample of $n = 64$ customers. Find $\mathrm{SE}(\bar{X})$ and $P(\bar{X} > 2.7)$. Explain why the normal distribution can be used.

Problem 6 [B] A sample of 20 customers: $\bar{x} = 48$ min, $s = 9$. (a) Construct a 95% CI for the mean. (b) Test $H_0 : \mu = 45$ against $H_1 : \mu \neq 45$ ($\alpha = 0.05$). (c) Are the results of (a) and (b) consistent?

Problem 7 [B] In a sample of 250 customers, 62% use the app. (a) Construct a 95% CI for the proportion. (b) How many customers must be surveyed for the margin of error to be 4 percentage points (with the proportion unknown)?

Problem 8 [B] Processing time (min) for 7 employees before and after a new system was introduced: before $30, 28, 35, 40, 25, 33, 38$; after $27, 26, 33, 36, 25, 30, 35$. Test whether processing time decreased (two-sided test, $\alpha = 0.05$) and construct a 95% CI for the mean change.

Problem 9 [B] A survey of 350 customers: age group and app use.

	Use the app	Do not use
Under 30	60	40
30–45	50	70
Over 45	40	90

Test independence ($\alpha = 0.05$). Which group contributes most to χ^2?

Problem 10 [B] ANOVA for 4 groups of 5 observations each: $SSB = 84$, $SSW = 160$. Complete the table and draw a conclusion ($\alpha = 0.05$).

Problem 11 [B] For $n = 10$ observations: $\bar{x} = 20$, $\bar{y} = 50$, $S_{xx} = 400$, $S_{xy} = 600$, $S_{yy} = 1200$.

(a) Find the regression equation, R^2 and s_e.
(b) Test the significance of the slope and construct a 95% CI for β_1.
(c) Predict y at $x = 25$ and construct a 95% prediction interval.

Problem 12 [C] A researcher found a positive significant relationship between the number of training sessions held by a bank and the mean income of customers in the region ($r = 0.62$, $p < 0.01$) and proposed increasing the number of training sessions to raise income. Give a critical assessment of the conclusion and suggest how causality could be tested.

Problem 13 [C] A manager wants to know whether customer satisfaction (rating 1–5) differs across three branches and chooses to "run three pairwise t-tests A–B, A–C, B–C at the 0.05 level". Why is this wrong? What should be proposed? What value of α should be used for each pairwise comparison under the Bonferroni correction?

Problem 14 [C] Carry out a mini-analysis. In data on 200 customers the mean spending is $\bar{x} = 4.8$ million, $s = 1.5$. (a) Construct a 95% CI for the mean. (b) The bank claims $\mu = 5$. Test it. (c) What is the effect size $d = (\bar{x} - \mu_0)/s$? (d) Is $n = 200$ sufficient to detect a deviation of 0.2 million with 80% power?